Thesis proposal: Sociological constraints on action planning and interpretation

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1 Introduction

The relationships we are in, the roles we occupy, and the groups we belong to shape both how we act and how we interpret other people's behavior. The same everyday action – sharing food from a single container, doing a small favor, or choosing a particular word – can feel ordinary in one social context but uncomfortable, rude, or surprising in another. Yet standard cognitive accounts of action planning and interpretation often leave this sociological structure unspecified: models and tasks typically focus on anonymous, interchangeable agents acting in carefully specified physical environments. This thesis expands these accounts by formalizing how sociological structure shapes the utilities that guide decision making in everyday social life.

In standard computational accounts of theory of mind, observers infer mental states by inverting a model of how rational agents plan actions to maximize utility, given their beliefs and desires, under the constraints of the physical environment (C. L. Baker et al., 2009, 2017; Jara-Ettinger et al., 2016). These models have been remarkably productive: they have been extended to capture cooperation (Kleiman-Weiner et al., 2025), collaboration (Wu et al., 2021; Xiang et al., 2023), referential communication (Goodman & Frank, 2016), metacognition (Berke et al., 2023), higher-order goals like reputation (Yoon et al., 2020), and emotional and moral learning (Houlihan et al., 2023; Radkani et al., 2025). Their power lies in formally unifying behaviors that look very different on the surface – taking a longer path, waiting longer, choosing one food over another – as expressions of the same underlying computation over costs and rewards. This means that, in these models, sparse observations can support joint inference over multiple (often seemingly unrelated) variables via rational belief updating.

These models specify the *physical* environment in detail. Walls, distances, terrain, and time determine the effort required to act, and are typically hand-specified as variables that shape the utility landscape in which agents plan. The *sociological* environment, by contrast, is largely left out. When it has been incorporated, it has usually been through two complementary but partial ways. The first way is modeling relationships as patterns of "adopted utility": a positive weight on another's outcomes captures cooperation, and a negative weight captures competition (Jern & Kemp, 2014; Kleiman-Weiner et al., 2017; Powell, 2022; Ullman et al., 2009). The second is to model social roles – barista and customer, parent and child – by enumerating role-specific actions and utilities (A. Baker et al., 2025; Davis et al., 2026; Jara-Ettinger & Dunham, 2024). Each approach captures an important special case of social behavior. Adopted utility characterizes relationships along a single cooperation-competition axis, capturing states of individuals that may be more or less likely in particular relationships. Role-based accounts handle a fixed inventory of roles by stipulating additional utilities that may be associated with role fulfillment. However, neither approach offers a general way of describing how the social environment shapes the costs and benefits of action within an integrated planning framework.

A more general approach is suggested by a long-standing idea from social psychology. Lewin (1943) argued that behavior is a function of person and environment, where "environment" in-

cludes the full set of physical and sociological forces acting on the individual. This view suggests that these forces should be treated within a common framework: just as physical structure – walls, distances, terrain, and other object features – shapes which actions are possible and costly, sociological structure shapes which actions are appropriate, expected, or expressive of social relationships and groups. At the time, this proposal did not specify how these forces might combine, and models of rational action planning did not yet exist, so the proposal could not be systematically tested. Cognitive models of rational decision making now make it possible to formalize this idea and test it in experimental settings where its influence can be observed.

This thesis incorporates sociological structure into the same utility-landscape framework that already organizes physical structure, beliefs, and goals. New components of people's utilities arise from social outcomes, such as social harmony and the avoidance of awkwardness or discomfort. Existing utility components, such as the value of an action, can also be reshaped by social context. In this view, relationships, hierarchies, and group memberships are both constraints on action and targets of inference: they shape what agents do, and observers can also recover them from sparse behavior in the same way they can jointly infer beliefs, goals, and physical constraints. Capturing these sociological aspects in an integrated framework shows how observers' beliefs about some variables can constrain their beliefs about others, in ways analogous to how inverse-planning models already handle physical structure.

Testing this proposal requires experiments that capture aspects of naturalistic social interaction where social structure is expected to matter, as well as more expressive models that can generate predictions in these settings. Recent methodological advances – including frameworks for rapidly developing large-scale synchronous experiments, statistical models that map continuous stimuli into the representational spaces of cognitive models, and advances in natural language processing – allow us to make progress in both directions.

We develop our proposal through three case studies, each pairing a well-characterized sociological dimension with a domain of action where that dimension has long been argued, in other disciplines, to play a central role in decision making. The first case study examines *relationship intimacy* in the domain of food sharing and other vulnerable or boundary-crossing actions (Chapter 2). Anthropological and social-psychological work has long observed that the comfort of sharing or exchanging substances, self-disclosing, and maintaining physical boundaries tracks the closeness or intimacy of a relationship (Fiske, 1991; Miller et al., 1998; Thomas et al., 2022). Here, we model how relationship intimacy modulates the discomfort, or cost, of actions that cross interpersonal boundaries. This domain is an ideal initial test case for an integrated social inverse-planning model: food is one of the primary settings in which inverse-planning models have been developed and tested, and the mechanics and properties of food make it possible to experimentally separate standard utilities, such as the value of eating the food and the cost of obtaining it, from relationship-focused utilities, such as the discomfort of sharing food in a particular way. If relationship intimacy reshapes the utility landscape of food-related and other vulnerable actions, then observers should be able to use intimacy to predict other people's actions. They should also be able to use inverse

planning to recover intimacy, as well as the standard targets of inverse planning, from sparse observations. We show that an integrated social inverse-planning model is needed to capture behavior in these contexts.

The second case study examines *relationship hierarchy* in the domain of generous and prosocial action. Classical theory and laboratory experiments have emphasized reciprocity: how a generous act creates an expectation that the recipient will return the favor. Daily-life and ethnographic observation, however, suggest a different pattern: in hierarchical relationships, generosity often sets a precedent for the same actor to be generous again. These expectations have not been directly tested experimentally. In this research, we show how the hierarchy of a social relationship affects when people expect reciprocity. First, we systematically test the conditions under which people have different expectations for sequences of generous acts (Chapter 3). Then, we will test whether these expectations, and corresponding inferences from behavior, reflect the same underlying utility-theoretic logic operating over different relational structures, and whether this logic can capture people's interpretations of the meanings and moral judgments of generous acts (Chapter 4). We propose that formalizing this process requires a model that incorporates how, in addition to straightforward weights on others' welfare, the degree of hierarchy or asymmetry in a relationship can change the psychological cost of accepting inequality. Extending this framework to divisions of tasks and labor, we then examine how relationship structure and the value of actions must be continually inferred and managed together. This mutual dependence predicts when the same observed arrangement supports a shared relationship interpretation and when it remains underdetermined, allowing partners with different priors to rationally reach different conclusions about whether the relationship is equal, hierarchical, fair, or sustainable, from the same observations (Chapter 5).

The third case study examines *social groups and affiliation* in the domain of referential language. Existing cognitive accounts of referential language treat speakers as balancing informativity and efficiency: speakers want to help listeners identify the right object or concept without saying more than necessary. Sociolinguistic accounts, however, suggest that referential language is also diagnostic of social identity: utterances convey information not only about the world, but also about the speaker. Speakers in sociological contexts may therefore have an additional utility for strategically signaling their group identity to listeners. We first show that a model with a social-signaling component is needed to capture how people actively and flexibly trade off the referential utility of being understood against the social utility of being identified as an in-group member (Chapter 6). Then, we test how active social-signaling motivations affect population-level aspects of language through large interactive experiments in experimental microsocieties, asking how these motivations shape the emergence and maintenance of linguistic diversity (Chapter 7). Together, these studies argue that social identity signaling actively shapes language use and language change, at both the local and population levels.

Overall, this thesis expands cognitive accounts of action planning and understanding by treating sociological structure as both a constraint on action and a target of inference. It situates cognitive accounts of rational behavior within everyday social interaction, where sociological structure

affects how people plan and interpret actions.

2 Thesis contents

2.1 Chapter 1: The sociological landscape of action planning and interpretation

Authors: TBD

Progress: Not started.

Plan: This will be a review article / theory piece expanding the concepts in the introduction above.

2.2 Chapter 2: Inverse planning of social interactions in relationships

Authors: Alicia Chen, Ashley Thomas, Josh Tenenbaum, Rebecca Saxe

Status: In progress. Some experiments and early versions of the analyses will be published as a conference paper in the 2026 CogSci proceedings and is attached. An early draft of the manuscript for the full journal version is also attached.

Abstract: Cognitive models of inverse planning typically explain actions in terms of physical costs and rewards: agents choose goals by trading off the value of the goal against the effort needed to achieve the goal. Yet many everyday actions are also shaped by relationship-dependent utilities. We extend inverse-planning models by incorporating how social relationship intimacy modulates the discomfort (a kind of cost) of actions that cross interpersonal boundaries. Across experiments, human judgments about actions, desires, effort, and relationships are best captured by models that include this relationship structure. Joint inference over these variables allows ordinary actions to reveal not only what people want and what the physical world permits, but also what kinds of relationships they have. This work situates inverse planning within everyday social interaction, and provides a formal cognitive account of how social structure is recognized, sustained, and constituted through action.

2.3 Chapter 3: Expectations of reciprocal generosity are specific to equal relationships

Authors: Alicia Chen, Rebecca Saxe

Status: Completed (in press at a journal). The manuscript is attached.

Abstract: When someone acts generously, what expectations does this create? Classic theories and experiments emphasize reciprocity: the idea that generosity will be returned by the recipient. Yet daily-life and ethnographic observations often show a different pattern: generosity sets a precedent, leading people to expect the same person to be generous again. Across six online behavioral experiments (total $N = 599$ U.S. adults) using third-party vignette judgments and first-person incentivized economic games, we test when generosity creates expectations of reciprocation versus a precedent. We found that people expected reciprocity only in equal or symmetric relationships. Otherwise, they expected generosity to continue from the same actor. These expectations generalized across roles, contexts, and cost structures. Taken together, the results suggest that classic evidence on reciprocity and turn-taking may capture expectations among strangers or equals, but not the wider set of relationships that structure much of social life.

2.4 Chapter 4: Meanings and interpretations of sequences of generosity in social relationships

Authors: Alicia Chen, Rebecca Saxe

Status: In progress. Early parts and versions have been part of different manuscripts. Some early experiments were part of a preprint that was later split in two (the first part became part of Chapter 3). An early version of a cognitive model was published in the 2024 Cognitive Science Society proceedings. We are reworking the model and scope of this project.

Plan: This project will use formal model comparison to test how people weigh different considerations when they are deciding whether to act generously. Standard approaches to modeling generosity consider how people weigh their own utility (either directly or indirectly through expectations of future generosity), the other person's utility, and the cost of unequal outcomes (inequity aversion). Here we test whether people have an intuitive theory that the cost of unequal outcomes is modulated by the social relationship (where, specifically, a hierarchical relationship decreases the cost of unequal outcomes), and whether such a model can generate people's inferences about actions and relationships better than a model that does not include relationship-based utilities. The total utility in the full social model is:

$$U_{\text{tot}} = U_{\text{self}} + w_{\text{other}} \cdot U_{\text{other}} - w_{\text{IA}} \cdot \frac{U_{\text{IA}}}{h} \tag{1}$$

where h corresponds to the degree of hierarchy in the relationship. As h increases, the cost of unequal outcomes approaches 0.

This project will tie together various unpublished/unsubmitted experiments and/or alter features of existing experiments to make them more amenable to testing the features of the model. For example, Study 2, Study 3, and Study 4 in https://osf.io/preprints/psyarxiv/6kfwr_v1 are unpublished; the results of those studies show that sequences of generous action are informative for relationships, that people use relationships and the past sequence of actions to judge how moral/fair/satisfied/annoyed they expect people to be from observing a single generous act, and that people expect that a primary motivation for alternating generosity is to communicate about an intended equal relationship. Furthermore, Studies 5 and 6 in Chapter 3 have some features that do not make those studies directly amenable to model comparison (the payoff structure needs changes because it doesn't correspond to the costs and benefits of the generous acts themselves), so the plan is to use a similar design but different payoff structure that allows us to use values for U_{self} and U_{other} directly from the task.

A preliminary implementation of the model using the data from Chapter 3 (Study 6) shows that the full social model is needed to capture the full pattern of results. The plan right now is to (1) run a version of Study 6 above with different payoffs and re-fit the model, (2) implement the

inverse version of the model and see how much it can explain the pattern of results in Study 2 in the preprint linked above, (3) explore ways to fit the model to the Study 2 data (i.e., how to estimate the costs and benefits of each generous act and incorporate it in the model to try to explain scenario-specific variance), (4) explore what features of this model might be able to capture people's moral judgments (Study 3 in the preprint above).

2.5 Chapter 5: How divergent interpretations from divisions of labor arise from rational inference

Authors: TBD

Status: In progress. A UROP is currently working on this.

Plan: Real-world divisions of labor – household chores, lab tasks, splits of work between collaborators – are common areas of disagreement. Partners who jointly observe the same pattern of who-does-what can come away with drastically different beliefs about what the arrangement says about their relationship: whether it is equal or hierarchical, and whether the way tasks are divided is fair or sustainable. A standard approach in psychology is to attribute this disagreement to motivated reasoning or self-serving biases (Ross & Sicoly, 1979; Thompson & Loewenstein, 1992). This project tests a different explanation: that divergent interpretations of a relationship can arise rationally from the same observations, as a direct consequence of how people jointly infer and manage relationship structure and the value of actions.

We treat relationship R and the action value of a task V_i as variables that are jointly inferred from observed sequences of shared tasks. Partners may begin with different priors about V – about whether two tasks are comparable in cost, importance, or kind. They may also begin with different priors about R – how equal and hierarchical a relationship is. Crucially, the value of an action is constructed and communicated about within the relationship in which it is performed, and R and V have to be repeatedly managed and inferred. Consider the case where there are two tasks, with corresponding values V_1 and V_2:

$$P_{t+1}(R, V_1, V_2 \mid \text{observation}) \propto P(\text{observation} \mid R, V_1, V_2) \cdot P_t(R, V_1, V_2) \tag{2}$$

This account supports numerous predictions that may be generated through a single assumption about the likelihood (that symmetric relationships are expected to yield equal payoffs over a sequence of actions). First, two people observing the same arrangement of tasks, but holding even slightly different priors over whether the tasks are equal in value, may rationally reach different conclusions about whether the relationship is symmetric or asymmetric. Second, the function of alternating individual tasks can 'close' the channel through which divergence may enter: alternation guarantees equal payoffs without requiring agreement on the value of the underlying actions, so observing it can instead lead to inferences of an equal relationship even when priors over V don't align.

A 'division of labor' arrangement, by contrast, where one person always does task 1 (with value V_1) and the other person always does task 2 (with value V_2) leaves the inference underdetermined precisely to the extent that priors over V (and R) differ. This can predict when, and how much, partners' interpretations of the same arrangement should diverge — and how that divergence

should grow or shrink if new evidence about action value is observed.

To test these predictions, we plan a sequence of behavioral experiments that elicit relationship and task value inferences from third-party observers across division-of-labor scenarios. **Experiment 1** tests the model's core likelihood assumption directly: given a described relationship type, what sequences of task assignments do people expect? Participants are given symmetric or asymmetric relationships and asked to predict the divisions of tasks they would expect to observe. **Experiment 2** tests how priors over V shape inferences about R, in a 3×2 design crossing the described values of the two tasks (equal, unequal, uncertain) with the observed sequence (alternating, division-of-labor). The model predicts that division-of-labor arrangements yield more asymmetric relationship inferences when value priors are unequal or uncertain than when they are equal, while alternating arrangements yield convergent symmetric relationship inferences regardless of value priors.

Experiment 3 extends the same design to within-dyad divergence. Participants reason about two characters who hold different priors over V and judge what each character infers about the relationship, as well as how stable or satisfying they think the relationship is. The model predicts predictably divergent posteriors over R from the same observed division-of-labor sequences, but not for for alternating arrangements. **Experiment 4** tests the reverse inference direction. Using alien-planet scenarios in which participants have no external source of value information, we manipulate R and measure inferred V. This tests how priors about relationships affect how people learn the values of actions.

Each experiment uses both remote-society scenarios (to control prior beliefs about action value) and US-context scenarios (where priors on action values can be rated by participants and by language models). Experiments can also perturb one partner's beliefs about V midway through a sequence and tracing how their posterior over R diverges from their partner's. Overall, the goal of this project is to show that a familiar puzzle – that people often look at the same arrangement of tasks or labor and disagree about what it means – can arise from rational processes. More broadly, the project connects cognitive models of social inference to a longer tradition in sociology and linguistic anthropology on how social meaning is constructed within relationships (Bucholtz & Hall, 2005).

2.6 Chapter 6: Signaling social identity in referential communication

Authors: Alicia Chen, Robert Hawkins, Rebecca Saxe

Status: Completed (under revision at a journal). The manuscript is attached.

Abstract: Any choice of words simultaneously conveys information about the world and, at the same time, conveys information about the speaker, revealing aspects of their social identity. In this paper, we investigate how speakers strategically modify referential language to signal group membership. Across four experiments using a minimal referential communication paradigm, we find that speakers with the explicit goal of signaling social affiliation (1) choose more concise utterances, (2) preferentially select group-specific referents and descriptions, and (3) resist the otherwise strong tendency to be understood by everyone in the audience. Standard models of referential communication that focus on the trade-off between informativity and efficiency cannot explain these patterns; we argue instead for a model where speakers trade off the referential utility of being understood against the social utility of being identified as an in-group member.

2.7 Chapter 7: Disentangling passive and active origins of linguistic diversity using experimental microsocieties

Authors: Alicia Chen, Rebecca Saxe, Robert Hawkins

Status: In progress. The design, hypotheses, and pilot data have been submitted to a journal as a State 1 Registered Report. The manuscript is attached.

Abstract: Why does human language exhibit so much diversity? Linguistic diversity may emerge passively from efficient reference within socially isolated groups, or actively from motivations to signal group identity. Although both mechanisms are well documented, their relative contributions are difficult to disentangle using observational data alone. Here, we use controlled, real-time online experiments to test how referential and social-signaling pressures differentially shape the emergence and use of linguistic conventions. In a multiplayer reference game, small groups of participants first develop novel naming conventions for abstract shapes, and then interact in a larger mixed population under conditions that orthogonally manipulate interaction patterns and social-signaling goals. We predict that social-signaling motivations contribute to lexical variation in two ways: (1) by accelerating differentiation during convention formation, and (2) by resisting the convergence pressures introduced by inter-group mixing. This research is designed to provide causal evidence for how social-signaling motivations sustain linguistic diversity, and to quantify their contribution relative to standard referential pressures.

3 Timeline

- May-June 2026: finish, write up and submit intimacy and food-sharing project
- July-Oct 2026: Work on divisions of labor / modeling prosocial actions in hierarchical relationships
- Nov-Dec 2026 (based on status of Stage 1 RR): collect data for lexical variation project, write Stage 2 manuscript
- Jan-Mar 2027: Work on theory/review article
- March-May 2027: Work on thesis
- May 2027: Defend thesis

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4 Other work done during PhD

These are projects I started before my PhD that I worked on during the first few years of my PhD. They will not be included in the thesis.

4.1 A hierarchical Bayesian model of adaptive teaching

Authors: Alicia Chen, Andrew Palacci, Natalia Vélez, Robert Hawkins, Sam Gershman

Status: Completed (published in a journal)

Abstract: How do teachers learn about what learners already know? How do learners aid teachers by providing them with information about their background knowledge and what they find confusing? We formalize this collaborative reasoning process using a hierarchical Bayesian model of pedagogy. We then evaluate this model in two online behavioral experiments (N = 312 adults). In Experiment 1, we show that teachers select examples that account for learners' background knowledge, and adjust their examples based on learners' feedback. In Experiment 2, we show that learners strategically provide more feedback when teachers' examples deviate from their background knowledge. These findings provide a foundation for extending computational accounts of pedagogy to richer interactive settings.

4.2 Discreteness and systematicity in a continuous signal-meaning space

Authors: Alicia Chen*, Matthias Hofer*, Moshe Poliak, Roger Levy, Noga Zaslavsky

Status: Completed (published in a journal)

Abstract: Human spoken language uses a continuous stream of acoustic signals to communicate about continuous features of the world, by using discrete forms—words—that segment the world into categories. Here we investigate how discreteness (the segmentation of a continuous signal space into discrete forms) and systematicity (the consistent alignment of these forms with what they refer to in the world) can emerge under communicative pressure. In an exploratory study, participants were paired with one another and played a game in which they varied the pitch of auditory signals to communicate about a continuous color space, generalizing from a small, shared set of signal-color pairings. The emergent systems exhibited both discreteness and systematicity, but only systematicity robustly predicted successful communication. These findings offer insight into the cognitive strategies that could support the creation and evolution of language, highlighting how pressures for effective communication can shape continuous signal spaces into structured, learnable systems.

A social inverse-planning model captures how relationship intimacy constrains action planning and interpretation

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Abstract

Cognitive models of inverse planning typically treat a planner's utilities as functions of physical properties. However, many everyday actions are also shaped by utilities that are based on sociological properties, like social relationships. In this research, we extend inverse planning models to include social relationships as entities that change the utilities of actions. Using food sharing as a case study, we show that models require this sociological structure to capture human observers' judgments about actions, desires, and relationships. This work situates inverse planning models within everyday social interactions, where ongoing relationships constrain how people plan and interpret actions.

Keywords: inverse planning; theory of mind; social relationships; social cognition

Introduction

John and Sarah walk 16 blocks across town to get a bowl of noodles from Ramen House, passing dozens of restaurants along the way. From this simple scenario, human observers can infer that John and Sarah already *know* about Ramen House (its location, hours and menu) and *prefer* Ramen House noodles to other food options. How observers infer internal states like *knowledge* and *desire* from action is the classic problem of theory of mind reasoning (Dennett, 1989; Gopnik & Meltzoff, 1997; Wellman, 2014). A standard and effective cognitive model of this reasoning is "inverse planning" (C. L. Baker et al., 2009, 2017; Jara-Ettinger et al., 2016). Inverse planning assumes that agents choose actions rationally, given their beliefs and desires, to maximize reward and minimize costs, and uses observed actions to infer unobserved beliefs and desires. In the standard examples, rewards arise from outcomes like eating a desired food, and costs arise from properties like distance traveled. That is, the agents' internal states all directly refer to properties of the physical world.

Yet these examples leave out a fundamental class of rewards and costs that influence all social actions, including any time two people eat together. These rewards and costs arise from the sociological world of relationships and are not reducible to physical properties. Just as distance, effort, and outcomes shape which actions are feasible or worthwhile, social relationships shape which actions are comfortable, rewarding, or even conceivable (Davis et al., 2026; Jara-Ettinger & Dunham, 2024; Thomas, 2024).

For example, imagine that when they get to Ramen House, John and Sarah share a single bowl of noodles using the same chopsticks. From this simple scenario, observers can infer not only how much John and Sarah like *noodles*, but also how much they like *each other*. This is because when people plan whether and how to share food, key aspects of utilities depend on their relationship. Sharing food from the same plate, drinking from the same cup, and biting from the same part of the food, all feel comfortable only in relatively intimate relationships (Miller et al., 1998; Thomas et al., 2022). These actions are common between family members, romantic partners, or close friends, but not between students and teachers, colleagues, or casual neighbors.

These observations suggest that constraints on social actions arise from both the physical environment (e.g., distance, obstacles, effort) and the sociological environment (e.g., relationship structure). Existing inverse planning models capture how agents select actions given beliefs about physical constraints, and how observers jointly infer agents' internal states along with unobserved physical properties such as invisible barriers or food trucks that have closed. Here, we extend this framework by developing a social inverse planning model that treats sociological structure as a parallel source of constraint and inference. Incorporating sociological structure in this way enables joint inference over both physical and sociological constraints, capturing a broader range of everyday reasoning. It also explains how sociological structure systematically shifts inferences about the standard targets of inference – agents' desires – in inverse planning models.

Over three experiments, human observers make inferences about agents' expected actions, desires, and relationships. Our results show that standard inverse planning models that consider only physical costs and constraints cannot capture the full patterns of human behavior. Instead, people's inferences are closely captured by a social inverse planning model that considers how intimacy modulates the costs of sharing food.

Computational framework

We model social actions as taken by a joint "we-agent," treating the dyad as a collective actor with shared goals and mutual understanding (Chater et al., 2022; Tomasello et al., 2005). This approach contrasts with prior work that segments eating food into distinct roles (a sharer and a receiver) to model how a food receiver infers a sharer's beliefs about the relationship (Hung et al., 2022). By treating the dyad as a joint actor, our approach captures the shared background assump-

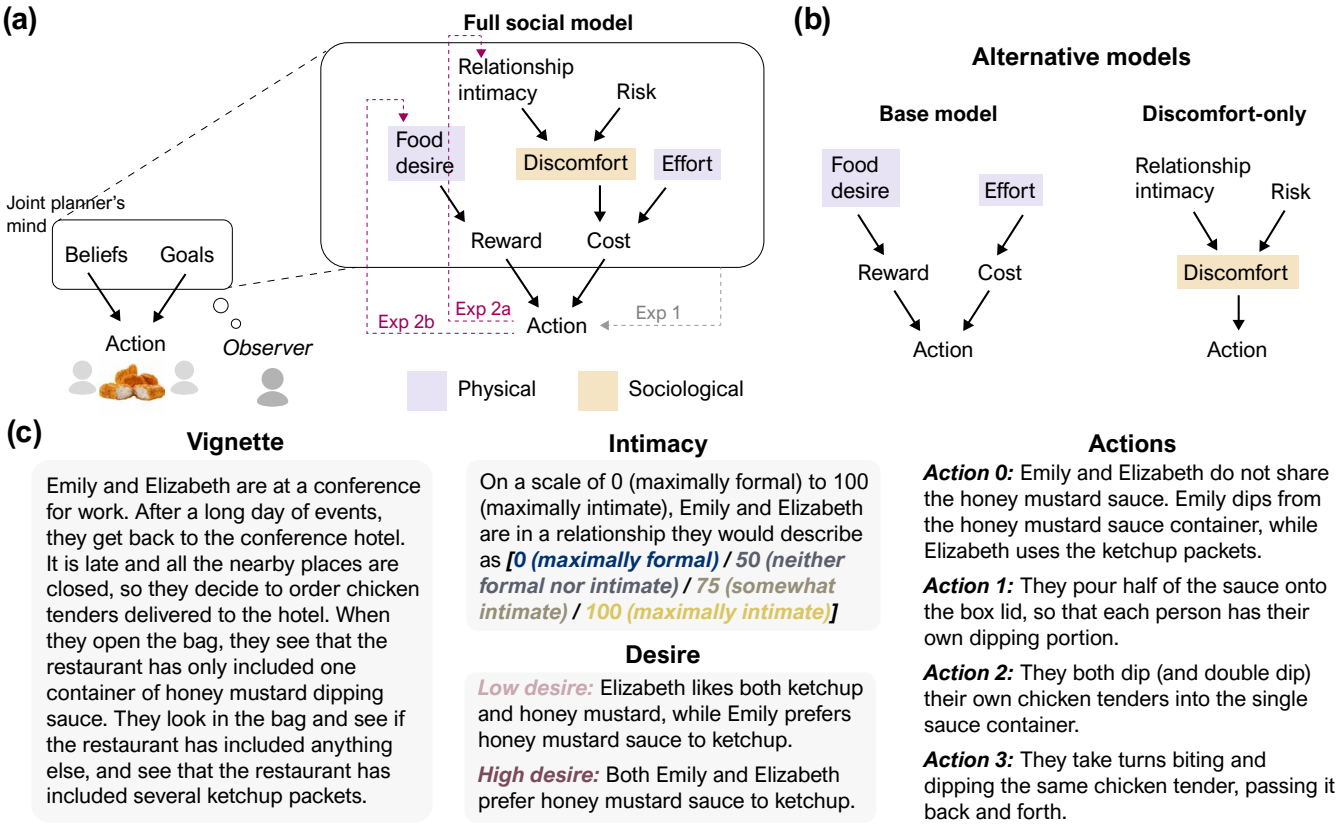


Figure 1: Overview of the social inverse planning framework and experimental design.

tions characteristic of stable dyadic relationships and abstracts away from explicit bargaining processes (Clark, 1996; Fiske, 1991). In this way we incorporate relational structures analogous to physical structure in the standard models – as entities that shape utility components and are part of common knowledge. We first describe the general framework here, and then provide experiment-specific details in later sections.

Standard inverse planning models assume that agents choose actions to maximize expected utility, defined as the reward of achieving a goal minus the costs incurred (C. L. Baker et al., 2009, 2017; Jara-Ettinger et al., 2016). The actor chooses an action proportional to its total utility, which is a weighted sum of the individual utility components. Observers can then use Bayesian inference to recover the agent's internal states from observed actions.

In standard applications, rewards are determined directly by the agent's desires, and costs arise from physical properties. For example, rewards include eating a preferred food (C. L. Baker et al., 2017; Jern & Kemp, 2014), obtaining or lifting an object (Xiang et al., 2023; Zhi-Xuan et al., 2020), or getting to a specific location (C. L. Baker et al., 2009). Costs include traveling further and across more challenging or aversive terrain (C. L. Baker et al., 2009, 2017; Jara-Ettinger et al., 2020), lifting a heavier object (Xiang et al., 2023), or jumping higher (Jara-Ettinger et al., 2020; Liu et al., 2017). In prior models, these rewards and costs are manually specified by researchers, under the assumption that such features are common knowledge.

We extend this framework by incorporating how the utilities of actions also include aspects related to social relationships (Figure 1a). As a case study, we incorporate in the models how relationship intimacy changes the *discomfort* – a kind of cost – associated with performing risky or vulnerable actions like sharing saliva. This discomfort depends both on the risk of saliva transfer and on the intimacy of the relationship.

We test the model in the domain of eating and sharing food, which is a familiar domain where the effects of physical and sociological utilities can be experimentally dissociated (we can separate *whether* the two people eat the food, from *how* they eat it). This enables direct comparison with the standard models, which have also primarily been tested in the domain of food preferences. The resulting framework can recover both desires and relationships from observed actions.

Transparency and openness

Preregistration All procedures and alternative models were preregistered[1]. The reported model comparison deviated

[1]Experiment 1: https://osf.io/9ze34/ Experiment 2a: https://osf.io/74pg8/ Experiment 2b: https://osf.io/jgvyq/

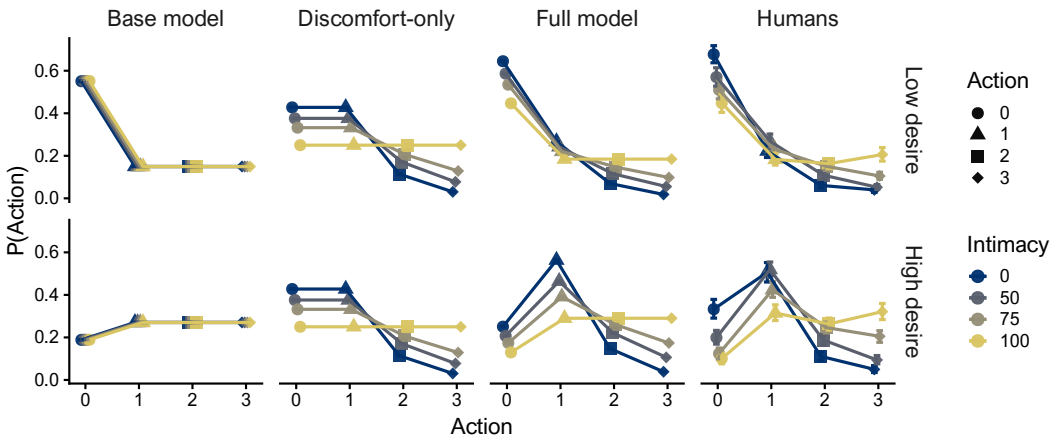


Figure 2: Experiment 1 results and best-fitting model predictions. Error bars are 95% bootstrapped confidence intervals.

slightly from the preregistrations; see the individual preregistration links for more detail.

Implementation details The experiments were created using jsPsych (de Leeuw, 2024; de Leeuw et al., 2023). The cognitive models were implemented in memo (Chandra et al., 2025). The cognitive models were implemented manually; AI tools (Claude Code and Cursor) were used to aid the generation of other code for the project. All AI generated code was validated and tested by the first author. All code and anonymized data are available at https://github.com/aliciamchen/social-inverse-planning_cogsci-26.

Experiment 1: Forward planning

In Experiment 1, we tested how observers use sociological and physical properties to predict others' actions.

Methods

Participants For all experiments, participants were recruited on Prolific and pre-screened to be adult fluent English speakers from the United States. Participants were excluded if they failed an attention check and two memory checks. For Experiment 1, we recruited 118 (70 female, 46 male, 1 nonbinary; ages 18–77, M(SD) age = 42.3(13.7)) participants, and 3 participants were excluded. Participants were paid $6.25 for completing the experiment, which took an estimated 25 minutes. Participants completing one experiment were excluded from subsequent experiments. Participants gave informed consent, and all procedures were approved by the MIT Committee for the Use of Humans as Experimental Subjects.

Materials We wrote 16 scenarios describing situations in US cultural contexts where two people might plausibly share food (e.g., cake at a birthday party, a hot dog at a basketball game, drinks at a house party; see Figure 1c for an example; all scenarios available in the project repository). The gender pairing of the characters' names varied across scenarios.

Procedure Participants read vignettes describing food-sharing scenarios and rated the probability that two characters will take each of four possible actions, ranging from not sharing the food at all to sharing in ways that involve increasing risk of saliva transfer.

We manipulated two factors: how much the two characters want the food, and the intimacy of their relationship (Figure 1c). In the 'high desire' condition, both characters want to eat the food (e.g., both like the food, both are hungry). In the 'low desire' condition, one character wants the food less (e.g., prefers another food, is not hungry). Relationship intimacy was described on a 0–100 scale, ranging from maximally formal (0) to maximally intimate (100), and sampled at four levels (0, 50, 75, 100). The resulting design was thus 2 (desire: low vs. high) x 4 (intimacy: 0, 50, 75, 100). Each participant saw all 16 scenarios; the assignment of scenario to condition was balanced within and across participants.

On each trial, participants allocated probability mass across four actions. Action 0 describes the two characters not sharing the food (e.g., one person eats the food while the other eats something else). Action 1 describes them sharing the food in a way that does not risk saliva transfer (e.g., pouring separate portions). Action 2 and 3 are ways to share the food with increasing risk of saliva transfer (see Figure 1c).

Model specification

In the full social model (Figure 1a), a joint actor chooses an action $a \in \{a_0, a_1, a_2, a_3\}$ proportional to its total utility, following a softmax choice rule with inverse temperature α. The total utility of an action a given motivational state (desire) $s \in \{\text{low}, \text{high}\}$ and intimacy $I \in [0, 1]$ is:

$$U(a|s, I) = w_v \cdot V(a|s) - w_d \cdot d(a|I) - w_e \cdot e(a) \quad (1)$$

To generate model predictions, we assigned each action values based on the experimental design, reflecting qualitative distinctions that are explicitly encoded in the scenarios. We describe each component of the utility below.

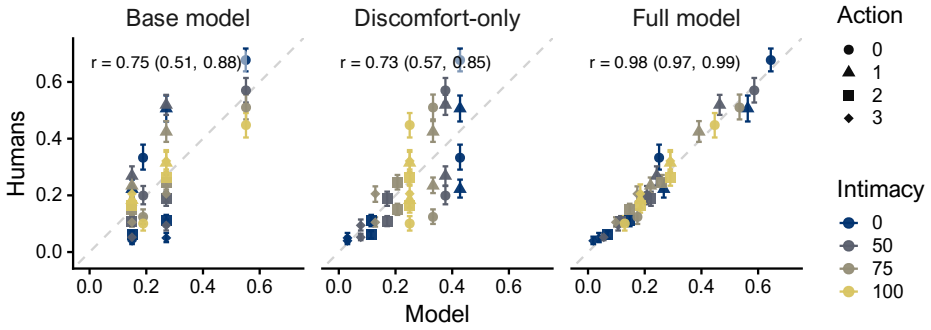


Figure 3: Model-human correlations in Experiment 1. Error bars are 95% bootstrapped confidence intervals.

Reward The reward $V(a|s)$ of an action depends on the action and the desire state. In the 'high desire' state, sharing actions (a_1–a_3) serve the goal of jointly eating the food: $V(a|s = \text{high}) = [0, 1, 1, 1]$ for actions a_0 through a_3. In the 'low desire' state, the same sharing actions have negative utility: $V(a|s = \text{low}) = [0, -1, -1, -1]$.

Discomfort The *discomfort* of taking an action depends on both the risk of transferring saliva via the action and the intimacy of the relationship. We set a saliva-transfer risk $\rho = [0, 0, 1, 2]$ for a_0 through a_3. Intimacy reduces the discomfort caused by risk, and the shape is controlled by γ:

$$d(a|I) = \rho(a) \cdot (1 - I)^{\gamma} \tag{2}$$

Effort Finally, the effort $e = [0, 1, 1, 1]$ captures the baseline physical cost of sharing food versus not sharing.

Alternative models

We considered two alternative models (Figure 1b), each inspired by part of the prior literature:

- A 'Base' model that incorporates only the reward for eating the food and the effort of sharing.

$$U_{\text{base}}(a, s) = w_v \cdot V(a|s) - w_e \cdot e(a) \tag{3}$$

- A 'Discomfort-only' model that captures only how intimacy mitigates discomfort from risky actions (Miller et al., 1998).

$$U_{\text{discomfort-only}}(a|I) = -w_d \cdot d(a|I) \tag{4}$$

Results

Table 1: Experiment 1 model comparison. ΔAIC is computed relative to the Full model (positive = worse than Full).

Model	AIC	ΔAIC	r (95% CI)
Full	4349.11	0	0.98 (0.97, 0.99)
Discomfort-only	4624.28	275.17	0.73 (0.57, 0.85)
Base	4707.94	358.83	0.75 (0.51, 0.88)

We fit each model to participants' probability judgments by maximum likelihood using stochastic gradient descent, fixing $\alpha = 1$ for identifiability. We generated predictions using leave-one-scenario-out cross-validation, refitting on 15 scenarios and predicting the held-out scenario. We compared models by in-sample AIC and out-of-sample Pearson correlation between predictions and human cell means, pooled across CV folds.

The experimental design allows us to test two predictions: whether food desire reverses preferences between not sharing (Action 0) and no-risk sharing (Action 1), and whether relationship intimacy selectively increases tolerance for saliva-transfer risk, raising the likelihood of higher-risk sharing (Actions 2 and 3).

The Base model captures the desire-based reversal between Actions 0 and 1 but is insensitive to intimacy. The Discomfort-only model captures how intimacy increases risk tolerance, but cannot represent desire and therefore does not distinguish between sharing and not sharing. Only the full social model (best-fitting params $w_v = 0.85$, $w_d = 1.34$, $w_e = 0.03$, $\gamma = 0.86$) captures both patterns (Figure 2, Figure 3). These results show that representing relational structure is necessary to capture forward planning judgments in this domain.

Experiment 2: Inverse planning

In Experiment 2, we tested whether the fits from Experiment 1 generalize to inverse planning. Two complementary experiments test both inferential targets, intimacy (Experiment 2a) and food desire (Experiment 2b).

Methods

Participants For Experiment 2a, we recruited 117 participants (56 female, 56 male, 3 non-binary, 2 abstain to respond; ages 19–78, M(SD) age = 44.3(13.9)) and excluded 3 participants. For Experiment 2b, we recruited 120 participants (62 female, 51 male, 5 non-binary; ages 19–74, M(SD) age = 41.1(13.2)) and did not exclude any participants. Participants were paid $5 for completing the experiment, which took an estimated 20 minutes to complete.

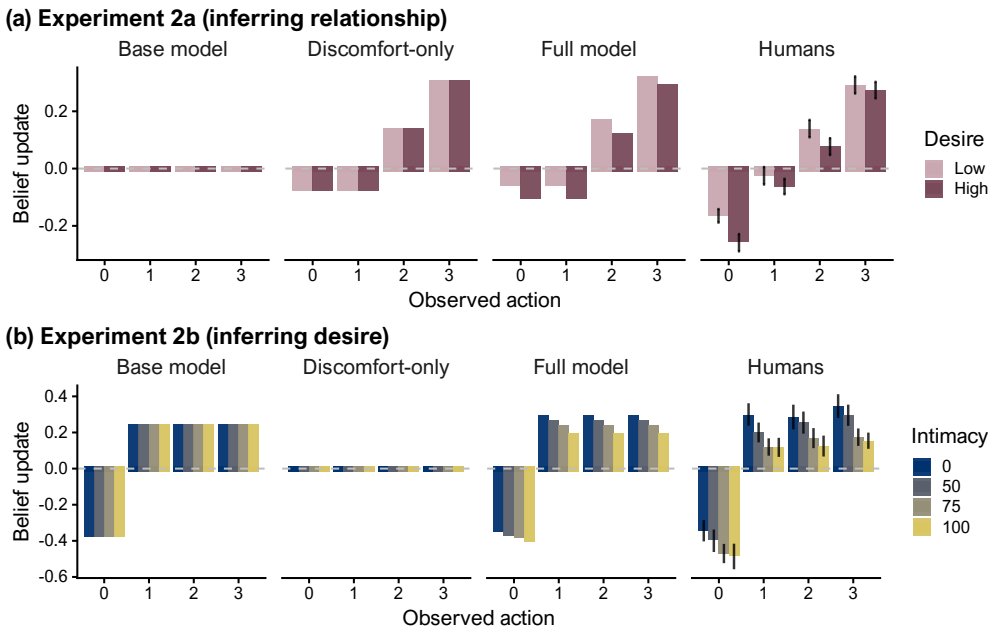


Figure 4: Experiment 2 results and best-fitting model predictions. Error bars are 95% bootstrapped confidence intervals.

Procedure The scenarios were the same as in Experiment 1 (Figure 1c), but the experiments followed the inverse structure. On each trial, participants provided ratings both before and after observing the characters take a specific action; the primary dependent variable is the belief update (posterior minus prior). In Experiment 2a, participants rated relationship intimacy (0-100 scale) given desire (low vs. high) and observed action (0-3), yielding a 2×4 design. In Experiment 2b, participants rated the relative probability of the low-desire versus high-desire state, given intimacy (0, 50, 75, or 100) and observed action (0-3), yielding a 4×4 design.

Model specification

Upon observing an action, an observer who knows one latent variable (desire state or intimacy) can infer the other by inverting the forward model:

$$P(I|a,s) \propto P(a|s,I) \cdot P(I|s) \quad (5)$$

$$P(s|a,I) \propto P(a|s,I) \cdot P(s|I), \quad (6)$$

where $P(a|s,I)$ is the forward planning likelihood from Experiment 1. Because the dependent variable is belief update, we set uniform priors so that predicted belief update directly reflects the diagnostic value of the observed action.

For each model, we froze the best-fitting actor parameters obtained from Experiment 1 to generate the inverse predictions, and fit only an additional parameter, α_{obs} (so that the effective likelihood term is $P(a|s,I)^{\alpha_{\text{obs}}}$), by minimizing the mean squared error between the model's predicted belief update and the human belief update. This parameter captures how sharply people weigh higher-likelihood states. The reported fits and predictions are out of sample, following the cross-validation procedure of Experiment 1.

Results

Table 2: Experiment 2 model comparison: out-of-sample per-trial ΔMSE relative to the Full model, pooled across folds.

Model	ΔMSE (Exp 2a)	ΔMSE (Exp 2b)
Full	0	0
Discomfort-only	0.001	0.080
Base	0.030	0.003

Across both experiments, observers' belief updates reflected joint reasoning about reward, cost, and relationship context, in ways that the alternative models could not capture (Figure 4). The Base model captures only the basic inference of how sharing (actions 1–3) increases inferred desire and non-sharing (action 0) decreases it. Conversely, the Discomfort-only model predicts greater intimacy from higher-risk sharing (actions 2–3), but cannot account for desire or how it interacts with intimacy.

By contrast, the social model captures systematic 'explaining-away' effects between desire and intimacy. Risky sharing is informative about intimacy primarily when desire is low; when desire is high, the same behavior is less diagnostic because motivation to eat already provides an alternative explanation for accepting discomfort. On the other hand, choosing not to share despite really wanting the food (high desire) is strongly diagnostic of a more formal relationship (Figure 4a). A parallel pattern emerges for desire inference. When intimacy is low, risky sharing strongly increases in-

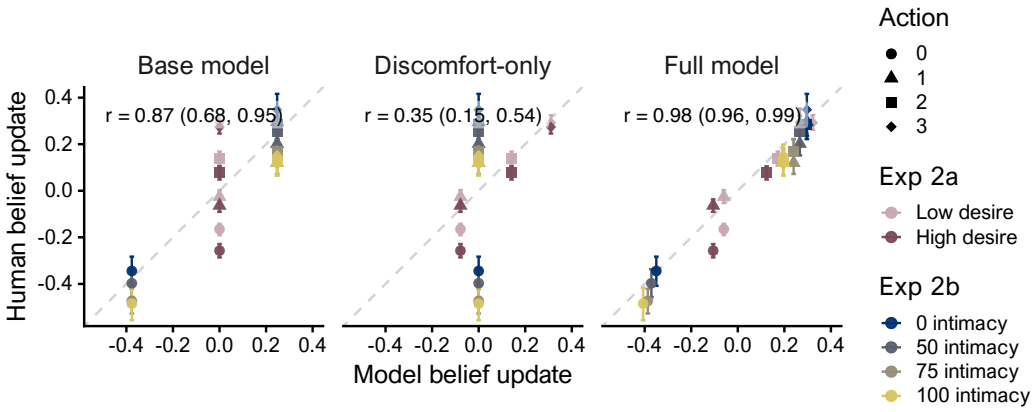


Figure 5: Model-human correlations for Experiment 2. Error bars are 95% bootstrapped confidence intervals.

ferred desire; when intimacy is high, the same risky actions are less informative because the relationship already explains why someone would be comfortable sharing (Figure 4b).

Across both Experiment 2a and Experiment 2b, the full social model achieved the lowest MSE, though the margin over the next-best alternative was small. The full model is, however, the only model that qualitatively captures the explaining-away pattern across both inferential targets (Figure 5).

Discussion

The present work extends inverse planning models by treating sociological structure – specifically, relationship intimacy – as a factor that systematically shapes the utilities of action. Much of everyday social reasoning involves observing people who are acting together, whether on a date, at a dinner party, or in a meeting. In these contexts, the utilities of the actions depend not only on physical properties but also on the relationship between the actors.

Prior work has incorporated relationships into models of social reasoning in several ways, including by representing how agents weigh others' utilities (Jern & Kemp, 2014; Kleiman-Weiner et al., 2017; Powell, 2022; Ullman et al., 2009), their attitudes toward others (Wang & Jara-Ettinger, 2025), or by specifying behaviors consistent with particular roles or relationships (A. Baker et al., 2025; Davis et al., 2026). These approaches capture important aspects of social reasoning, but typically treat relationships as separate objects of reasoning rather than structures that directly constrain existing utilities. By contrast, our approach allows relational structure to interact with – and be compared directly to – the physical variables that have traditionally constituted inverse planning models.

Modeling two characters as a joint "we-agent" reflects the shared assumptions that support coordination in stable relationships. This approach is well suited to modeling how third-party observers reason about interacting dyads in contexts where relational structure can be treated as stable and mutually known. At the same time, it does not capture intra-relationship dynamics such as negotiating roles or managing uncertainty about the relationship (Chuey & Gweon, 2025; Hung et al., 2022; Wu et al., 2021; Xiang et al., 2023). These processes likely play an important role in contexts where relationships are uncertain or evolving.

Although we studied food sharing as a case study, the same framework should generalize to other social actions where relationship structure shapes what actions feel appropriate, comfortable, or desirable. These include asking for help, self-disclosure, gift giving and generosity, and teasing (Chen & Saxe, 2025; Fiske, 1991; Keltner et al., 1998; Nadler et al., 1985; Taylor & Altman, 1975). Intimacy is also only one dimension of social structure: group membership, status and hierarchy, or culturally specific roles may similarly influence how agents trade off rewards and costs (Fiske, 1991; Jara-Ettinger & Dunham, 2024; Thomas, 2024). These factors can be incorporated within the same framework by specifying how they alter utility components.

More broadly, these results suggest that inverse planning models should explicitly represent how social context shapes utilities. When models succeed without relational variables – even in minimal grid-world paradigms – this may reflect the implicit incorporation of relational assumptions into other utility terms, or the selective omission of it when it is not relevant for the current judgment (Wong et al., 2025). We argue that making these assumptions explicit is necessary for understanding both sociological structure and the commitments of existing models.

Acknowledgments

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Expectations of reciprocal generosity are specific to equal relationships

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Abstract

When someone acts generously, what expectations does this create? Classic theories and experiments emphasize reciprocity: the idea that generosity will be returned by the recipient. Yet daily-life and ethnographic observations often show a different pattern: generosity sets a precedent, leading people to expect the same person to be generous again. Across six online behavioral experiments (total $N = 599$ U.S. adults) using third-party vignette judgments and first-person incentivized economic games, we test when generosity creates expectations of reciprocation versus a precedent. We found that people expected reciprocity only in equal or symmetric relationships. Otherwise, they expected generosity to continue from the same actor. These expectations generalized across roles, contexts, and cost structures. Taken together, the results suggest that classic evidence on reciprocity and turn-taking may capture expectations among strangers or equals, but not the wider set of relationships that structure much of social life.

Keywords: social cognition, generosity, reciprocity, social relationships

Expectations of reciprocal generosity are specific to equal relationships

Two colleagues, Igor and Marie, go out for coffee, and Igor picks up the check. Who will pay the next time they get coffee? For many theorists in social and moral psychology (Cialdini, 1993; Earp et al., 2021; Gouldner, 1960; Le Pargneux & Cushman, 2024; Molm et al., 2007; Thibaut & Kelley, 1959), the answer is obvious: Marie. Acts of generosity are assumed to generate strong norms and expectations of reciprocity. When one person acts generously, the other is expected to return the favor. Such expectations can strongly influence future behavior, even outweighing personal preferences and self-interest (Bicchieri, 2005; Cialdini & Goldstein, 2004; Prentice & Paluck, 2020).

Here, we suggest that this classic assumption may not generalize to the broader landscape of human social relationships. Drawing on relational theories from anthropology and social psychology (Clark et al., 2017; Fiske, 1992; Graeber, 2012; Levinson, 2025; Yan, 1993), we argue that direct reciprocity only governs a subset of social interactions: those between equals or strangers. Otherwise, generosity creates a precedent, rather than a debt that needs to be repaid. In other words, if Igor is Marie's boss, his paying once may intuitively set the expectation that he will pay the next time as well. The present research examines when and why people might have these contrasting expectations, by applying a cognitive science approach to causally manipulate and measure how social-relational structure shapes the expectations created by a single generous act.

Decades of economic and psychological theories assume that a generous act typically elicits expectations of direct reciprocation. People who act generously anticipate repayment, and recipients often feel obligated to reciprocate (Blau, 1964; Cialdini, 1993; Flynn & Yu, 2021; Gouldner, 1960; Kardas et al., 2018; Trivers, 1971). Reciprocation alleviates indebtedness (Cialdini, 1993), fosters trust and positive affect (Molm et al., 2007), and maintains cooperation in repeated interactions (Axelrod & Hamilton, 1981; Fehr & Fischbacher, 2003; Rand & Nowak, 2013). In experimental incentivized games, partners often alternate giving and receiving roles (McKelvey & Palfrey, 2001; Prisbrey, 1993), a pattern observed even in children (Grueneisen & Tomasello, 2017; Melis et al., 2016; Olson & Spelke, 2008) and preverbal infants (Tatone & Csibra, 2020).

Yet anthropological and sociological accounts suggest that acts of generosity create expectations of reciprocation only in a narrow subset of social contexts – specifically, among social equals. For example, Graeber (2012) argues that reciprocal exchange implies the potential and desire for formal equality between

partners, creating and sustaining relationships between people of equal rank in society. As a result, directly repaying generosity may be socially *inappropriate* when it could disrupt the norms that maintain a social hierarchy (Parry, 1986; Yan, 1993). Hierarchical relationships, Graeber speculates, instead function based on a logic of precedent: when someone acts generously, it creates the expectation that the same person will act generously again. For instance, ethnographic work from rural China shows that villagers routinely offer gifts to superiors with no expectation of return, to preserve the relationship and fulfill socially defined roles (Yan, 1993). In other words, in relationships that are already unequal, repeated acts of generosity may sustain the relationship precisely by *not* generating expectations of alternation or reciprocation.

More generally, anthropologists and sociologists conceptualize generous acts, like other social interactions, as a means to create and sustain different kinds of social relationships (Carrier, 1991; Fiske, 1992; Graeber, 2012; Mauss, 1925; Rai & Fiske, 2011; Sahlins, 2013; Schwartz, 1967; Weiner, 1992). Indeed, relationship structure shapes expectations about prosociality from early in development: infants and children typically expect reciprocation (Tatone & Csibra, 2020, 2024) but also coordinate along dominance asymmetries (Grueneisen & Tomasello, 2017), and both adults and children associate hierarchy with distinct patterns of responsibility and prosociality (Gopnik et al., 2025; Stavans & Diesendruck, 2021; Terrizzi et al., 2020). Yet the specific prediction that hierarchical relationships evoke expectations of *precedent* – articulated most explicitly by Graeber (2012), who supports it with observational evidence but which is really a speculation about the underlying human cognition – has not been experimentally tested.

Here we systematically tested this prediction, by examining the minimal conditions that influence the expectations created by a single generous act, for the next similar interaction between the same people. Specifically, when do people expect that the same individual will perform the generous role again (precedent), rather than expecting the recipient to play the generous role in return (reciprocity)? Consistent with most previous experimental research (McKelvey & Palfrey, 2001; Prisbrey, 1993), we recruited participants from a large-scale society with high market integration, the United States, where people have strong norms of balanced exchange (Henrich, Ensminger, et al., 2010; Henrich et al., 2005). By contrast to most previous experiments, we additionally provided minimal verbal descriptions contextualizing the interactions in hypothetical structured social relationships that were either equal or hierarchical. Across experiments, relationships were described as “symmetric” versus “asymmetric,” as one party having “equal,” “more” or “less power, status or influence” than the other, or using familiar concrete relationship

labels like “cousins” or “friends,” versus “uncle-nephew” or “boss-employee.”

We hypothesized that expectations of direct reciprocity would be restricted to relationships with the potential or desire for equality – such as friends, neighbors, co-citizens, or anonymous strangers. In four experiments, we systematically measured third-party observers’ expectations for everyday interactions, and isolated the contribution of minimal relational information to these expectations. Then, in two experiments, we tested whether these findings generalize to participants’ own strategies in incentivized experimental games.

The verbal descriptions of relationships exerted strong and systematic effects on people’s expectations in all experiments. That is, even in large-scale societies where expectations of reciprocity and balanced exchange have been treated as a default, the expectations created by a single act of generosity differ substantially in structured social relationships.

Transparency and Openness

All data and code for running the experiments and reproducing the analyses and plots are available on GitHub at https://github.com/aliciamchen/generosity-actions and also deposited on Zenodo at https://doi.org/10.5281/zenodo.17477355. The experiments and analyses were all preregistered unless otherwise stated. Study 1 was preregistered (https://osf.io/fxwbh/) and collected on January 30, 2023. Study 2 was preregistered (https://osf.io/jvnca/) and collected on April 12, 2023. Study 3 was preregistered (https://osf.io/vt4kp) on April 30, 2025 and collected on April 30 and May 1, 2025. Study 4 was preregistered (https://osf.io/dfv24/) and collected on August 5, 2023. Study 5 was preregistered (https://osf.io/ag9v2/) and collected on May 16, 2025. Study 6 was preregistered (https://osf.io/zn6wh/) and collected on May 17th, 2025. Note that the numbering of the studies was changed from as stated in the preregistration titles. There were minor deviations from the preregistrations, reported in the Supplementary Material (Section A1). AI tools were used to assist with code development and editing the paper. All AI-assisted output was reviewed, edited, and verified by the authors, who take full responsibility for the final work.

Implementation and analysis details

Experiments were implemented using the jsPsych library (de Leeuw et al., 2023), and analyses were conducted in R using the lme4, lmerTest, and emmeans packages (Bates et al., 2014; Kuznetsova et al., 2017; Lenth et al., 2018). For studies 3, 5, and 6, we used the DataPipe tool to automatically send the

data to the Open Science Framework (de Leeuw, 2024).

For Studies 1–3 (linear mixed-effects models), we report Cohen's *d* for pairwise contrasts, computed using the residual standard deviation from the model as the standardizer (Lenth et al., 2018). For Studies 4–6 (logistic mixed models), we report odds ratios with 95% Wald confidence intervals.

Where key theoretical conclusions are based on null findings, we supplement the frequentist tests with Bayes factors (BF_{01}), computed using the BayesFactor package (Morey & Rouder, 2024) with the default JZS prior (Cauchy scale $r = \sqrt{2}/2$ for fixed effects; Rouder et al., 2012), to quantify the evidence in favor of the null hypothesis. BF_{01} expresses how many times more likely the data are under the null hypothesis than the alternative; values above 3 are conventionally interpreted as moderate evidence for the null, and values above 10 as strong evidence (Lee & Wagenmakers, 2013).

Sample sizes were determined in advance based on visual inspection of pilot data: once a sample size was chosen for Study 1, it was kept consistent for Studies 1–4, and similarly for Studies 5–6. Sensitivity analyses computing the minimum detectable effect sizes at 80% power for all studies are reported in the Supplementary Material; these use a paired-comparison framework as a conservative approximation to the full mixed-model structure.

Study 1: Being generous creates a precedent in hierarchical relationships

First, we tested participants' third-party evaluations of scenarios describing characters playing the generous role in daily-life interactions (e.g. buying coffee during a work meeting, preparing a meal and cleaning after, deferring to the other's preference; see Supplementary Table S2 for all the 18 scenarios), given information about a social relationship between the two characters. Here we use the term 'generous' to describe any social interaction between two people in which one person bears more cost, or receives less benefit, than the other. This definition includes interactions that benefit only one partner (altruistic actions, like babysitting someone else's children so they can go on a date; Batson, 2011; Fehr & Fischbacher, 2003) and interactions that benefit both parties unequally (like one person buying two tickets to attend a concert together; Andreoni, 1990; Bekkers & Wiepking, 2011). All scenarios involved dyadic interactions in which one person actively and voluntarily performs a generous act for another, which are contexts that should evoke strong expectations of direct reciprocity (Tatone & Csibra, 2020; Widlok, 2016).

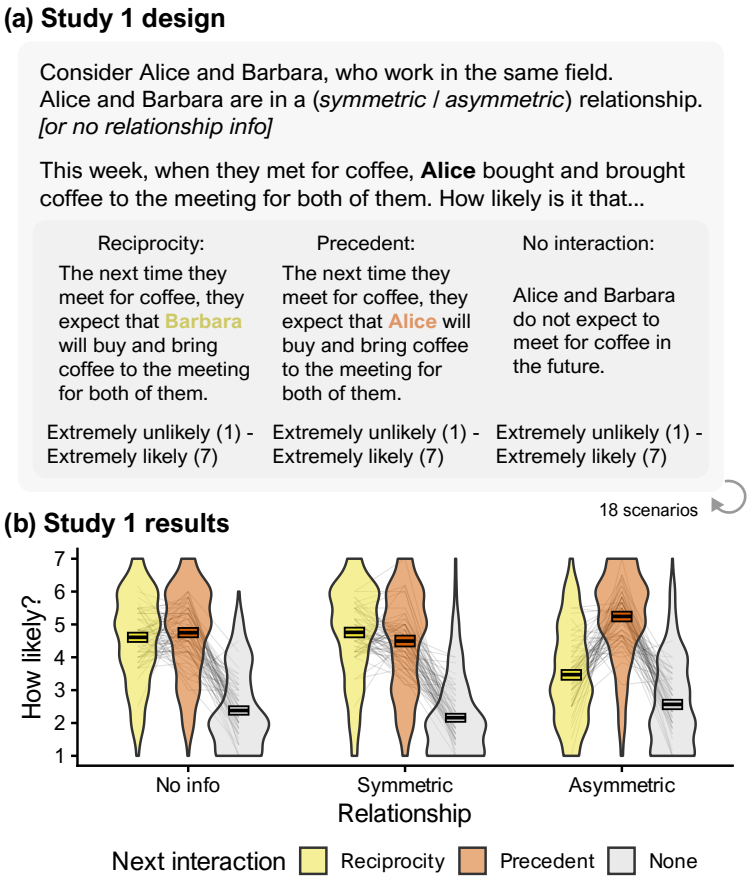


Fig. 1

*Study 1. Participants **(a)** saw information about a social relationship and one generous act, and predicted the next action. Results are shown in **(b)**. Violins show the distribution of individual trial ratings. Black boxes show bootstrapped 95% confidence intervals of the mean. Thin lines connect per-subject condition means across the three expected-action responses.*

Methods

Participants

Convenience samples of adult fluent English speakers from the United States were recruited on the online crowdsourcing platform Prolific. Sample sizes were all determined in advance and preregistered (see Transparency and Openness for details). Participants who completed an experiment were excluded from the subsequent experiments. Data were excluded from participants who did not pass an attention check or indicated at the end of the experiment that they did not understand the instructions. Participants gave informed consent, and all procedures were approved by the MIT Committee for the Use of Humans as

Experimental Subjects.

For Experiment 1, we recruited 59 participants (26 female, 30 male, 3 nonbinary; ages 20–62, *M*(*SD*) age = 36.2(12.5)). For Experiments 1-4, participants were paid $5 for completing an experiment, for an estimated pay of $15/hour. Participants who completed an experiment were excluded from subsequent experiments.

Procedure

In Study 1 (Figure 1), the scenarios either omitted information about the relationship between the two characters, or described that the two characters were in an "asymmetric" relationship where one character has more power/status/influence than the other person, or a "symmetric" relationship where the two characters have equal power/status/influence. In the asymmetric condition, participants were not told which character was higher-ranked; Study 2 was designed to address this by manipulating the direction of the asymmetry. For each scenario, after one character did a generous act, participants used a 7-point Likert scale to evaluate how much the characters expected (1) the other character would do that act the next time ('reciprocity'), (2) the same character would do the act the next time ('precedent'), and (3) the interaction would not happen again (Figure 1a).

For all experiments, participants read each of the scenarios, and the assignment of scenarios to conditions was counterbalanced across participants. The trial sequence was randomized. The two characters interacting were always the same gender, as indicated by names and pronouns.

Results and discussion

To test our predictions for Studies 1-3, we used linear mixed-effects regression models to predict participants' responses, including categorical fixed effects of relationship condition and next action (i.e. the three Likert responses) along with their interaction, and random intercepts for each scenario and participant, followed by planned contrasts between the estimated marginal means. Because these contrasts test specific, pre-specified hypotheses, *p*-values are reported without adjustment for multiple comparisons[1].

When participants were given no information about the relationship between the dyads, neither reciprocation ($M = 4.61$, $SE = 0.09$, 95% CI [4.43, 4.78]) nor following a precedent ($M = 4.75$, $SE = 0.09$, 95% CI [4.57, 4.93]) was judged to be more likely ($\Delta M = 0.14$ [−0.06, 0.34], $t(3095) = 1.38$, $p = 0.168$,

[1] As a check for robustness, for Studies 1-3 we also re-analyzed our data normalizing participants' responses on each trial to sum to 1; all results reported here are consistent with this check.

$d = 0.10$). People had strong expectations that generous acts lead to future social interactions: Either partner being generous a second time was judged more likely than no future interaction (ΔM = 2.30 [2.12, 2.47], $t(3095) = 25.91$, $p < 0.001$) (Figure 1b, left).

Minimal verbal descriptions of the structure of the relationship changed people's expectations for who would be generous. In a relationships described as 'symmetric', people expected the dyad to alternate the generous role (M = 4.75, SE = 0.09, 95% CI [4.58, 4.93]) rather than follow a precedent (M = 4.49, SE = 0.09, 95% CI [4.31, 4.67]; $\Delta M = -0.26$ [−0.47, −0.06], $t(3095) = -2.58$, $p = 0.010$, $d = -0.19$). In a relationship described as 'asymmetric', people expected the dyad to follow a precedent (M = 5.25, SE = 0.09, 95% CI [5.07, 5.43]) rather than alternate who was generous (M = 3.48, SE = 0.09, 95% CI [3.30, 3.65]; ΔM = 1.77 [1.57, 1.97], $t(3095) = 17.29$, $p < 0.001$, $d = 1.30$) (Figure 1b). This pattern was robust across scenarios (Supplementary Figure S1) and when using within-subject z-scored ratings or ordinal regression to address potential ceiling effects (Supplementary Material, Section A.3).

Overall, observing the same action created systematic and contrasting expectations, depending on minimal abstract information about the social relationship. In symmetric equal relationships, one partner playing the generous role created the expectation that partners would alternate, generating reciprocity. In asymmetric hierarchical relationships, the same generous interaction created the opposite expectation: that the same person would be generous again.

Reciprocity was only expected over a precedent in equal relationships. The presence of a minimal cue about an asymmetry led to strong shifts in expectations, toward following a precedent. What drives this preference for precedent, and how can such minimal information so dramatically reshape expectations, even against established patterns from existing experimental studies? We address these questions in the studies that follow.

Study 2: Generous acts set a precedent in both directions of hierarchical relationships

Study 1 showed that asymmetric relationships lead to expectations of precedent, but did not specify the direction of the asymmetry. In hierarchical relationships, the relative ranking of the generous actor – whether they are lower-ranked or higher-ranked – could matter. In Study 2, we tested whether the direction of the hierarchy matters in people's expectations.

One possibility is that a generous act in a hierarchical relationship sets a precedent in both directions: whether generosity flows upward or downward in a hierarchy, the same person is expected to be

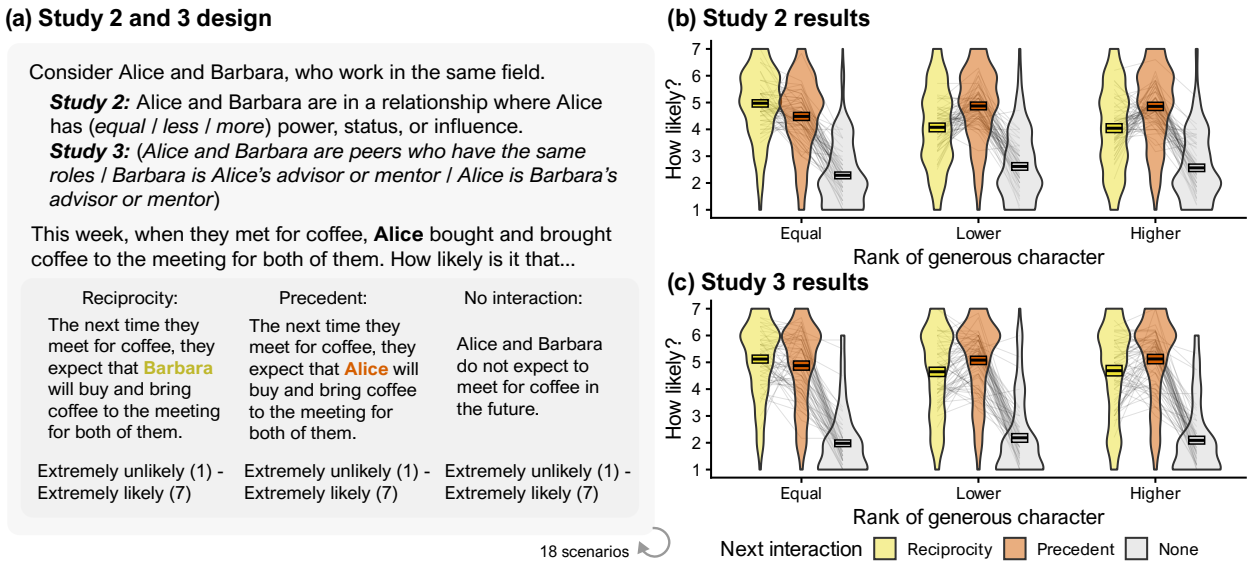


Fig. 2

*Study 2 and 3. Participants **(a)** saw information about a social relationship and one generous act, and predicted the next action. In Study 2, participants were shown information about the relative power, status, or influence between the two characters. In Study 3, participants were shown a description of a concrete relationship. Results for Study 2 are shown in **(b)** and results for Study 3 are shown in **(c)**. Violins show the distribution of individual trial ratings. Black boxes show bootstrapped 95% confidence intervals of the mean. Thin lines connect per-subject condition means across the three expected-action responses.*

generous again. Alternatively, people might expect a precedent primarily when lower-ranked individuals benefit higher-ranked individuals, given previous research suggesting that higher-ranked people are typically expected to benefit from uneven interactions (Malinowski, 1922; Emerson, 1964; Grueneisen & Tomasello, 2017; Le Pargneux & Cushman, 2024; but see Yan, 1993). We tested these possibilities in Study 2.

Methods

Participants

For Study 2, we recruited 60 participants (28 female, 27 male, 5 nonbinary; ages 18–65, *M*(*SD*) age = 33.0(10.9)), and excluded 1 participant.

Procedure

We gave participants information about the *direction* of the asymmetry between the two characters interacting, specifically whether the character performing the initial generous act had more or less "power, influence and status" than the person receiving the initial act. The rest of the procedure was the same as in Study 1.

Scenario validation experiment

Generous behavior has been defined as an act where one person incurs a cost, to benefit another (Fehr & Fischbacher, 2003), and these measures have been shown to affect adult and children's evaluations of generosity (Radovanovic et al., 2023; Sommerville et al., 2018). Thus, to test whether there is separable variance in the scenarios that could explain variance in the actual experiments, we also ran two experiments measuring the relative cost and benefit ($N = 58$ for benefit, $N = 59$ for effort, Supplementary Figure S3) between the two people interacting in each scenario.

In each experiment, we showed participants all 18 scenarios, each scenario displaying one generous action that one character performs for the other. In the 'benefit' experiment, we asked participants how much each of the two characters in the scenario benefits from the interaction, compared to if they don't interact. In the 'effort' experiment, we asked participants how much effort each character puts into the interaction, compared to if they don't interact. Thus, participants answered two questions on each trial, corresponding to the two characters in the scenario, each using a 7 point Likert scale ('not at all' (1) - 'extremely' (7)).

The scenarios in Supplementary Figure S3 are ordered based on the differential benefit between the two people interacting. In general, people expect the recipient of the generosity to benefit more than the generous actor, and that the generous actor puts in more effort than the recipient (Supplementary Figure S3a). Additionally, relative benefit generally corresponds to relative effort (scenarios where the generous actor puts in relatively more effort also correspond to the scenarios where they benefit less) (Supplementary Figure S3a). Some scenarios are more uneven in either benefit or effort: For example, in the 'restaurant' scenario, participants expect that the generous actor (who offers to go to the recipient's preferred restaurant) puts in comparatively less effort, compared to how much the recipient benefits from the interaction.

Overall, the results of these experiments suggest that, in general, our scenarios display uneven benefit or uneven effort (one character puts in more effort, or benefits more, than the other), and that they

cover a wide range of benefit and effort.

Results and discussion

The results from Study 1 replicated: When the relationship was hierarchical, participants expected generous acts to set a precedent (M = 4.86, SE = 0.08, 95% CI [4.71, 5.02]) rather than elicit reciprocity (M = 4.06, SE = 0.08, 95% CI [3.90, 4.22]; $\Delta M = -0.80$ [−0.95, −0.66], $t(3090) = -11.04$, $p < 0.001$, $d = -0.59$). Participants had the opposite expectation in equal relationships, expecting reciprocity (M = 4.98, SE = 0.09, 95% CI [4.79, 5.16]) over a precedent (M = 4.48, SE = 0.09, 95% CI [4.30, 4.67]; ΔM = 0.49 [0.29, 0.69], $t(3090) = 4.76$, $p < 0.001$, $d = 0.36$) (Figure 2b).

Participants were equally likely to expect a precedent from lower-ranked (M = 4.86, SE = 0.09, 95% CI [4.67, 5.04]) and higher-ranked (M = 4.87, SE = 0.09, 95% CI [4.69, 5.06]) characters (ΔM = 0.02 [−0.19, 0.22], $t(3098) = 0.16$, $p = 0.871$, $d = 0.01$; $BF_{01} = 11.79$, indicating strong evidence for the null) (Figure 2b). The expectations of a precedent held in both cases.

There was variation in people's responses across scenarios (Supplementary Figure S2). As an exploratory analysis, we measured the influence of the degree of asymmetry of costs and benefits incurred by the specific kind of generous interaction in the scenario, on participants' expectations. For each direction of hierarchical relationship, we conducted two separate linear mixed-effects regressions predicting how much people expected a precedent (probability of choosing a precedent vs. reciprocity, computed from the Likert responses) from differential benefit and from differential effort, as measured in the scenario validation experiment. Costs and benefits varied across scenarios (Supplementary Figure S3a) but had very little influence on expectations of reciprocity versus precedent setting. When the lower-ranked character played the generous role, the relative costs or benefits of the action did not affect how much people expected a precedent (effort $b = 0.00$, $t(16.04) = 0.21$, $p = 0.835$; benefit $b = -0.03$, $t(16.01) = -1.47$, $p = 0.162$) (Supplementary Figure S4). When the higher-ranked character was generous, cost did not influence expectations ($b = 0.02$, $t(16.03) = 0.58$, $p = 0.567$) (Supplementary Figure S4b), although for larger differential benefits, participants did consider following a precedent by the higher-ranked character slightly less likely ($b = -0.06$, $t(15.94) = -2.50$, $p = 0.024$) (Supplementary Figure S4a).

In sum, replicating Study 1, participants expected generous acts to elicit reciprocity in equal relationships, but to set a precedent in hierarchical relationships. Study 2 revealed that these results generalize across both directions of a hierarchical relationship. Generous acts set a precedent regardless of

whether the generous person is lower- or higher-ranked.

Study 3: Concrete labels for relationships generate the same systematic expectations as abstract labels

Archetypal symmetric and asymmetric social relationships are remarkably consistent across a large range of cultures and time periods of human history (X. Cheng et al., 2025). Studies 1 and 2 provided only minimal abstract descriptions of relationships. Study 3 (Figure 2ac) aimed to test whether people's vernacular intuitions about specific familiar roles map onto the more abstract distinction between symmetric and asymmetric relationships used in the previous two studies.

Methods

Participants

For Study 3, we recruited 60 participants (28 female, 32 male, 0 nonbinary; ages 18–79, $M(SD)$ age = 40.4(15.2)), and excluded 3 participants.

Procedure

The vignettes in Study 3 contained concrete labels for archetypal symmetric and asymmetric relationships (e.g., coworkers vs. advisor and advisee, cousins vs. uncle and nephew; see Figure 2a, and Supplementary Table S3 for a list of the specific relationships). The procedure was otherwise the same as for Study 2.

Results and discussion

The results replicated Studies 1 and 2 (Figure 2b). In equal relationships, participants expected reciprocity (M = 5.13, SE = 0.11, 95% CI [4.91, 5.34]) over precedent (M = 4.88, SE = 0.11, 95% CI [4.66, 5.10]) (ΔM = 0.25 [0.03, 0.46], $t(2982) = 2.29$, $p = 0.022$, $d = 0.18$). In hierarchical relationships, people expected precedent (M = 5.10, SE = 0.10, 95% CI [4.91, 5.29]) over reciprocity (M = 4.67, SE = 0.10, 95% CI [4.47, 4.86]) ($\Delta M = -0.43$ [-0.58, -0.29], $t(2982) = -5.70$, $p < 0.001$, $d = -0.31$). The relative direction of the hierarchical relationship did not alter these expectations: There was no difference in how much participants expected a precedent, when the generous character was lower-ranked (M = 5.10, SE = 0.11, 95% CI [4.88, 5.32]) versus higher-ranked (M = 5.11, SE = 0.11, 95% CI [4.89, 5.33]) ($\Delta M = -0.01$ [-0.22, 0.20], $t(2991) = -0.10$, $p = 0.922$, $d = -0.01$; $BF_{01} = 11.10$, indicating strong evidence for the null) (see Supplementary Figure S5 for the scenario-specific responses). Neither differential cost nor

benefit of the interaction influenced expectations when the lower-ranked character was generous (cost: $b = -0.00$, $t(16.78) = -0.22$, $p = 0.831$, benefit: $b = -0.02$, $t(16.44) = -1.11$, $p = 0.281$) (Figure S4cd). When the higher-ranked character was generous, differential cost did not influence expectations, but, as in Study 2, high ranked people were judged slightly less likely to repeat generous acts with large differential benefits (cost: $b = -0.01$, $t(16.61) = -0.49$, $p = 0.633$; benefit: $b = -0.05$, $t(15.89) = -3.21$, $p = 0.006$) (Figure S4cd).

Study 4: Expectations for who will be generous are based on the specific interaction and the general precedent

Studies 2 and 3 showed that people overall expected a generous act to set a precedent in hierarchical relationships, regardless of whether the generous actor was lower- or higher-ranked. However, expectations for who would be generous also varied across the specific interaction scenarios (Supplementary Figures S2, S5) in a way that could not be explained by relative cost or benefit. In Study 4, we investigate another plausible source of the variation: people's prior expectations for specific types of interactions in hierarchical relationships.

Specific scenarios could evoke different scripts for the roles different types of people should play (Jara-Ettinger & Dunham, 2025). Given these scripts, people might have prior expectations for who should be generous, even without observing any social interactions (implicit coordination; Schelling, 1980). For example, in many US contexts, it is expected that the lower-ranked person walks to the higher-ranked person's office to have a meeting, whereas the higher-ranked person is expected to conduct detailed checks on important tasks. These priors could be local and arbitrary, but might also reflect generalizable associations with different kinds of relationships (Nisbett & Miyamoto, 2005). If these scenario-based priors are strong, then people might expect roles to repeat just because those roles are likely in each interaction taken separately. This mechanism could explain scenario-specific variance in Study 2 and 3: Observing an action that aligns with one's prior beliefs could lead to strong expectations that the same action will happen again. In Study 4 (Figure 3), we explicitly measure these scenario-specific priors and assess their influence on people's expectations for when others will follow precedents.

In addition to scenario-specific priors, people could also be applying an abstract rule that people should repeat their previous roles in coordinated actions. If an abstract rule governs expectations of precedent-following, then people would predict repeated generous roles based on observed behavior alone.

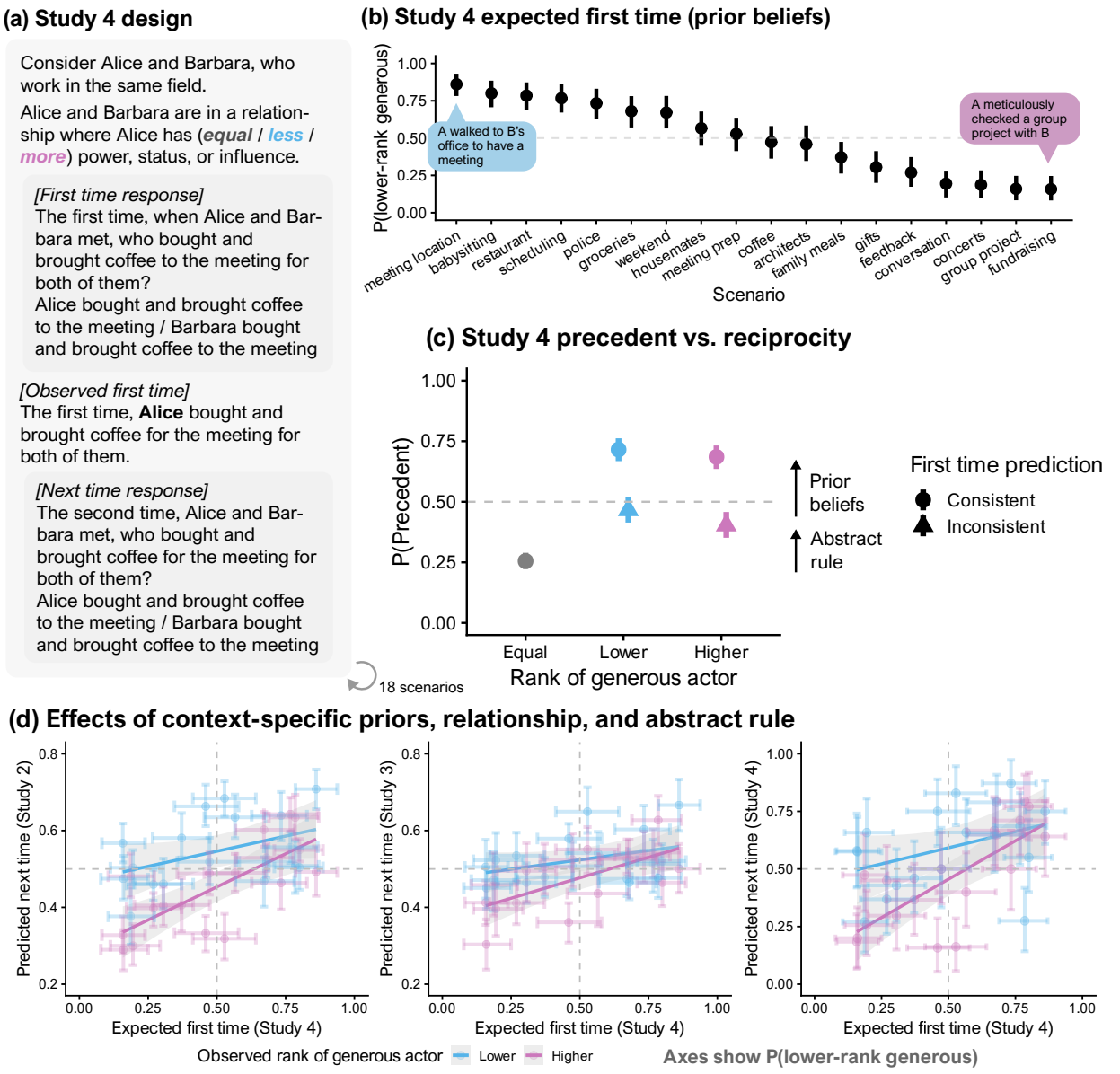


Fig. 3

Study 4. Participants ***(a)*** *reported their priors for who would be generous, and after observing who was actually generous (assigned by the experiment), predicted who would be generous the next time. Participants'* ***(b)*** *priors cover a range of expectations.* ***(c)*** *shows aggregate expectations that the next interaction will follow the precedent set by the first interaction (y-axis), based on whether the observed first interaction was consistent (circle) or inconsistent (triangle) with the participant's prior beliefs. In hierarchical relationships,* ***(d)*** *shows how people's prior expectations for each scenario (x-axis; as shown in (b)) and the experimentally manipulated observations of the first interaction (color) predict who they expect to be generous in the second interaction (y-axis). Each point corresponds to one scenario. Error bars are bootstrapped 95% confidence intervals.*

For example, if a specific higher-ranked person was observed walking to the lower-ranked partner's office for a meeting, contrary to the typical cultural script, then on the next interaction they would be expected to repeat their previous role and walk to the lower-ranked partner's office again.

Methods

Participants

For Study 4, we recruited 120 participants (59 female, 60 male, 1 nonbinary; ages 18–85, *M*(*SD*) age = 37.8(12.0)), and excluded 7 participants.

Procedure

In Study 4, participants were given minimal information about the relative rank of the two people interacting, as in Study 2, and then we explicitly dissociated the contributions of scenario-specific priors and an abstract rule by measuring expectations at two time-points. First, participants predicted who would be generous (two-alternative forced choice) based solely on the vignette, reflecting their prior beliefs. These beliefs should be probabilistic and open to revision. Then, after observing who was actually generous in the experimental manipulation (either confirming or violating their prior beliefs), participants predicted who would be generous in the next interaction (Figure 3a).

Results and discussion

Participants' prior beliefs[2] are shown in Figure 3b. Unlike abstract transfers of money or points, the vignettes evoke a wide range of expectations for which character in a hierarchical relationship would be generous in their first interaction. In some scenarios, participants expected the lower-ranked character to act generously, and in some scenarios, participants expected the higher-ranked character to act generously. This variation was not predicted by the differential costs or benefits in each scenario (two separate logistic mixed-effects regressions predicting whether the lower- or the higher-ranked character was initially generous, with random intercepts for scenario and participant; from relative benefit $b = 0.38$, $z = 1.50$, $p = 0.134$, OR = 1.47 [0.89, 2.42]; and from relative effort $b = -0.11$, $z = -0.43$, $p = 0.667$, OR = 0.89 [0.53, 1.50]) (Supplementary Figure S4e).

We then examined how expectations shifted after observing a social interaction between the two characters. We ran a logistic mixed-effects regression predicting the expected sequence of actions

[2] Note that these are only meaningful in an asymmetric relationships, where the roles differ.

('reciprocity' if the predicted next action was the opposite of the observed first action; 'precedent' if the predicted next action was the same as the observed first action) from the observed first action ('generous lower-rank' vs. 'generous higher-rank') and whether the observation was consistent with expectations, along with their interaction; including random intercepts for scenario and participant. In equal relationships, observing one person act generously led to expectations of reciprocity – that the opposite partner would be generous next ($M = -1.41$, $SE = 0.22$, 95% CI [-1.84, -0.98]). By contrast, in hierarchical relationships, people had stronger expectations that the observed partner would act generously again, following a precedent; although the size of the effect depended on whether the observed first interaction matched or violated participants' prior beliefs (Figure 3c). When the observed action aligned with participants' prior beliefs (i.e., when both the persistence of prior beliefs and an abstract rule could contribute to precedent-following), people were very likely to expect that the action would repeat, regardless of whether the generous actor was higher- or lower-ranked (lower: M = 1.16, $SE = 0.24$, 95% CI [0.70, 1.63]; higher: M = 0.87, $SE = 0.24$, 95% CI [0.40, 1.33]; $BF_{01} = 2.46$, indicating anecdotal evidence for the null) (both $p < 0.001$). Even when the observed action contradicted participants' prior beliefs (i.e., when only an abstract rule could contribute to precedent-following), participants still expected the observed actor to be generous again, more than in equal relationships (lower: M = -0.20, $SE = 0.23$, 95% CI [-0.66, 0.26]; higher: M = -0.39, $SE = 0.23$, 95% CI [-0.85, 0.07]) (both $p < 0.001$). In other words, given a minimal description of an asymmetric relationship, participants were less likely to expect reciprocity and more likely to expect a precedent – even when observed interactions violated participants' prior beliefs. Thus both scenario-specific priors and an abstract rule, activated based on knowledge of a hierarchical social relationship, contribute independently to expectations of precedent-following.

We examined how this behavioral pattern was shaped by expectations for the first action, across vignettes. We ran a logistic mixed-effects regression predicting 'expected next time' responses from 'expected first time' and 'observed first time', along with their interaction, and random intercepts for each scenario and participant. Across individual vignettes (Figure 3d, right), observers' predictions for the second interaction were additively shaped by both their scenario-specific prior beliefs (x-axis) and their experimentally manipulated observations of the first interaction (color): there were strong main effects of both the expected actor ($b = 0.51$, $z = 8.34$, $p < 0.001$, OR = 1.66 [1.47, 1.87]) and the observed actor ($b = 0.29$, $z = 5.14$, $p < 0.001$, OR = 1.34 [1.20, 1.50]) from the first interaction, on expectations for who

would be generous in the next interaction (y-axis).

One concern might be that participants' prior beliefs influenced expectations for the second interaction only because the experiment required participants to explicitly report these priors. To check this possibility, we conducted exploratory analyses using data from Studies 2 and 3. The regression model followed the same structure, but the response variable was instead relative probabilities computed from the Likert responses (specifically, the probability of the lower-ranked character acting generous next). All of the results replicated (Study 2: 'expected first time' $b = 0.25$, $t(16.18) = 9.46$, $p < 0.001$; 'observed first time' $b = 0.09$, $t(685.45) = 7.54$, $p < 0.001$; Figure 3d, left) (Study 3: 'expected first time' $b = 0.15$, $t(16.51) = 3.84$, $p = 0.001$; 'observed first time' $b = 0.05$, $t(663.17) = 4.21$, $p < 0.001$; Figure 3d, middle). Additional variance in Study 3 was explained by including features of the concrete relationship labels derived from a large-scale study (see Supplementary Material, section A7 for detail).

In sum, Study 4 replicated the key findings of the earlier studies, and explained scenario specific variance by dissociating the effect of scenario-specific prior beliefs. Predictions of precedent-following in hierarchical relationships are shaped by both these priors and an abstract rule that what happened last time should happen again. In everyday life, these inferences are difficult to separate because culturally specific scripts reflect what usually happens. Naturalistic measurements of social experience would rarely capture interactions in which expected social roles are violated. By experimentally dissociating prior beliefs from experimentally manipulated observations, we showed that people rely on both. Participants' prior expectations for who would be generous were strengthened when aligned with observed actions, and weakened when they were violated. In both cases, however, people activated an abstract rule that previous roles in an interaction should repeat in hierarchical relationships. A minimal cue to hierarchy was sufficient to trigger these expectations.

Studies 5-6: Costly first person choices reflect the same expectations

Studies 1–4 measured third-party observers' predictions of who will be generous, using vignettes reflecting middle-class U.S. contexts. In Studies 5 and 6, we tested whether these expectations also shape people's own choices in incentivized games. Incentivized coordination and bargaining games are the paradigm that has produced some of the strongest evidence for reciprocity and turn-taking as default strategies (Camerer, 2011; McKelvey & Palfrey, 2001). The predominance of turn-taking in these games may itself reflect the fact that they are typically played between strangers, who are implicitly equal in status

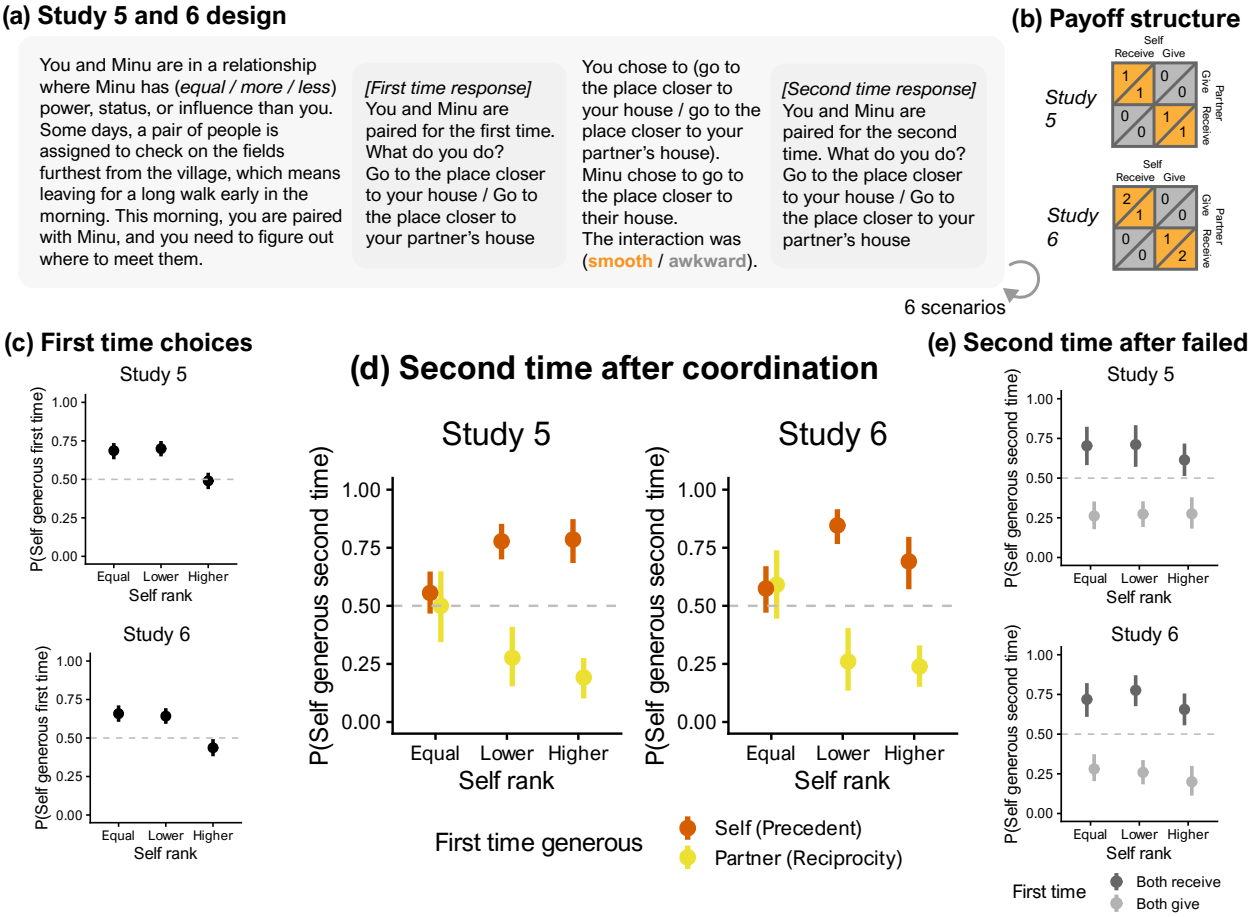


Fig. 4

Studies 5 and 6. Participants ***(a)*** *chose what they would do the first time and the second time they interact with their partner. The explicit incentives in the experiments are shown in* ***(b)****: Participants were incentivized based on successfully coordinating and choosing complementary actions in both studies, but accepted a personal cost for being generous in Study 6. Participants* ***(c)*** *were generally generous in their first choices, except for when they were higher ranked. Following a successful coordination, participants* ***(d)*** *repeated their actions if they were lower or higher ranked, but switched their actions if they were in an equal relationship. Following an unsuccessful coordination, they* ***(e)*** *chose the complementary action to what their partner chose the first time. Error bars are bootstrapped 95% confidence intervals.*

(Hoffman et al., 1994). If minimal relationship cues can shift behavior even in this setting – where there are explicit financial stakes and minimal culturally specific priors – it would provide a strong test of whether the expectations documented in Studies 1-4 generalize beyond hypothetical vignette judgments. In sum, Studies 5 and 6 (Figure 4) test whether minimal relational information is sufficient to alter behavior even in a setting designed to maximize strategic thinking.

Methods

Participants

For Study 5, we recruited 156 participants (79 female, 76 male, 1 nonbinary; ages 19–77, *M*(*SD*) age = 39.2(11.8)) and excluded 3 participants. For Study 6, we recruited 162 participants (76 female, 84 male, 2 nonbinary; ages 19–78, *M*(*SD*) age = 40.7(13.7)) and excluded 4 participants. There were two attention checks and participants were excluded if they missed both attention checks. In each of Studies 5 and 6, participants received $3 for completing the experiment, with a bonus of up to $3 based how many points they earned in the task, for an estimated pay of $15/hour.

Procedure

Participants were told to imagine that they were visiting a remote society with unknown norms (to minimize the effect of participants' culturally specific priors), where they would interact with multiple individuals who understood these norms (Figure 4a). Participants interacted twice each with simulated partners described as having either more, equal, or less power/status/influence, across six scenarios involving asymmetric interactions with one generous role (e.g., serving food first or carrying heavier objects; see Supplementary Table S4 for all scenarios). Participants chose a first role (two-alternative forced choice between giving and receiving generosity) and then received immediate feedback of their partner's choice and whether the coordination was "smooth" (complementary choices, which are required for successful coordination in these role-differentiated scenarios) or "awkward" (non-complementary choices). Our key hypotheses concerned what participants chose to do during the second interaction, based on the the outcome of the first interaction and the minimal description of the relationship.

Each of the six scenarios appeared in one of the six conditions (partner rank: 'equal' vs. 'lower' vs. 'higher'; partner action: 'give' vs. 'receive'). Like in the previous experiments, the assignment of conditions to scenarios was balanced within and across participants. The trial sequence was randomized,

with the constraint that the 'first time' interaction for a certain partner always happens before the 'second time' interaction with that partner.

The two studies differed only in participant incentives (Figure 4b). In Study 5, participants received one point for smooth interactions (coordination: the partners chose complementary roles) and no points for awkward interactions (miscoordination: partners chose the same role). Although the financial payoffs for giving and receiving were identical in Study 5, the vignettes described one role as performing a generous act and the other as receiving it, preserving the narrative cost asymmetry of the vignette studies. Study 5 thus serves as an intermediate between the vignette paradigm of Studies 1–4 and a fully incentivized paradigm. In Study 6, the incentives corresponded to a Leader-Follower game (archaically named Battle of the Sexes; Luce & Raiffa, 1957): two points for smoothly receiving a generous act, one point for smoothly doing a generous act, and no points for awkward interactions. Thus, in Study 6, participants incurred a direct financial cost when choosing to act generously compared to receiving generosity.

Results and discussion

People expect others to follow a precedent, except after successful interactions in equal relationships. In these games, participants expressed predictions for others' actions behaviorally, by making complementary choices. To test our predictions, we ran a logistic mixed-effects regression predicting participants' likelihood of expecting a precedent (versus reciprocity), from the social relationship and whether the first interaction was smooth or awkward, along with their interaction; with random intercepts for scenario and participant; followed by post-hoc contrasts between the marginal means. Consistent with our key hypothesis, participants' choices for the second social interaction varied depending on the minimal description of their relationship with the partner (Figure 4d). After a smooth first interaction in a hierarchical relationship, participants continued choosing the complement of their partner's first choice. That is, participants expected to coordinate with both lower-ranked and higher-ranked partners by following a precedent. This was the case even in Study 6, where participants followed a precedent even when they were in the disadvantaged role (i.e. repeatedly giving at a personal cost) (Study 5 lower: M = 1.60, *SE* = 0.24, 95% CI [1.12, 2.07]; higher: M = 1.48, *SE* = 0.22, 95% CI [1.04, 1.91]) (Study 6 lower: M = 1.18, *SE* = 0.20, 95% CI [0.78, 1.57]; higher: M = 1.65, *SE* = 0.24, 95% CI [1.18, 2.13]). By contrast, after a smooth first interaction in an equal relationship, participants were comparatively more likely to reciprocate their partner's initial action (all $p < 0.002$), accepting the risk of miscoordination in service of

balanced exchange. This pattern was robust across both Studies 5 (symmetric payoffs) (M = 0.23, SE = 0.20, 95% CI [−0.17, 0.62]) and 6 (asymmetric payoffs) (M = 0.14, SE = 0.19, 95% CI [−0.24, 0.51]). Even in Study 5, where the financial payoffs were identical for both roles, participants treated the narrative distinction between giving and receiving as meaningful – sacrificing coordination efficiency for turn-taking, but only when the relationship was described as equal.

If the first interaction resulted in miscoordination, participants chose the opposite of what they and their partner initially chose, giving if they had chosen to receive and receiving if they had chosen to give, in all three relationships (Figure 4e).

What contributed to people's choices to be generous? For the first interaction (Figure 4c), participants chose to initiate generosity when they were equal-ranked and when they were lower-ranked (potentially because of reputational or status-related benefits; Flynn & Yu, 2021), but not when they were higher-ranked. This was the case even in Study 6, when there were explicit financial disincentives for choosing to act generously.

For the second interaction, several factors could influence participants' choices in addition to relationship structure. People might rely on habits and simply repeat their own previous actions (Wood & Rünger, 2016). People might overall feel motivated to reciprocate generous acts they received (Fehr & Gächter, 2000; Gouldner, 1960). People might apply a simple learning rule to repeat their action after successful coordination and change actions after unsuccessful coordination (the win-stay lose-shift strategy; Homans, 1958; Nowak & Sigmund, 1993). Finally, each person could have an overall baseline motive to act prosocially and benefit others (Aknin et al., 2018; Rand & Nowak, 2013).

We predicted participants' choice to be generous in the second interaction, based on all of the factors above, plus an additional interaction term between the partner's first choice and the social relationship, with random intercepts for scenario and participant. An effect of the participant's first choice corresponds to habit (whether participants are likely to give when they chose to give the last time), an effect of the partner's first choice corresponds to straightforward reciprocity (whether participants are more likely to give when their partner chose to give the last time), an interaction effect corresponds to an effect of a 'win-stay lose-shift strategy' (complementary choices leading to a smooth interaction; the levels are effect-coded, so the negative of the interaction effect directly corresponds to WSLS), and an intercept corresponds to baseline generosity.

This full model explained the results better than a simpler model that did not include the social relationship (likelihood-ratio test; Study 5 $\chi^2(4) = 11.68$, $p = 0.020$; Study 6 $\chi^2(4) = 26.64$, $p < 0.001$). The interaction term was significant (Type III ANOVA; Study 5 $\chi^2(2) = 9.38$, $p = 0.009$, Study 6 $\chi^2(2) = 17.98$, $p < 0.001$). The only other predictor across the two experiments was a negative effect of simple reciprocity (Study 5 $b = -0.86$, $z = -11.39$, $p < 0.001$, OR = 0.42 [0.37, 0.49]; Study 6 $b = -0.91$, $z = -12.20$, $p < 0.001$, OR = 0.40 [0.35, 0.47]): With the exception of a successful first interaction in an equal relationship, participants generally chose the opposite of their partner's first choice whether the first interaction was successful or not, indicating that participants expected their partner to perform the same actions again, following a precedent. There was a small negative effect of habit in Study 6 ($b = -0.16$, $z = -2.06$, $p = 0.039$, OR = 0.86 [0.74, 0.99]) but not in Study 5 ($b = -0.01$, $z = -0.17$, $p = 0.863$, OR = 0.99 [0.85, 1.15]). There was no evidence of a heuristic rule like win-stay-lose-switch (Study 5 $b = -0.09$, $z = -1.18$, $p = 0.237$, OR = 0.91 [0.79, 1.06]; Study 6 $b = -0.12$, $z = -1.61$, $p = 0.107$, OR = 0.89 [0.76, 1.03]) or baseline generosity in the second interaction (Study 5 $b = -0.07$, $z = -0.97$, $p = 0.330$, OR = 0.93 [0.80, 1.08]; Study 6 $b = 0.01$, $z = 0.15$, $p = 0.882$, OR = 1.01 [0.86, 1.18]).

In sum, participants' own financially incentivized behaviors in repeated asymmetric coordination games reflect their prediction that partners are likely to follow a precedent across interactions. Participants were less likely to expect a precedent and more likely to expect reciprocity, only in equal relationships after a successful first social interaction. As in Studies 1-4, minimal context about relationships and daily-life social interactions was sufficient to induce these expectations.

General Discussion

We found that people form distinct and systematic expectations for sequences of generous acts. In most situations, a single generous act established a precedent: if the context arose again, the same person was expected to be generous in the same way again. Yet people's predictions and behavior were substantially altered by minimal descriptions of social relationships. In asymmetric hierarchical relationships, following a precedent became more likely. Expectations of reciprocity were restricted to equal relationships. We replicated this pattern six times, across third-party hypothetical judgments and first-person coordination games with financial payoffs. Precedents were anticipated in the absence of any relationship information (Study 1), in asymmetric relationships (Studies 1-6), both up and down a

hierarchy of power or status (Studies 2-6), in concrete familiar asymmetric relationships like uncle-nephew or boss-employee (Study 3), and across a range of specific vignettes (Study 4) and action costs and benefits (Studies 2-4). These expectations influenced people's own choices (Study 5), even when following a precedent meant repeatedly being generous and thus repeatedly receiving a lower payoff than the partner (Study 6).

Our results contrast with many studies of lab-based coordination games between strangers, in which players typically expect direct reciprocity, and use partner choice to avoid partners who try to repeatedly benefit (Barclay, 2016; Lau & Mui, 2008; McKelvey & Palfrey, 2001). The predominance of reciprocity in these paradigms may reflect a default assumption of equal rank among strangers (Hoffman et al., 1994; Tatone & Csibra, 2020). By contrast, our results suggest that this assumption is brittle, and can be disrupted by even minimal social context and framing (DeScioli & Krishna, 2013; Hoffman et al., 1994). Compared to standard paradigms, we made two key methodological changes: our vignettes described meaningful everyday asymmetric social interactions (giving gifts, household labor, carrying a heavy object) rather than simple exchanges of money; and we situated the social interactions in minimal but meaningful structured social relationships. These changes led to systematic shifts in expectations. The preference for reciprocity in equal relationships was also consistently smaller in magnitude than the strong shift toward precedent in hierarchical relationships, suggesting that reciprocity among equals may be a weaker baseline expectation rather than the robust default norm assumed by much prior work.

This research directly tests the prediction, articulated most explicitly by Graeber (2012) and grounded in relational theories (Clark et al., 2017; Fiske, 1992), that expectations for sequences of generosity are context-sensitive expectations governed by the perceived structure of social relations. This view challenges one way economists traditionally conceptualize human behavior. Classic economic accounts are grounded in the assumption that human social and moral life is motivated by a sense of justice in the form of reciprocity and fairness (Homans, 1958; Lévi-Strauss, 1952). According to this view, the psychological principle that actions and debts should be reciprocated, is not only used as a descriptive theory of social behavior but also as a normative justification for large-scale market-based economies, where the payoffs of actions are explicitly quantified and meant to be balanced (Fehr & Gächter, 2000; Sugden et al., 2004). By contrast, Graeber draws on extensive ethnographic evidence describing everyday generous acts, to argue that people only have these expectations in social relationships that are actually or

potentially between social equals. This theoretical claim is ultimately a psychological one, yet it has not been directly tested. In this paper, we addressed this gap by empirically testing the minimal conditions that evoke these context-dependent expectations.

Our research has limitations. Notably, our experiments recruited native English speakers from the United States – the kind of population typically sampled for existing experimental work demonstrating expected norms of reciprocity – to test whether minimal cues about structured social relationships could evoke contrasting expectations similar to those described in ethnographic studies. Even for this population from a society with high mobility, loose norms (Gelfand et al., 2011), and high market integration (Henrich, Ensminger, et al., 2010), minimal relationship information shifted expectations. Generalizing the current claims beyond this cultural setting would require cross-cultural research (Henrich, Heine, & Norenzayan, 2010; Rad et al., 2018). We speculate that scenario-specific cultural norms vary across cultures and historical contexts, and yet that the principle – linking a hierarchical relationship to an abstract rule that past actions should repeat, and an equal relationship to reciprocity – may be shared across societies.

Our studies focused on direct, short-term reciprocity in dyadic interactions involving active, voluntary giving – in Fiske (1992)'s framework, "equality matching" (balanced turn-taking) and "market pricing" (quantified exchange). We did not probe longer-term or indirect forms of reciprocity (Bekkers & Wiepking, 2011; Fiske, 1992; Nowak & Sigmund, 2005), communal or need-based giving where benefits are provided without bookkeeping (Clark & Mills, 2012; Hruschka, 2010) – what anthropologists call "promiscuous sharing," characteristic of high-intimacy relationships (Fiske, 1992) – or passive transfers such as taking or leaving goods for others, which may evoke weaker reciprocal obligations (Tatone & Csibra, 2020; Widlok, 2016). None of the relationships in our studies were at the extreme ends of intimacy (e.g., spouses, parent-infant), and our response options asked about the next specific interaction rather than longer-term patterns of balance. How closeness, transfer mode, and timescale interact with hierarchy to shape exchange norms are important questions for future work.

There are at least two cognitive mechanisms that could contribute to people's strong expectations of precedents in our experiments. First, acting predictably, as in following a precedent, is a way to simplify social coordination, increasing collective benefits. All else being equal, people expect each other to choose the same action if the same context arises again, and prefer others who are predictable in their social actions (Theriault et al., 2021). Precedents and conventions exert a powerful force, stabilizing expectations across

many types of social interactions (Hawkins et al., 2019; Henrich, 2016; Lewis, 1969; Millikan, 2005). In iterated asymmetric coordination games in particular, repeating the same roles is a good strategy because each player's policy is simple, reducing demands on memory and risks of miscoordination (Hammerstein, 2003; Stevens & Hauser, 2004). Indeed, following an asymmetric precedent is often in every individual's own strategic best interest, including those who are repeatedly disadvantaged by the precedent and even though it sustains and entrenches unequal payoffs (O'Connor, 2019). Thus people follow precedents when the probability and costs of miscoordination are high (Prisbrey, 1993; Yu & Thompson, 2024).

Second, in our studies, expectations were influenced by described and implied social relationships (Thomas, 2024). People may expect precedent particularly in hierarchical relationships because each asymmetric interaction affirms and reinforces the overall social order (Rai & Fiske, 2011) – when hierarchy is mutually recognized as legitimate, the unequal payoffs of following a precedent may be a feature rather than a bug. A notable feature of our results is that expectations of precedent held in both directions of a hierarchical relationship: whether the generous actor was lower-ranked or higher-ranked, participants expected the same person to be generous again. The most parsimonious interpretation is that status asymmetry itself activates a precedent norm, irrespective of directionality. One might speculate that upward generosity (lower-ranked benefiting higher-ranked) signals dominance, while downward generosity signals prestige (J. T. Cheng et al., 2013; Henrich & Gil-White, 2001), and that both lead to precedent through different mechanisms. However, our data do not distinguish these possibilities, and the mapping between directionality and hierarchy type may not be straightforward, since upward voluntary giving (e.g., tributary gifts) can also occur in prestige-based or culturally legitimate hierarchies (Fiske, 1992; Henrich & Gil-White, 2001; Mauss, 1925; Yan, 1993). Whether people represent these as a single mapping (hierarchy $\rightarrow$ precedent; Graeber, 2012) or two distinct ones (dominance $\rightarrow$ lower gives, prestige $\rightarrow$ higher gives; Henrich & Gil-White, 2001) is an open question for future work.

In summary, we studied people's social cognitive expectations of repeated sequences of generous acts. We found experimental evidence that expectations of direct reciprocity emerge specifically in the context of perceived or desired equality between individuals. In other common social situations, including interactions between people in hierarchical relationships, one generous act created the opposite expectation, that the same person would be generous in the same way again.

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Author contributions

A.M.C and R.S. designed research, A.M.C. performed research and analyzed data, R.S. provided funding and supervision, and A.M.C. and R.S. wrote the paper.

Data availability statement

All data and code for running the experiments and reproducing the analyses and plots are available on GitHub at https://github.com/aliciamchen/generosity-actions and also deposited on Zenodo at https://doi.org/10.5281/zenodo.17477355.

Competing interests

The authors declare no conflict of interests.

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Signaling social identity in referential communication

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Abstract

Any choice of words simultaneously conveys information about the world and, at the same time, conveys information about the speaker, revealing aspects of their social identity. In this paper, we investigate how speakers strategically modify referential language to signal group membership. Across four experiments using a minimal referential communication paradigm, we find that speakers with the explicit goal of signaling social affiliation (1) choose more concise utterances, (2) preferentially select group-specific referents and descriptions, and (3) resist the otherwise strong tendency to be understood by everyone in the audience. Standard models of referential communication that focus on the trade-off between informativity and efficiency cannot explain these patterns; we argue instead for a model where speakers trade off the referential utility of being understood against the social utility of being identified as an in-group member.

Keywords: social cognition, communication, audience design, computational modeling

Significance Statement

When a group of people interacts repeatedly – from friend groups, to classrooms or schools, to larger online communities – they develop shared ways to refer to concepts and objects. Consequently, speakers can use these expressions as social identity markers, signaling group membership to other group members. In controlled experiments and a formal model, we study how speakers strategically use group-specific terms, similar to slang or jargon, to identify themselves as insiders. This research helps to explain how people use linguistic choices to create and maintain social boundaries between different social groups.

Signaling social identity in referential communication

Human communication is shaped by multiple considerations. Perhaps the most basic consideration is referential efficiency: Speakers want to be clear enough so that listeners can identify what they are talking about, while also not saying more than they need to say. A balance between these two goals – maximizing referential informativity, while minimizing cognitive effort – can explain many aspects of language evolution, processing, and use (e.g., Frank & Goodman, 2012; Gibson et al., 2019; Jara-Ettinger & Rubio-Fernandez, 2022; Kanwal et al., 2017; Kemp & Regier, 2012; Zaslavsky et al., 2018; Zipf, 1949). The influence of these goals is especially apparent in the idiosyncratic group-specific conventions that emerge when people interact with each other repeatedly (Boyce et al., 2024; Hawkins et al., 2023). Over time, conversational partners tend to develop efficient and shared ways to refer to things, that work because everyone in the group knows what they mean (Brennan & Clark, 1996; Clark & Marshall, 1981; Clark & Wilkes-Gibbs, 1986; Krauss & Weinheimer, 1964). When addressing listeners who lack this shared history, however, speakers shift to longer and more descriptive expressions so that outsiders can understand (Wilkes-Gibbs & Clark, 1992).

Notably, when people *interpret* language, they make inferences about the speaker that go beyond identifying the intended referent. Linguistic variation is an especially reliable and observable signal of social groups and communities (Eckert, 1989; Kinzler, 2021; Labov, 1973; Lupyan & Dale, 2016), and so one important type of inference concerns the speaker's group membership or social identity (e.g., Bucholtz & Hall, 2005; Gumperz, 1982; Kinzler et al., 2007; Rhodes et al., 2018; Smaldino, 2022). While social identity influences many aspects of language (including syntax, grammatical constructions, and accent; Bucholtz & Hall, 2005; Camp & Nowak, 2025; Fedzechkina et al., 2023; Giles et al., 1991; Goffman, 1956; Labov, 1966; Scherer & Giles, 1979), one way listeners make inferences about speaker group membership is from the speaker's word choice in reference. After hearing only brief verbal descriptions from a speaker, and even without explicitly referring to group membership, listeners infer the kinds of communities speakers belong to (Isaacs & Clark, 1987; Walker et al., 2025). For example, in one experiment, people described New York City landmarks to each other. New Yorkers could reliably identify a compatriot New Yorker from a single description (e.g., "Washington Square Park" vs. "the fountain with the arch in the background"; Isaacs & Clark, 1987). The social diagnosticity of words extends past locally specific referents like city landmarks (Eckert, 2018): Popular discourse, for instance, claims that New

Yorkers can be recognized by their use of the word “train” rather than “subway” or “metro.”

The way social identity affects language can be unconscious, or it can be deliberate and strategic (Bell, 1984; Gumperz, 1982; Myers-Scotton, 1998; Woolard, 2004). Just as listeners can infer social identity from language, speakers can intentionally and strategically shape their word choice to signal and emphasize these identities (Burnett, 2023; Gibson et al., 2019; Le Page, 1968; Pleyer et al., 2025). For instance, a speaker might use a concise term like “rizz” (short for “charisma”) for efficiency, but also to reveal their membership in the communities that use it (Labov, 1973). Field-specific jargon or group-specific slang can signal social affiliation at the expense of being broadly understood (Brown et al., 2020; Damirjian, 2025). Speakers may choose topics or referents that are uniquely discussed with their in-group, or concepts that are especially salient within their community (Eckert & Brown, 2006; Holmes & Meyerhoff, 1999). In this way, signaling social identity may compete against the tendency to ensure that everyone in a mixed audience can understand what the speaker is saying (S. O. Yoon & Brown-Schmidt, 2018, 2019). Linguistic choices that emphasize social identity can thus lead to negative consequences such as misunderstandings or the exclusion of non-group members (Bullock et al., 2019; Cruz & Lombrozo, 2025; Eckert, 1989; Martínez et al., 2022), as well as broader polarization in society (Walker et al., 2025).

In this paper, we characterize the influence of strategic social goals in referential language, by integrating two previously separate research traditions. The standard *referential communication* experimental framework in which a speaker describes a target object for the listener to identify, has been used to study how groups develop arbitrary conventions for talking about abstract shapes (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986), and how speakers change their language based on different kinds of communicative goals (S. O. Yoon et al., 2012). Formalizing the underlying decision-making process in this paradigm has enabled researchers to quantify the role of referential efficiency in convention formation (Hawkins et al., 2023). Yet this framework has largely ignored social signaling motivations.

Conversely, sociolinguistic research has richly documented social identity signaling in everyday speech, but has not used controlled experiments to quantify its influence compared to other factors in referential word choice. We bridge these traditions by extending the reference game paradigm used by the former tradition, to explicitly manipulate speakers’ social-signaling goals. We use this paradigm precisely because it provides the minimal conditions for a tension to arise between goals in word choice. If social affiliation systematically competes with referential efficiency even in minimal experimental contexts, then

the existing models, which formalize the strategic decision-making process underlying how people choose referential language (Goodman & Frank, 2016; Hawkins et al., 2023), are missing a core driver of language production. By quantifying this trade-off experimentally and formalizing it in a computational model, we can begin to measure how referential and social signaling goals jointly shape even the most basic settings of language use, providing a foundation for understanding more complex real-world cases where these pressures are amplified by more deeply-held social identities and consequences.

To test how speakers navigate these pressures, we developed a computational model that formalizes utterance choice as a graded trade-off between referential efficiency and social identification. The experiments are designed to measure the model's predictions. Participants first observe different groups developing their own linguistic conventions in an online reference game where people type freely into a chat box (Boyce et al., 2024); next, the participants themselves must decide what to refer to and how to refer to it, based on their audience (people in the observed groups and/or naive players) and their goal (to refer effectively, or signal social affiliation). This design preserves experimental control – we experimentally assign which groups talk about which referents, what words they use, and what the speakers' goals are – while maintaining ecological validity, as participants observe real conversations produced naturally by human speakers interacting in groups over time.

In Experiment 1, we find that people use information in the observed conversations to selectively choose what to talk about and what to say based on their audience and goal. Then, in Experiments 2-4, we isolate specific comparisons to measure the precise contributions of social signaling on people's choices. The results show that people with a social signaling goal choose referents that afford group-specific labels (Experiment 2), select more concise group-specific labels (Experiment 3), and resist the otherwise strong tendency to prioritize the least informed audience member (Experiment 4). We compare a baseline referential model – capturing the standard trade-off between informativity and effort – to an augmented social signaling model that includes a utility for in-group identification. While the baseline model accounts for referential efficiency-driven choices, the social model is needed to explain behavior when participants choose utterances to achieve a social goal.

Transparency and openness

All de-identified data, materials, and code are available at https://github.com/aliciamchen/code-switching and also deposited on Zenodo at

https://doi.org/10.5281/zenodo.17905933. The data were collected between October 2024 and July 2025. The sample sizes, design, and analyses for all experiments were all preregistered before data collection. The preregistration for Experiment 1 is available at https://doi.org/10.17605/osf.io/uzsc2, the preregistration for Experiment 2 is available at https://doi.org/10.17605/osf.io/jyktz, the preregistration for Experiment 3 is available at https://doi.org/10.17605/osf.io/z27eb, and the preregistration for Experiment 4 is available at https://doi.org/10.17605/osf.io/h4trx. All statistical model specifications and outputs are available in the project repository. There were minor deviations from the preregistrations, described in Appendix B. AI tools were used to aid the generation of analysis and plotting scripts and to improve the readability of the manuscript.

Experiment 1

In Experiment 1, we conducted an initial test of whether people track and deploy group-specific referential conventions for social signaling, in the context of experimental reference games. In the first phase of the task (the 'observation phase'), participants passively watched two groups taking turns writing descriptions to help the other members of their group identify abstract tangram shapes (Figure 1; Clark and Wilkes-Gibbs, 1986). As in prior work, group-specific conventions naturally emerged over the sequence of interactions (Boyce et al., 2024), such that the two groups differed in which tangrams they discussed and the expressions they developed for those tangrams. In the second phase of the task (the 'selection phase'), we introduced the critical manipulation of communicative goals. Participants were shown an audience composed of members of one observed group along with naive people who had not participated in the game. They were asked either to pursue a referential goal (help their audience identify the intended tangram) or a social-signaling goal (help the group members in the audience identify the speaker as a member of their group). This design allows us to address three key questions about how social signaling operates in referential communication.

First, to what extent do participants spontaneously encode and track group-specific linguistic patterns during passive observation, and can they use this knowledge for referential purposes? Prior work suggests that overhearers often acquire less information than active participants in reference games (Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992), making it important to establish that our paradigm provides a strong enough foundation to reproduce the basic referential behaviors seen in prior studies. Demonstrating that participants can use passively observed conventions for reference validates the task and

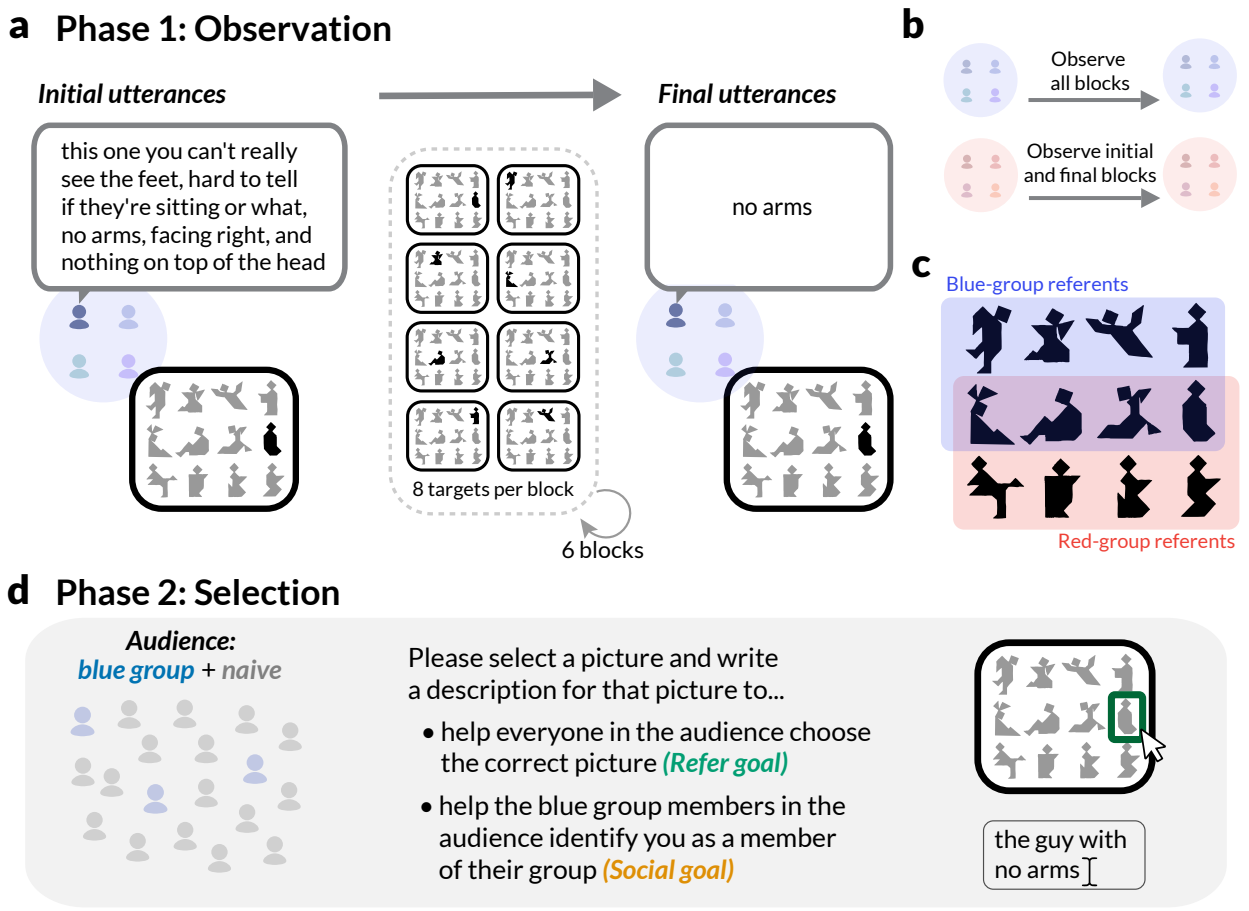


Figure 1

Experiment 1. ***(a)*** *In the first phase of the task, participants were third-party observers viewing a group of four other people playing a multiparty reference game.* ***(b)*** *Participants watched the full convergence process of the blue group, and only watched the initial and final blocks for the red group.* ***(c)*** *Each group referred to eight of the twelve tangrams. Four of the tangrams were only referred to by the blue group, four only by the red group, and four by both groups.* ***(d)*** *In the second phase of the task, participants were shown an audience and a goal, selected a tangram to refer to, and wrote a description for that tangram.*

provides a baseline against which we can compare the predictions of a social-signaling model.

Second, when given a social-signaling goal, how do participants determine which specific linguistic choices – among an open-ended set of possibilities – would effectively signal group membership? Because the task provides no explicit instruction during the passive observation phase, participants must infer this from the observed interactions. We test whether participants identify referents and expressions with high social-signaling value, and selectively deploy them when their goal is to

communicate group affiliation.

Third, how do speakers weigh social signaling against referential clarity when these goals conflict? By manipulating the proportion of naive people in the audience, we introduce situations where strategies that best serve one goal might not be ideal for the other. When addressing an all-in-group audience, there is no goal conflict, because the short utterances that were created in the group simultaneously support effective reference and social identification. By contrast, when there are naive people in the audience, purely referential accounts would predict that participants would 'aim low' to the least-informed members of the audience, so that everyone can understand (S. O. Yoon & Brown-Schmidt, 2018, 2019). A social-signaling account predicts the opposite: speakers might preferentially use insider language to affiliate with in-group members, even at the cost of making naive listeners confused. Thus, any divergence between referential and social-signaling choices should systematically scale with the proportion of naive audience members.

Critically, our focus is not simply on whether participants can follow instructions to signal group identity, but rather to understand *how* they do so. We ask whether participants independently identify the socially meaningful aspects of the communication system they observed, assess their relative signaling value, and calibrate their use based on audience composition. If participants make systematic, goal-dependent trade-offs even without explicit guidance about the social meaning of different choices, this would provide strong evidence that people spontaneously track the language of minimal groups and strategically draw on this information when communicating.

Methods

Participants

For all experiments, participants were recruited on the crowdsourcing platform Prolific and pre-screened to be adult fluent English speakers from the United States, and were excluded if they indicated at the end of the experiment that they did not understand the instructions. For Experiment 1, we recruited 185 (93 female, 90 male, 2 non-binary; ages 20-72, M(SD) age = 40.2(11.7)) participants, and no participants were excluded. Participants were paid $12 for completing the experiment, which took an estimated 40 minutes to complete. Participants gave informed consent, and all procedures were approved by the MIT Committee for the Use of Humans as Experimental Subjects.

Design and stimuli

Phase 1: Observation phase. In the *observation phase* (Figure 1a), participants first learned each group's referential conventions through passive observation, without being told what they would do in the second phase of the task. They watched videos showing interleaved rounds of communication by two groups (arbitrarily named the 'red' and 'blue' groups) composed of four people each (see https://osf.io/f9dxb for an example round). The groups played the reference game within their group. In each round of the reference game, one player (the 'speaker') described the target tangram in a communal chat; the three other players (the 'listeners') each made a guess by selecting one tangram. Players received feedback on each of the listeners' guesses at the end of the round. Each video corresponded to one round of gameplay and ranged in duration from approximately 5 to 30 seconds.

The gameplay in each group consisted of 6 blocks (matching the four-person 'full feedback' condition described in Boyce et al., 2024). In each block, one speaker described each of the eight tangrams once, in randomized order, before the speaker role rotated to the next player and the next block began. Over the blocks, participants observed expressions converge to shorter, more abstract, and group-specific labels (e.g. "this one you can't really see the feet, hard to tell if they're sitting or what, no arms, facing right, and nothing on the top of the head" becomes "no arms"; see Figure 1a). The task focused on one of the groups (always set to the blue group for simplicity): Participants viewed the full convergence history (6 blocks x 8 targets per block = 48 rounds) for the blue group, but only saw the red group's initial and final blocks (2 blocks x 8 targets per block = 16 rounds; see Figure 1b). Of the total 12 tangrams, four of the tangrams were talked about only in the blue group ("blue-specific referents"), four of the tangrams were talked about only in the red group ("red-specific referents"), and four of the tangrams were talked about in both groups ("common referents"; see Figure 1c). Within each set, the two groups converged on distinct, group-specific labels for all tangrams, ensuring that no overlap in referential labels occurred between groups. There were three different tangram sets (assignments of tangrams and conversations to groups). Across participants we also counterbalanced the assignment of individual tangrams to the "red" versus "blue" groups.

Videos were created by extracting individual conversations about specific tangrams from a large corpus of online multiplayer reference games (Boyce et al., 2024) and splicing them together to satisfy the task's design criteria. Because each round of the repeated reference game is a self-contained conversation centered on a single referent, these spliced sequences appeared natural and coherent to participants.

Phase 2: Selection phase. Participants then moved to the *selection phase*, in which participants selected a tangram and freely produced a referring expression (Figure 1d). On each trial, participants were given a goal and audience, chose one tangram out of the 12 possible tangrams, and then wrote a description of that tangram in a text box. The audience and goal were manipulated fully within-subject. The audience varied continuously in size and composition of in-group members, and was shown to participants as an iconic depiction from 0 to 4 blue-group avatars and 0, 1, 2, 4, 8, or 16 naive avatars. For the goal, participants were told what kind of question the audience would be answering (the 'referential goal' of choosing the best tangram given the label, or the 'social signaling goal' of being identified as a member of their group; Figure 1d)[1].

There were 53 trials in total. To incentivize performance, participants were nominally offered a bonus of $0.10 for each 'correct' response; in practice, all participants were given a $6 bonus and debriefed after the experiment that there were no objectively 'correct' answers. To avoid the trivial strategy of repeatedly selecting the same tangram – which would reduce the variety of communicative choices we could observe – participants were told that they could not select the same tangram they had chosen on the immediately preceding trial.

Implementation details

The experiments were created using jsPsych (de Leeuw et al., 2023) using the DataPipe tool to automatically send the data to the Open Science Framework (de Leeuw, 2024). Data analysis was performed in R using the lmer, lmerTest, and emmeans packages (Bates et al., 2014; Kuznetsova et al., 2017; Lenth et al., 2018).

Results

The purpose of Experiment 1 was to test how people strategically adjusted their referential language based on their audience and goal, in an unconstrained setting where participants could freely choose any tangram to talk about and also freely generate their own descriptions in a text box. We analyzed three dependent measures: (1) participants' choice of tangrams ('in-group' tangrams that the audience group talked about, versus 'out-group' tangrams that only the other group talked about), (2) the number of

[1] In practice, in all experiments we omitted the conditions where the social-signaling goal was directed to only naive or out-group members because using language to identify in-group members isn't meaningful when there aren't any in-group members in the audience.

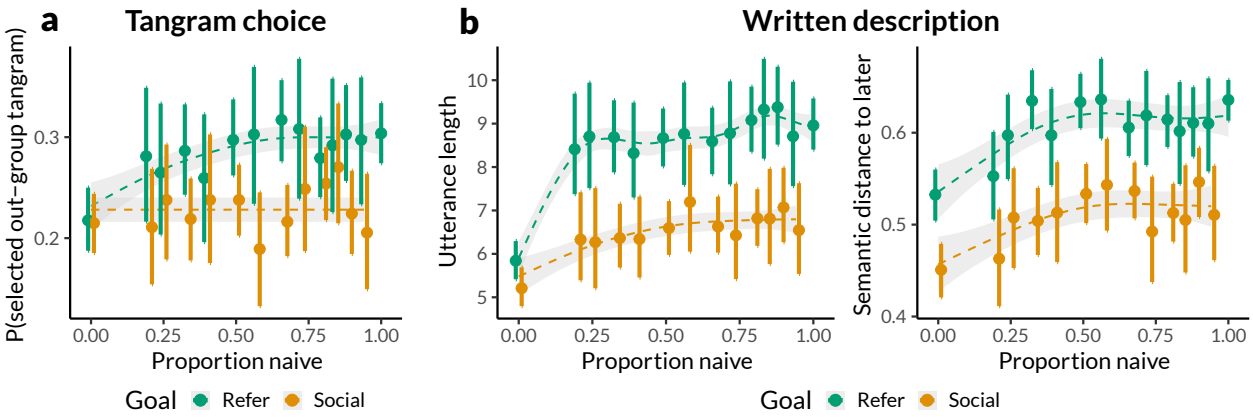


Figure 2

Experiment 1 results. Given a goal of social identification, participants were more likely to ***(a)*** *choose tangrams that the in-group group talked about, and* ***(b)*** *write shorter descriptions that were also more similar to the descriptions that the in-group used. Error bars are bootstrapped 95% confidence intervals computed over trial-level means.*

words in participants' descriptions, and (3) how semantically different their descriptions were from the converged labels produced by the in-group during the observation phase. The semantic distances were computed by embedding the utterances using the `paraphrase-MiniLM-L6-v2` model (Reimers & Gurevych, 2019) to transform each description into a 384-dimensional vector, and then taking the complement of cosine similarity between utterances. We used mixed-effects regression models to test our predictions, predicting each of the above measures from audience composition (proportion of naive audience members) and goal ('refer' vs. 'social'), along with their interaction, with random effects for participant, tangram set, and selected tangram. We also included a quadratic term for proportion naive, to account for nonlinear effects of audience composition.

We first examined whether the paradigm reproduces the standard referential efficiency findings. Standard accounts of referential efficiency predict that people want to be both informative and concise, and consider the knowledge level of their audience when computing the value of producing an utterance (Brown-Schmidt et al., 2015; Goodman & Frank, 2016; Hawkins et al., 2023). This account predicts that participants are more likely to tailor their expressions to naive people if their audience has a larger proportion of them (S. O. Yoon & Brown-Schmidt, 2018, 2019). Consistent with these predictions, when addressing an audience of only in-group members, participants were more likely to choose tangrams that

the in-group had talked about (analogous to choosing group-specific specialized concepts, like New York landmarks for New Yorkers). Participants were also more likely to produce shorter descriptions that were similar to the words that the in-group used; and as the number of naive people in the audience increased, so did participants' likelihood of choosing out-group referents and writing longer descriptions that were less similar to the in-group expressions (estimated linear trends in the 'refer' condition: tangram choice $b_x = 0.31$, $z = 2.44$, $p = 0.015$; utterance length $b_x = 2.14$, $z = 5.52$, $p < 0.001$; semantic distance $b_x = 0.07$, $z = 4.21$, $p < 0.001$) (Figure 2). As predicted, we also observed nonlinear effects of audience composition (quadratic trends: tangram choice $b_{x^2} = -0.86$, $z = -2.67$, $p = 0.008$; utterance length $b_{x^2} = -5.12$, $z = -8.28$, $p < 0.001$; semantic distance $b_{x^2} = -0.14$, $z = -3.60$, $p < 0.001$): our results show that, compared to when there are no naive people in the audience, the presence of even a few naive people induced substantial shifts toward longer, more descriptive utterances, consistent with previous studies showing that people 'aim low' to the least knowledgeable members of their audience (Bell, 1984; S. O. Yoon & Brown-Schmidt, 2018, 2019).

Thus participants spontaneously picked up on the observed utterances through passive observation, and strategically adjusted them based on audience composition. Despite previous research showing that observers typically acquire less information than those who actively engage in the reference game (Schober & Clark, 1989), our results showed that participants were able to use passive exposure to reliably encode the tangrams and expressions each group used, to adapt their utterances during the selection phase. The paradigm thus elicits the standard phenomena in reference games related to common ground and coordination, even when participants don't actively engage in the dialogue themselves (Barr, 2004).

We next asked how participants' behavior changed when their explicit goal was to signal in-group identity. For all three dependent measures (tangram selection, utterance length, semantic distance), we observed main effects of the 'social' goal, suggesting that social-signaling instructions systematically shifted participants' choices. When they were told to explicitly signal social identity to the in-group members of their audience, people were more likely to choose tangrams that the in-group talked about ($b = -0.17$, $z = -3.97$, $p < 0.001$) (Figure 2a). They produced shorter utterances overall, counteracting the otherwise strong tendency to make sure everyone in the audience understood ($b = -0.98$, $t(7.14) = -5.95$, $p < 0.001$) (Figure 2b, left). The utterances were more semantically similar to the observed in-group utterances ($b = -0.04$, $t(17.8) = -4.52$, $p < 0.001$), for all audiences (Figure 2b, right).

Furthermore, there were significant interaction effects for the utterance measures (utterance length $p < 0.001$; semantic distance $p = 0.022$): The effect of the social-signaling goal was most pronounced when the in-group members were small minorities in the audience.

The influence of a social-signaling goal in this experiment depends on the extent to which participants learned and remembered the in-group's language in the observation phase. Because the observation phase was entirely passive and also relatively long (participants clicked through 72 videos), participants likely varied in their attention, learning, and memory for the group's labels. We therefore tested whether the effect of social-signaling goals would be more detectable in participants who had learned the group-specific expressions. We computed a 'learning score' for each participant, by comparing the average semantic similarity of participants' referential-goal descriptions for homogeneous in-group audiences, to the converged in-group labels they had observed in the first phase of the task. In other words, this metric captures whether people produce the same expressions that the in-group produced, for the same task that the in-group was observed doing. Indeed, participants with higher learning scores showed stronger effects of the social goal across all three dependent measures (tangram choice $r = 0.27$, $t(183) = 3.84$, $p < 0.001$; utterance length $r = 0.44$, $t(183) = 6.70$, $p < 0.001$; semantic distance $r = 0.42$, $t(183) = 6.26$, $p < 0.001$). The more participants remembered the group's labels, the more they leveraged them to signal social identity. Considering only the top 50% of participants, in terms of learning score, the effects of the social goal were even more pronounced (Figure A1).

In sum, participants spontaneously tracked group-specific linguistic content in minimal groups, and used this knowledge strategically in the selection phase. When their goal was referential, they adapted what they chose to talk about and what they said, based on the knowledge state of their audience. When their goal was to signal social identity, they preferentially used shorter, in-group-like expressions despite the presence of naive listeners who might not understand.

Experiment 2

Experiment 1 replicated extensive prior evidence that people select language by considering the referential informativity for the audience, and further showed that people strategically adjust their language based on whether their goal is referential or social signaling. However, because people freely chose any referent they wanted to talk about, and wrote any expression in a free-response format, participants' responses combined a number of decisions: which referent best affords social signaling, how conciseness

serves social goals, and how audience composition modulates these trade-offs.

In the subsequent experiments, we decompose these bundled decisions using a forced-choice paradigm. By systematically controlling what varies across the choices (referent vs. expression, group-specific vs. shared, early vs. late), we can formalize these trade-offs in a computational framework that specifies how social utility competes with referential efficiency. Experiment 2 specifically measured speakers' choice of referent – what to talk about – to reveal their social identity.

Methods

Participants

We recruited 65 (40 female, 22 male, 2 non-binary, 1 abstain; ages 20-70, M(SD) age = 35.8(12.4)) adult fluent English speakers from the United States on Prolific. No participants were excluded. Participants gave informed consent. They were paid $15 for completing the experiment, which took an estimated 60 minutes to complete.

Procedure

Participants observed two distinct groups playing the same referential communication game. Unlike in Experiment 1, both groups referred to all tangrams in the context. To keep the experiment to a manageable length under the fully within-subject design, the context contained 6 tangrams instead of 12. This reduction was necessary to ensure participants could experience all possible combinations during the critical selection-phase trials without the task becoming prohibitively long.

A key feature of Experiment 2 was that each tangram was assigned to have either a shared label or a group-specific label. For the 'shared-label' tangrams, both groups independently converged on the same label (e.g., both the red and blue groups calling one tangram "falling guy"). For the 'group-specific label' tangrams, each group developed its own unique label for the same tangram (e.g., one group calling a tangram "baby," while the other called it "no arms") (Figure 3a, right). Thus for the 'shared label' tangrams, both groups were observed as coincidentally having converged to the same labels for those tangrams. This design allows us to test whether participants track the cross-group statistics of referential expressions, and rule out simpler heuristics such as directly repeating the more frequently observed expressions.

Next, participants proceeded to the selection phase, which manipulated the audience and goal. The

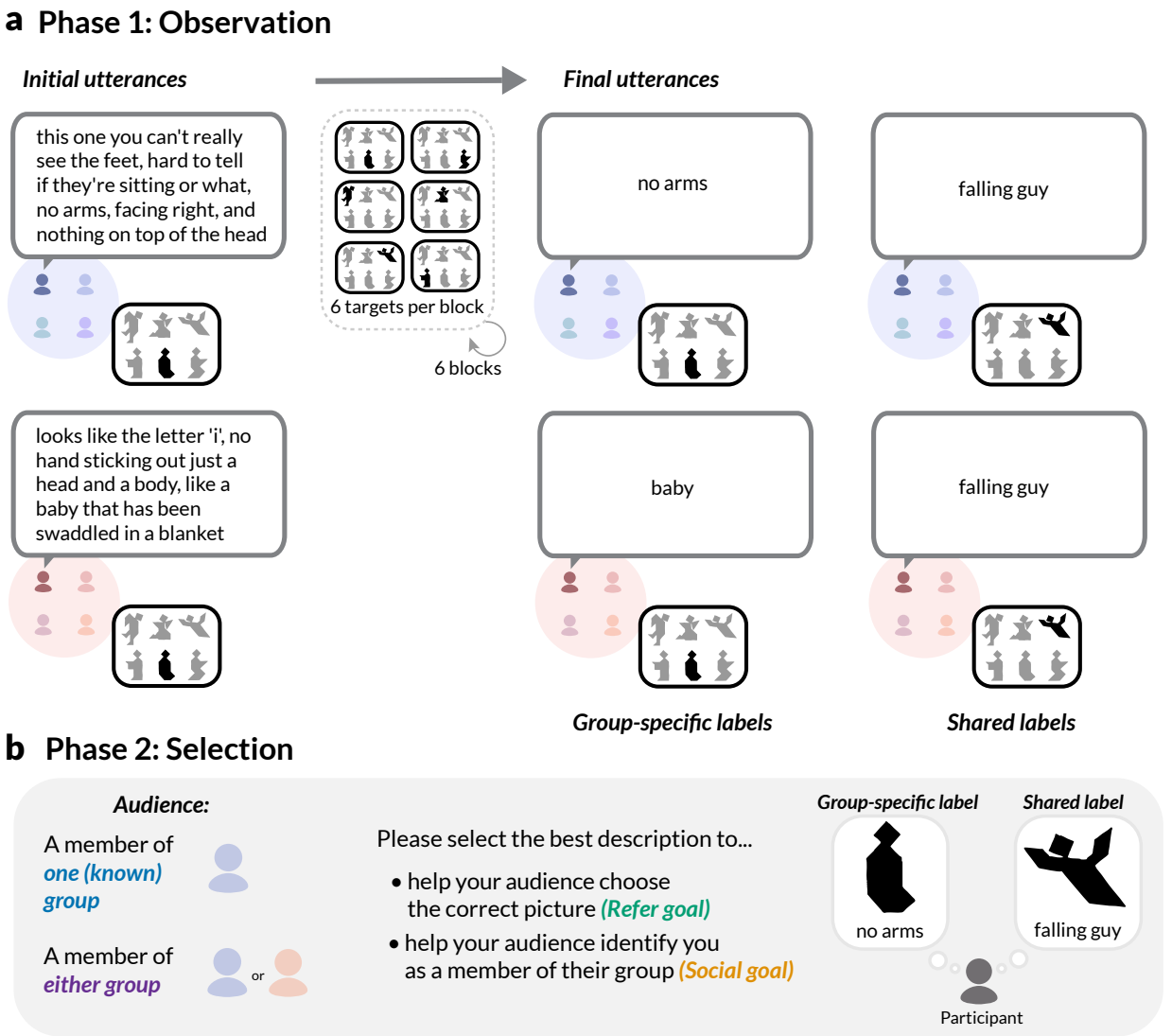


Figure 3

Experiment 2. ***(a)*** *In the first phase of the task, participants were third-party observers viewing two groups of people playing a multiparty reference game. Half of the tangrams converged to group-specific labels, and half of the tangrams converged to shared labels.* ***(b)*** *In the second phase of the task, participants were shown an audience and a goal, and selected between a group-specific label and a shared label.*

audience was described as as a member of a known, specified group (a member of the red group, or a member of the blue group), or an ambiguous member whose group identity was uncertain (a member of 'either the red group or the blue group'). The goal manipulation was the same as in Experiment 1: participants were either trying to refer (help the audience identify the target tangram) or signal social

identity (indicate their own group membership).

On each critical trial, participants chose between two converged tangram–label pairs: one associated with a group-specific label and one with a shared label (Figure 3b). This design allows us to simultaneously isolate the effect of a social-signaling goal on tangram choices (referential efficiency directed to the in-group is matched, so any difference in choices to the in-group is due to a social goal), while testing the standard referential efficiency predictions that should arise in the effect of audience identity (the 'shared-label' tangrams are referentially helpful for either group, while the 'group-specific label' tangrams are only referentially helpful for their specific group).

The assignment of tangram and label to condition (group-specific vs. shared) was counterbalanced across participants, ensuring that our results did not depend on pre-existing intuitions about which expressions seem more distinctive or more likely to be group specific (analogous to how some concepts or terms feel more "specialized" in everyday communication). There were also three separate sets of tangrams and labels. The design was fully within-subject (participants chose between all possible tangram-label pairs, for all audience conditions). Manipulation check trials were included to measure how well participants learned the observed tangram-label pairs and their group-specific sources. In the manipulation check trials, participants chose between two converged labels, each corresponding to the same tangram. For each of the 'shared label' tangrams, participants were given the 'refer' goal and asked to choose between a label that they saw in the gameplay, and a plausible label that they did not see. For each of the 'group-specific label' tangrams, participants were given the 'social' goal, and asked to choose between each of the group's labels based on the audience group (the 'red' group or the 'blue' group). There were 15 manipulation check trials and 54 critical trials, for a total of 69 trials. Performance was incentivized as in Experiment 1.

Results

We tested our predictions using logistic mixed-effects regression models predicting participants' binary responses, from the goal-audience condition ('refer' to a known group, 'refer' to either group, and 'social' goal to a known group), with random effects for participant, tangram set, and tangram. Hypothesis tests were conducted using the estimated marginal means, and all effects are reported on the log-odds scale. The z-statistics reported for individual marginal means test whether each value differs from 0 on the log-odds scale (i.e., whether performance is different from chance).

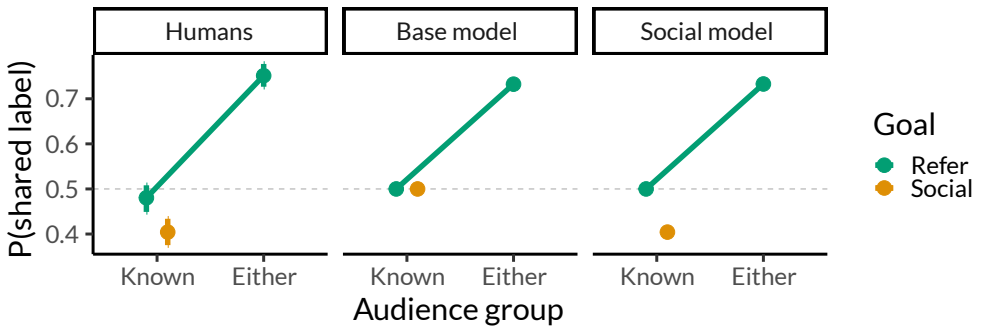


Figure 4

Experiment 2 results. The base model predicts adaptation based on audience, but only the social model also captures the additional pressure in social signaling to choose group-specific label (vs. shared-label) referents. Error bars are 95% bootstrapped confidence intervals computed over trial-level means.

We first examine the critical trials, in which participants chose between a 'shared label' tangram versus a 'group-specific label' tangram. When participants were uncertain about the audience, the label for the shared-label tangram provided higher referential efficiency, since it would be interpretable by members of either group. Consistent with this, when participants' goal was to refer, they more often chose the shared-label tangram when addressing 'either' group ($M = 1.60\,[1.14, 2.06]$, $z = 6.81$, $p < 0.001$; Figure 4 left, green).

Notably, the shared labels had also been experienced twice as often during the observation phase, potentially making them more accessible. If participants are using low-level strategies like picking the most familiar labels, then participants should be more likely to pick those labels regardless of condition. However, this was not the case. Participants only preferentially chose the shared-label tangrams when they were addressing 'either' group, which is when they would be more referentially informative. When addressing a specific, known group – where referential informativity was matched for both tangram-label pairs – participants selected shared and group-specific labels at similar rates ($M = -0.04\,[-0.46, 0.37]$, $z = -0.20$, $p = 0.840$; Figure 4 left, green). The observation that participants chose both options equally for a known audience, but then switched to the shared-label tangram for the uncertain audience, is therefore consistent with prior evidence that speakers are sensitive to their audience's knowledge state when considering referential informativity, and rules out the low-level strategy of simply picking the most familiar labels.

Next, we tested whether participants further adjust their selection based on an explicit social-signaling goal. When speaking to a known group, both choices are equally informative from a referential perspective, but only the group-specific label provided an opportunity to signal group membership. Consistent with this prediction, participants were more likely to choose the group-specific label to achieve this goal ($M = -0.50\,[-0.90, -0.10]$, $z = -2.47$, $p = 0.014$) (refer-known minus social-known: diff $= 0.46\,[0.19, 0.73]$, $z = 3.98$, $p < 0.001$) (Figure 4 left, orange).

We examined whether individual variability in the strength of the social-goal effect was related to their performance on the manipulation check trials. To assess performance on the manipulation checks, we ran logistic mixed-effects regressions predicting accuracy with a fixed intercept and random effects for participant, audience group, tangram set, and tangram. Participants were near-ceiling in distinguishing the shared labels from plausible unobserved labels (98.4% correct; intercept $= 4.12$, $z = 5.85$, $p < 0.001$). However, overall accuracy in identifying which group-specific labels originated in each group was moderate (72.6% correct; intercept $= 0.98$, $z = 4.63$, $p < 0.001$). Individual participants who more accurately remembered the origins of group-specific labels were better able to use tangram selection to achieve a social goal (correlation with proportion of group-specific labels selected in the social-known condition: $r = 0.45$, $t(63) = 4.04$, $p < 0.001$).

Computational framework

To better understand the mechanisms driving participants' choices, we tested whether a computational model incorporating social-signaling utility explained behavior better than a baseline referential model. Our approach provides a principled way to formalize how participants might integrate different goals (referential clarity vs. social identification), along with their audience, when selecting an utterance.

Consistent with previous models of referential efficiency (Frank & Goodman, 2012; Goodman & Frank, 2016), our computational model formalizes people's choice of utterances as a utility-driven decision process, where an agent selects an action from a set of alternatives in proportion to its total utility, modulated by the softmax temperature parameter α (where higher values concentrate more probability on the higher-utility choices). In our paradigm, agents select an utterance u, given an audience a and goal g:

$$P(u|a,g) \propto \exp(\alpha \cdot U_{\text{tot}}(u|a,g)). \tag{1}$$

In the *social signaling* model, the total utility of the action is the sum of referential informativity $U_{\text{inf}}(u|a,g)$ (how well the utterance allows the listener to identify the correct referent), the social-signaling utility $U_{\text{soc}}(u|a,g)$ (the extent to which the utterance signals in-group identity), and the cognitive cost of producing the utterance $c(u|a,g)$ (here, captured by utterance length), the balance of which is modulated by the weights w_r, w_s, and w_c[2]:

$$U_{\text{tot}}(u|a,g) = w_r \cdot U_{\text{inf}}(u|a,g) + w_s \cdot U_{\text{soc}}(u|a,g) - w_c \cdot c(u) \tag{2}$$

The difference between the referential baseline and social signaling models is that in the baseline referential model, there is no social utility (i.e., $w_s = 0$).

Model comparison

The experiments were designed to manipulate conditions that change the balance between referential and social-signaling utilities. Thus, to fit the two models, we stipulated the utility terms as binary values reflecting the structure of the experimental design. Encoding utilities as binary values allowed us to capture these intended contrasts cleanly and test whether differences in behavior aligned with the experimental manipulations.

When the goal was reference, we set the referential informativity of an utterance to 1 if the audience's group had used the utterance to successfully refer, and 0 otherwise. When the audience was 'either' group, we marginalized over the speaker's utilities for both groups. When the goal was social signaling, the social model evaluates utterances in terms of their social-identification value rather than their referential accuracy, following the design of the task. The referential model, which cannot represent social-signaling goals, continued to compute utility based only on referential accuracy. In this experiment, we stipulated the social-signaling utility under the assumption that it is based on the statistics of the

[2] We include the cost term here for theoretical completeness, though it does not influence predictions in this experiment because participants always select between converged descriptions. It becomes relevant in later experiments where participants choose between utterances of varying length.

observed labels: because group-specific labels are more diagnostic of group membership than shared labels, we assigned group-specific labels a social signaling value of 1, and shared labels a social signaling value of 0. Finally, for this experiment, we treated utterance cost as a fixed constant and did not estimate a weight for it, as all choices involved already-converged descriptions whose lengths did not meaningfully differ.

We implemented the computational models using memo, a probabilistic programming language for performing fast enumerative inference (Chandra et al., 2025). We fixed $\alpha = 1$ and fit the other parameters by maximum likelihood estimation, using stochastic gradient descent; all plots shown are predictions under the best-fitting parameters. Because the utilities are directly set based on the experimental conditions, the estimated utility weights should be interpreted as relative weights that help the model fit the difference between conditions, rather than standalone estimates for how much people balance each motivation. The social signaling model (best-fitting params $w_r = 3.34$, $w_s = 0.39$, negative log-likelihood $= 2257.7$), but not the baseline referential model (best-fitting params $w_r = 3.31$, negative log-likelihood $= 2279.3$) captured the patterns in participants' responses, specifically the difference between the 'social' and the 'refer to a known group' conditions, and explained the data better (likelihood-ratio test $\chi^2(1) = 43.24$, $p < 0.001$) (Figure 4).

In summary, in Experiment 2, we found that when referential informativity is matched, people can choose the option that reveals social group membership to an in-group member. A model that considers both referential and social-signaling utilities is needed to capture these patterns.

Experiment 3

In Experiment 2 participants simultaneously selected what to talk about and what to say, choosing between tangram-label pairs for two different tangrams. In Experiment 3, we shifted focus to a different but complementary trade-off: the decision between alternative labels for the same tangram. Specifically, participants chose between an earlier, longer label drawn from the group's initial conversation, which is more descriptive but less efficient – and a later, shorter label that emerged through repeated interactions within the group (Figure 5). This setup allows us to examine the standard contrast of how participants balance informativity and efficiency (e.g., Isaacs & Clark, 1987), and the role of a social-signaling goal in this trade-off.

Because participants are choosing between earlier and later labels, they now have to consider the balance of referential informativity and efficiency, depending on the audience. Earlier labels are more

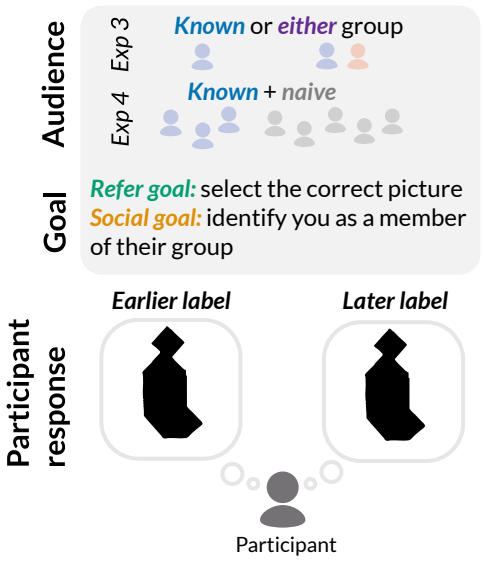


Figure 5

Experiments 3 and 4. In both experiments, participants chose between a longer label that was produced earlier in the group's conversations, versus a shorter label that was produced later in the group's conversations. The experiments differed in audience manipulation. Experiment 3 manipulated whether the audience was from a specified known group, or from either group. Experiment 4 manipulated the proportion of naive people in the audience.

descriptive and universally comprehensible but more effortful for both the speaker and the listener, whereas later labels are concise and efficient but only informative to a group that knows what they mean (Schober & Clark, 1989). The observation phase included (as above) both shared-label tangrams (generated independently by both groups) and group-specific label tangrams.

Methods

Participants

We recruited 61 (27 female, 32 male, 2 non-binary; ages 19-67, M(SD) age = 36.6(12.5) adult fluent English speakers from the United States on Prolific, and excluded 1 participant who indicated that they did not understand the instructions. Participants gave informed consent, and were paid $15 for completing the experiment, who took an estimated 60 minutes to complete.

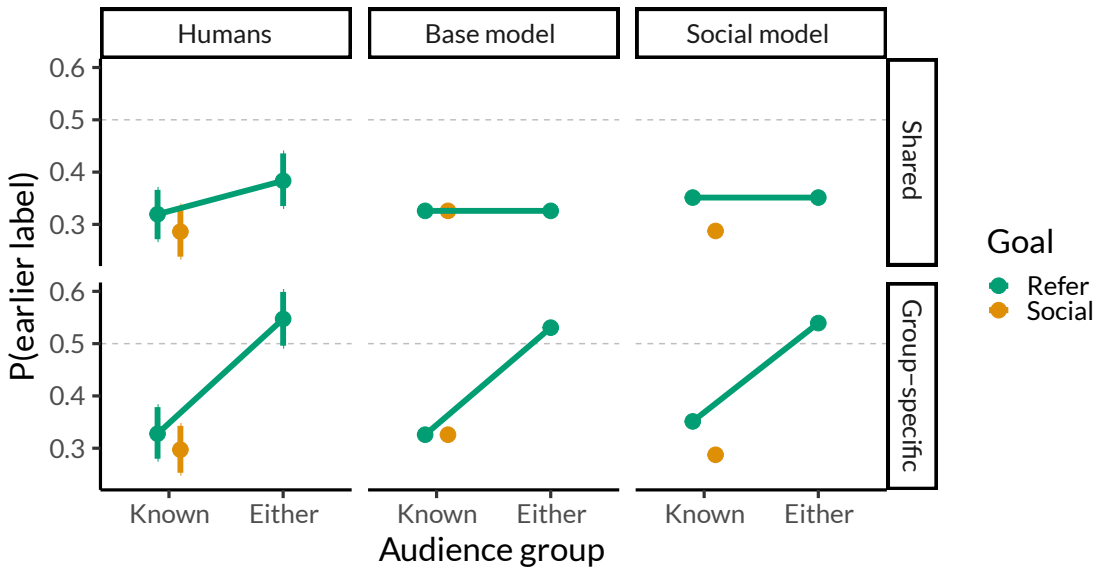


Figure 6

Experiment 3 results. Top row: results for the 'shared-label' tangrams. Bottom row: Results for the 'group-specific label' tangrams. Error bars are bootstrapped 95% confidence intervals computed over trial-level means.

Procedure

The observation phase procedure, the bonus structure, and the structure of the manipulation check trials were the same as in Experiment 2. However, in the critical selection phase trials (fully within-subject design), participants choose between two labels for the same tangram: a label that appeared earlier in the given group's conversation (e.g., "this one you can't really see the feet, hard to tell if they're sitting or what, no arms, facing right, and nothing on top of the head"), versus a label that appeared later in the same conversation (e.g., "no arms"). There were 15 manipulation check trials and 36 critical trials, for a total of 51 trials in the experiment.

Results

The baseline referential account predicts that people use later labels when addressing the in-group, but switch to earlier labels when addressing an uncertain audience (Hawkins et al., 2023; Wilkes-Gibbs & Clark, 1992). This is indeed what we saw with the 'group-specific label' tangrams ('refer to a known group' $M = -0.87\,[-1.26, -0.48]$, $z = -4.36$, $p < 0.001$; 'refer to either group' $M = 0.29\,[-0.27, 0.84]$, $z = 1.02$, $p = 0.310$) (refer-either minus refer-known: diff = 1.16 [0.56, 1.75], $z = 4.57$, $p < 0.001$)

(Figure 6, bottom row left). For the 'shared label' tangrams, the later labels are concise *and* informative to both groups, so participants chose them regardless of the audience ('refer to a known group' $M = -0.93\,[-1.34, -0.53]$, $z = -4.54$, $p < 0.001$; 'refer to either group' $M = -0.76\,[-1.37, -0.16]$, $z = -2.47$, $p = 0.014$) (refer-either minus refer-known: diff = 0.17 [-0.39, 0.73], $z = 0.72$, $p = 0.754$) (Figure 6, top row left). Thus our data confirm that people are sensitive to their audiences and adjust what they say based on anticipated referential efficiency.

Can a social signaling goal itself lead to the same pressure of producing shorter and concise utterances to the in-group? While both the earlier and later expressions *occurred* within specific groups, the later utterances were *generated* through within-group interactions. Knowing how to use these later expressions for effective reference could thus reveal a shared in-group history. Indeed, participants selected the concise converged expressions when social signaling to the in-group, compared to referring to an uncertain audience (refer-either minus social-known averaged over tangram: diff $= 0.87\,[0.37, 1.36]$, $z = 4.10$, $p < 0.001$) (Figure 6, left). Thus in our choice framework, when the audience is composed only of in-group members, the same later, shorter utterances could be used effectively to achieve two goals: reference and social signaling. When addressing the in-group, both kinds of goals created a pressure toward selecting the shorter, later labels.

The Experiment 1 results suggest that when speakers address an audience belonging to a homogeneous in-group, social-signaling goals may introduce a small additional pressure toward choosing shorter utterances, compared to referential goals (Figure 2b, left, 0% naive). Although the task in Experiment 3 was not designed to detect fine-grained differences in utterance length – participants choose between earlier and later utterances, and in this context both social-signaling and referential efficiency pressures favor the same option – we nevertheless tested whether people might be slightly more inclined to choose the shorter utterances under a social goal. The social signaling model, but not the base referential model, can capture this distinction. However, any such effect is expected to be small, since all utility components push toward the shorter choice when addressing fully homogeneous in-group audiences. Consistent with this interpretation, the contrast between social-signaling and referential goals was not significant in the preregistered dataset of Experiment 3 (refer-known minus social-known averaged over tangram: diff $= 0.20\,[-0.12, 0.53]$, $z = 1.47$, $p = 0.306$) (but see the model comparison below, and another test of this comparison in Experiment 4).

Model comparison

As in the previous experiment, we stipulated the utility values based on the experimental design to directly test the hypothesized trade-offs. Earlier labels were assigned referential value for all audiences, reflecting the assumption that they were produced for an audience that the speaker had not interacted with before. Later labels, by contrast, were assigned referential value only when directed to the in-group, reflecting the assumption those expressions would not be as interpretable by out-group or naive listeners[3]. For social-signaling utility, the later labels, but not the earlier labels, were assigned to carry social signaling value, reflecting the assumption that conventions formed through repeated social interaction are idiosyncratic to a group and therefore can signal group membership. Finally, we incorporated an efficiency cost for the earlier labels, which were longer and more complex, reflecting the assumption that even in a forced-choice paradigm, the cognitive cost term $c(u)$ captures the perceived or simulated effort associated with selecting longer labels. Participants may implicitly simulate producing or processing these labels, making them less attractive except when their referential or social utility is sufficient to offset that cost.

As in Experiment 2, the social-signaling model (best-fitting params[4] $w_r = 1.59$, $w_s = 0.29$, $w_c = 0.61$, negative log-likelihood = 1377.6), but not the baseline referential model (best-fitting params $w_r = 1.75$, $w_c = 0.73$, negative log-likelihood = 1380.3), captured the distinction between goals (likelihood-ratio test $\chi^2(1) = 5.39$, $p = 0.020$; Figure 6). Together, the model comparison identified a significant effect of social signaling value in participants' utterance selection.

Experiment 4

Experiment 3 showed that social signaling goals, like referential goals, lead to a pressure to produce concise, in-group-generated utterances. However, in the design of Experiment 3 the short utterances could achieve both goals efficiently and simultaneously, so that experiment did not manipulate any *tension* between these goals.

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[3] In practice, all labels carry some degree of referential utility. We revisit this in Experiment 4.

[4] Following Experiment 2, these parameter values, and particularly the social weight w_s, should be interpreted in terms of how the utility components are instantiated in the model. In the 'social' condition, two factors – social-signaling utility and utterance cost – favor the shorter, later utterances. Cost, however, applies across all conditions, while social utility is stipulated to apply only in the 'social' condition. Thus w_s reflects the additional social value needed – beyond cost (and the lack of referential utility in the 'social' condition) – to reproduce the observed separation between the goal conditions.

Experiment 4 was designed to create and measure this tension directly. As in Experiment 1, we asked whether speakers strategically trade off social signaling and referential clarity depending on their audience. Consider, for example, a scientist lecturing to a mixed audience of experts and novices. Prior research shows that when speakers in such situations pursue a referential goal, they tend to 'aim low,' adapting their language to the least-informed audience members (Bell, 1984; S. O. Yoon & Brown-Schmidt, 2018, 2019). Experiment 1 reproduced this pattern in an open-ended production setting, while also showing that social-signaling goals can override this tendency.

In Experiment 4, we used formal model comparison to test whether, as in Experiment 1, people make systematic trade-offs between social signaling considerations and referential clarity, based on their audience. The forced-choice paradigm allows us to not only formalize competing utilities in a computational model, but also to directly measure the referential success of each label type in a separate sample of naive listeners. Because here the set of possible utterances is fixed, we can empirically estimate how well naive listeners understand each label, quantifying the precise referential benefits of each choice and enabling a more precise evaluation of the trade-off between referential clarity and social signaling than was possible in Experiments 1-3.

Methods

Participants

We recruited 180 (95 female, 84 male, 1 non-binary; ages 18-72, M(SD) age = 37.4(11.9) adult fluent English speakers from the United States on Prolific, and excluded 1 participant who did not understand the instructions. Participants gave informed consent, and were paid $10 for completing the experiment, which took an estimated 30 minutes to complete.

Procedure

In the observation phase, participants viewed interactions from a single group, allowing them to learn that group's conventions. Then, in the selection phase, participants chose between earlier and later labels for a single tangram, as in Experiment 3. The audience composition varied continuously in size and proportion of group members, and was depicted iconically as in Experiment 1 (Figure 5). The audience-goal manipulation was within-subject. To keep the task length manageable given the number of sets of audiences, participants saw one pair of tangrams and labels for each condition manipulation ('refer'

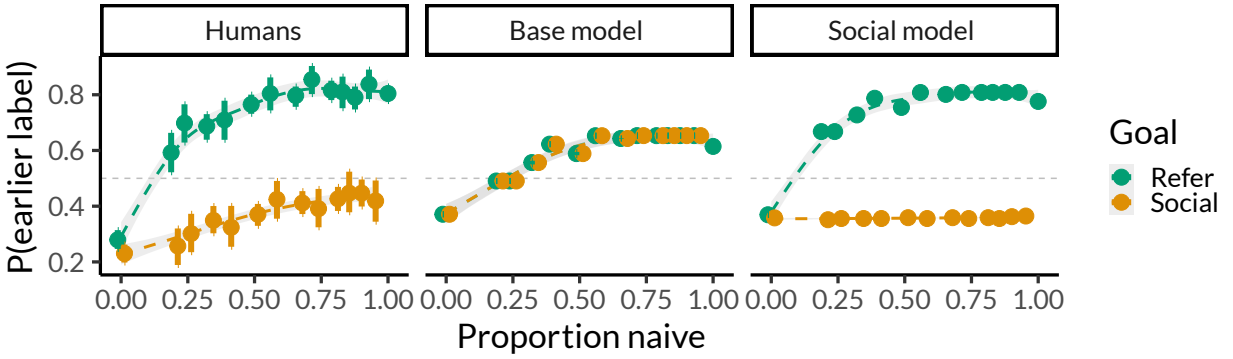


Figure 7

Experiment 4 results. Error bars are bootstrapped 95% confidence intervals computed over trial-level means.

vs. 'social'), with the appearance of each tangram-label pair balanced within and across participants. As in previous experiments, each participant saw one of three tangram-label sets in the observation phase. Manipulation-check trials tested whether participants could reliably distinguish between observed labels and plausible but unobserved alternatives. There were 53 critical trials and 6 manipulation check trials, for a total of 59 trials.

Eliciting transparency

When the audience includes naive people, we expected that participants' choices would partly depend on how relatively transparent the labels are (i.e. how interpretable they are to naive listeners). For example, if a later label is very opaque, participants might think that naive people won't understand what it is referring to, and instead be more likely to pick the earlier, longer label. To capture item-wise variability in referential transparency, a separate sample of participants ($N = 201$) on Prolific guessed the correct tangram given a label. This obtains a measure for how well naive participants can infer the correct referent from each label. Each participant provided one response per tangram per tangram set (18 trials total), with the assignment of label to participant balanced across participants. These judgments were used to generate an estimate of transparency for each of the labels. The item-wise transparency differences were incorporated into the main analysis to account for item-level differences in how informative the choices were for a naive audience.

Results

We tested our predictions using logistic mixed-effects regression models to predict participants' binary responses as a function of the proportion of naive people in the audience (coded as a quadratic term), the speaker's goal, and the interaction of these factors. We also included the measured transparency differences as predictors. First, participants remembered the converged labels from the observation phase (logistic mixed-effects regression predicting manipulation-check accuracy with a fixed intercept and maximal random effects: 94.0% correct; intercept $= 2.75$, $z = 10.71$, $p < 0.001$).

When speakers chose labels to 'refer', there were strong linear (estimated trends: $b_x = 3.83$, $z = 10.49$, $p < 0.001$) and quadratic (estimated trends: $b_{x^2} = -4.49$, $z = -10.00$, $p < 0.001$) effects of audience composition, replicating the 'aim low' effects showed in Experiment 1 and in previous research (S. O. Yoon and Brown-Schmidt, 2018, 2019; see Figure 7, left). The label transparency for each item influenced participant's choices: people were overall more likely to select the earlier label, when it was more transparent than the later label ($b = 1.00$, $z = 4.59$, $p < 0.001$). As in Experiment 1, we also observed a main effect of and interaction with goal ($b = 1.53$, $z = 9.18$, $p < 0.001$; both linear and quadratic interaction terms $p < 0.001$). As the in-group became more diluted, participants were comparatively less likely to switch to the earlier expressions for the 'social' goal than the 'refer' goal (Figure 7, right). That is, when given an explicit social-signaling goal, participants continued to use the in-group labels, despite the presence of many naive listeners who might not understand.

Transparency analyses

How much do naive listeners actually understand? The estimates from the transparency experiment allowed us to conduct exploratory analyses quantifying how the precise referential costs of each label affect people's choices. First, earlier labels were overall more transparent (83.0% correct; 95% CI: [81.3, 84.5]) than the later labels (70.2% correct; 95% CI: [67.7, 72.6]) (Figure 8a). Note that the transparency of both kinds of labels were well above chance (16.7% correct), consistent with previous estimates of transparency in tangram reference games (Boyce et al., 2025). Visualization of item-wise 2AFC pairs showed that earlier utterances were generally more transparent across tangram sets and counterbalancing assignments (Figure 8a).

Using these item-wise transparency estimates, we simulated expected listener success for each trial as a function of audience composition. In-group listeners were assumed to understand the labels (as

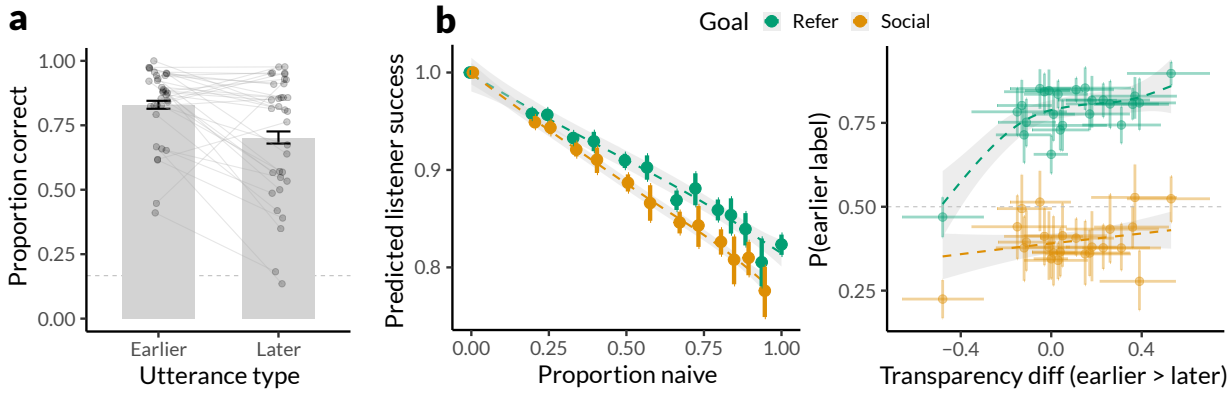


Figure 8

Item-wise differences in Experiment 4. ***(a)*** *Transparency was generally high for all utterances, but earlier utterances were more transparent than later utterances. Individual points connected by lines show 2AFC choice pairs.* ***(b)*** *Simulated listener success based on participants' choices. The predicted success decreases with the proportion of naive audience members, but decreases more quickly for the 'social' condition than for the 'refer' condition.* ***(c)*** *In the 'refer' condition, participants were more likely when they were more transparent than the later labels. Error bars are 95% bootstrapped confidence intervals computed over trial-level means.*

observed in the first phase of the task), while naive listeners' accuracy was estimated using the empirical transparency measures. Predicted success declined as the proportion of naive listeners increased, but declined more steeply for the labels chosen under the social-signaling goal (Figure 8b, left). Thus, participants systematically selected utterances that were less helpful for naive listeners when pursuing social goals. In other words, they accepted a real (although small) cost in referential informativity when they selected utterances for social signaling.

The transparency differences shaped participants' choices on an item level. Averaged across audience composition (excluding fully in-group audiences), participants were more likely to select the earlier label when it was comparatively more transparent (Figure 8b, right). However, this sensitivity differed by goal. Participants were more sensitive to transparency under the 'refer' goal, than under the 'social' goal. In fact, under the social-signaling goal, participants consistently preferred the later label even when the earlier label was substantially more transparent: For every single item, people were more likely to select the earlier label when the goal was reference, compared to when it was social signaling (Figure 8b,

right).

In sum, although later labels carried a measurable referential cost for naive listeners, participants accepted these costs when their goal was to signal group membership. Given a social-signaling goal, people's choices downweighted the differences in referential informativity, and favored utterances that were instead marked by their generation history within a group. These findings reveal a systematic trade-off between referential clarity and social signaling and show that speakers will override audience-oriented referential pressures when motivated to express social identity.

Model comparison

We implemented the computational models by considering each audience member as separately contributing to the total utility of producing an utterance (as in, e.g., Frank & Liu, 2018). Because the ceiling for participants' responses in the 'refer' condition asymptotes below 1, we fit an additional noise parameter ε corresponding to people's probability of choosing randomly on each trial. The baseline referential model (best-fitting params $w_r = 2.32$, $w_c = 2.45$, $\varepsilon = 0.69$, negative log-likelihood $= 6247.9$) successfully captured the nonlinear effect of the proportion of naive people (i.e. the 'aim low' results) (Figure 7 middle). However, only the social-signaling model (best-fitting params $w_r = 2.13$, $w_s = 0.04$, $w_c = 0.91$, $\varepsilon = 0.38$, negative log-likelihood $= 5609.5$) captured the distinction between the 'refer' and the 'social' goals, and was a better fit to the data (likelihood ratio test $\chi^2(1) = 1276.86$, $p < 0.001$; Figure 7 right).

In sum, Experiment 4 showed that participants adapted to their audience under referential goals, prioritizing clarity when naive people were present. But when signaling identity was the goal, participants maintained in-group expressions, even when speaking to mixed audiences for whom those labels would reduce overall comprehension. These findings highlight the trade-off between referential clarity and social signaling, demonstrating that speakers flexibly balance these competing pressures based on their audience and their communicative goals.

Discussion

Language is not only a way to transmit information, but also a powerful tool for expressing social identity. Across four experiments, we measured how people strategically adjust their referential language when their goal is to signal group membership. When people were given an explicit social signaling goal, they consistently selected referents and labels that maximized in-group identification (Experiments 1 and

2). They also favored concise, group-specific labels that emerged during repeated interaction, even when longer, more universally interpretable labels were available (Experiments 3 and 4). Finally, speakers resisted the well-documented tendency to adapt their language to accomodate the least knowledgeable audience members (Experiments 1 and 4), prioritizing identity signaling over universal comprehension. Formal cognitive modeling showed that extending referential efficiency models to include a social signaling term was essential for capturing these patterns.

Our findings align with decades of sociolinguistic research showing that speakers dynamically adjust their language to signal affiliation and solidarity with social groups (e.g., Bell, 1984; Eckert, 1989; Giles et al., 1991). Such language use is documented in many real-world contexts. Scientists and lawyers, for example, often use technical jargon as a deliberate marker of expertise or group membership, even when it's possible to explain the same concepts in simpler ways (Cruz & Lombrozo, 2025; Martínez et al., 2024). Political actors use dogwhistles and coded language to signal affiliation with particular constituencies (Henderson & McCready, 2017; Van Der Does et al., 2022). Online communities rapidly develop shared slang that functions both as an efficient shorthand and a marker of belonging (Damirjian, 2025). Our controlled experiments demonstrate that the cognitive pressures underlying these patterns can emerge even in minimal, low-stakes settings, suggesting that the drive to balance referential efficiency and social signaling is a fundamental feature of human communication.

To uncover the cognitive mechanisms behind these behaviors, we designed a tightly controlled experimental paradigm that isolates the mechanisms of interest. In standard accounts, repeated interaction gives rise to efficient, idiosyncratic, group-specific conventions only for efficiency; our paradigm builds on this foundation by allowing us to study not only how such conventions emerge, but also how people *use* them in social context. Participants observed arbitrary novel groups that developed linguistic conventions on the fly within minutes, creating the weakest conditions under which referential labels depend on social group identity.

By first exposing participants to the emergent conventions, and then manipulating their communicative goals, we could jointly assess how they balance the pressures of referential efficiency and social distinctiveness when choosing what to say. Even under these simplified conditions – and without any forewarning that the test phase would require social signaling – participants not only encoded group-specific language, but also deployed it strategically, adapting their language to their audience and

communicative goals. Because all of these components were assessed within one unified framework, the paradigm offers a minimal yet powerful way to observe the strategic use of referential language and to cleanly compare the effects of signaling social affiliation to standard referential efficiency accounts.

Overall, for a speaker to choose utterances that convey social information requires knowledge of how listeners will interpret this information. That is, for an utterance to function as a social marker, both speakers and listeners must recognize the potential social meaning of an utterance or label. Our models stipulated that utterances had social meaning if they were similar to the language that the group used (Experiment 1), emerged uniquely within one group (Experiment 2), or were generated through the process of repeated interaction within a group (Experiment 3, 4). In Experiments 2-4, which aimed to isolate the contribution of a social signaling goal, social information was experimentally assigned to one of the two response options on every trial and the social utility in our model was defined based on this experimental design.

Note that this experimental design and model operationalization does not test or express the recursive social reasoning processes that human minds likely use to reason about audience perceptions of speakers' social identity. For a speaker, reasoning about the social information of an utterance requires reasoning about listeners' beliefs (about the speaker's social identity), and how their beliefs would change given a produced utterance. Formalizing such a reasoning process would require representations of how listeners might interpret a variety of referential terms in general, within and across different social groups and contexts. For example, richer approaches could estimate utilities on a continuous scale, informed by measuring what speakers think about the population-level distinctiveness of utterances and which listeners have access to those utterances. When such granular knowledge of utterances and their alternatives is available, it could be incorporated into Bayesian models of recursive reasoning, which already capture different kinds of socially motivated communication (Burnett, 2017; Papineau & Degen, 2024; E. J. Yoon et al., 2020), and incorporate elements of audience adaptation based on existing and updated knowledge (Hawkins et al., 2023). Such formal models of recursive reasoning could simultaneously capture and test the reasoning process (i.e. about the referent the speaker is referring to, about the speaker, and about the broader population) that *generates* the social utility of utterances in the first place. Our approach lays a foundation for more expressive models that represent these reasoning processes in richer detail.

One discrepancy between our data and our formal social-signaling model concerns the implicit

goals people bring to referential language use. Our experiments were designed to isolate deliberate goals, explicitly prompting participants to focus either on referential efficiency or on social signaling, and the way we stipulated utilities in the model accordingly treated utility components as condition-specific. Under this structure, the model predicts that in the social-signaling condition, the proportion of out-group or naive people in the audience should have no effect on people's behavior, since the value of signaling in-group identity should not depend on how many out-group members are present. Empirically, however, participants' choices in both Experiments 1 and 4 were sensitive to the proportion of out-group members even when instructed *only* to signal social identity (Figure 2, Figure 7). This pattern suggests that speakers may spontaneously maintain a baseline of communicative clarity. In other words, they might be balancing multiple goals despite being explicitly tasked with only one. The current stipulation of utilities in the model does not capture these kinds of implicit goals. Future designs could capture these implicit goals by estimating the relative importance of referential and social-signaling goals *within* a single trial, using the same modeling framework.

Language can be used for social signaling because language usage patterns covary with the history of intragroup interaction. Why does this covariation arise in the first place? One possibility is that repeated intra-group interaction leads, purely for efficiency, to idiosyncratic expressions. Referential efficiency is sufficient to drive the emergence of idiosyncratic group-specific conventions, as shown in the selection phase (Hawkins et al., 2023). Once such variation arises, group-specific expressions could then be exploited for social signaling. This view is consistent with broader hypotheses that linguistic diversity arises by passive 'drift' when communities are relatively isolated from one another (Livingstone & Fyfe, 1999; Trudgill, 2004).

A second possibility is that social signaling is one key driving force behind the emergence of group-specific referring expressions in the first place (Roberts, 2012, 2013), even in minimal reference game contexts. Sociolinguistic and ethnographic studies suggest that people continuously create and use new terms in order to distinguish their own groups from other groups, and to identify themselves with certain social categories: either the ones they are in or the ones they want to be associated with (Labov, 1973; Milroy & Milroy, 1992; Nettle, 1999). In fact, adolescence, a stage where distinguishing social categories and affiliations is especially important, has been shown to be a primary time when new linguistic terms are created, leading to large-scale language change (Eckert, 1989). While our experiments cannot

distinguish between these possibilities, our results – which show that speakers continuously manage the trade-off between being understood and signaling in-group affiliation – highlight the mechanisms that allow local decisions about language use and social identity to shape broader patterns of language evolution and change.

Constraints on generality

In naturalistic settings, language is embedded within a wide variety of social contexts – ranging from school or hobby-based communities to regional, cultural, or ethnic groups – where different linguistic cues can signal social identity. Other properties of language that vary in these contexts include accent, prosody, and grammatical structure (Myers-Scotton, 1998; Poplack & Sankoff, 1984). Many social interactions are also conducted face-to-face, where prosody, gesture, facial expression, gaze, and backchannels are all informative for meaning (Floyd et al., 2025; Goodwin, 2000; Schober & Clark, 1989). All of these factors shape how speakers present themselves, how listeners infer social identities, and how people coordinate in real time. By focusing on linguistic conventions generated on the fly in a controlled reference game, testing what people picked up from a text-based chat, and restricting our sample to native English speakers from the United States, our experiments deliberately stripped away these broader cultural and multimodal factors to isolate the core cognitive mechanisms that guide social signaling in reference. Future work should examine whether and how similar trade-offs between social-signaling and referential goals emerge across different linguistic communities, cultural contexts, and interactional settings, particularly in communities with distinct norms around indirectness, hierarchy, or group identity.

Moreover, while our controlled paradigm enabled precise measurement and modeling, real-world communication involves more nuanced social dynamics, overlapping goals, and higher stakes. Our experiments establish that social-signaling goals influence referential choices, and how this influence operates at a computational level. A natural next question is *when* speakers spontaneously prioritize social signaling over referential clarity – which involves understanding the social contexts, motivations, and identity concerns that activate social-signaling goals. For example, when the costs of outsiders discovering one's social identity are high or when the audience is more diverse, people may develop more covert linguistic signals (Smaldino & Turner, 2022; Van Der Does et al., 2022), taking into account the knowledge level of outsiders and overhearers. Future work should investigate how social-signaling goals interact with other communicative priorities and constraints that shape language use in the complex dynamics of

real-world social identities.

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Appendix A

Supplementary figures

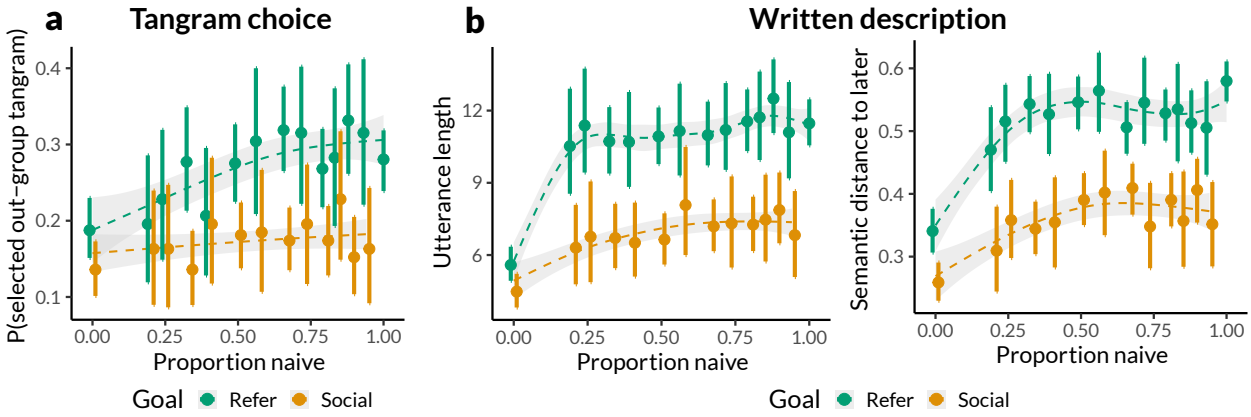


Figure A1

Experiment 1 results, for just the participants who scored above the median on the learning metric. Error bars are bootstrapped 95% confidence intervals.

Appendix B

Deviations from preregistration

The deviations from preregistrations, and their rationales, are stated below. All conclusions remain unchanged.

Experiment 1

For Experiments 1 and 4, we preregistered analyses testing for an overall main effect of the proportion of naive audience members. However, we determined that our primary theoretical interest was whether the proportion of naive people influenced behavior specifically within the 'refer' condition of the goal predictor. Focusing on the 'refer' condition allows for more precise comparison with existing research on referential efficiency. Consequently, we reported the marginal effects (estimated slopes within the 'refer' level of the factor) rather than focusing solely on the overall main effect, thereby providing a more targeted and theoretically relevant test.

Experiment 2

The preregistered model treated audience and goal as separate predictors. In the final analyses, these factors were combined into a single 'condition' variable, for consistency with Experiment 3. Additionally, given the unbalanced design (with the social goal only present in the 'uncertain' audience condition), combining these predictors provided a statistically more appropriate and coherent model.

Experiment 3

No deviations.

Experiment 4

The preregistered model did not include label transparency as a predictor. We incorporated this variable into the model to account for additional variance in participants' choices attributable to item-level differences in transparency. We also dropped the total amount of audience members from the model because it led to convergence issues and was not theoretically relevant.

Disentangling passive and active origins of linguistic diversity using experimental microsocieties

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Abstract

Why does human language exhibit so much diversity? Linguistic diversity may emerge passively from efficient reference within socially isolated groups, or actively from motivations to signal group identity. Although both mechanisms are well documented, their relative contributions are difficult to disentangle using observational data alone. Here, we use controlled, real-time online experiments to test how referential and social-signaling pressures differentially shape the emergence and use of linguistic conventions. In a multiplayer reference game, small groups of participants first develop novel naming conventions for abstract shapes, and then interact in a larger mixed population under conditions that orthogonally manipulate interaction patterns and social-signaling goals. We predict that social-signaling motivations contribute to lexical variation in two ways: (1) by accelerating differentiation during convention formation, and (2) by resisting the convergence pressures introduced by inter-group mixing. This research is designed to provide causal evidence for how social-signaling motivations sustain linguistic diversity, and to quantify their contribution relative to standard referential pressures.

Introduction

Across human societies, people use thousands of mutually unintelligible languages, along with countless dialects, registers, and slang variants, to talk about many of the same concepts and experiences. This variation may arise through two qualitatively different mechanisms: passive differentiation related to patterns of social contact and isolation, and active differentiation driven by social identity goals. Here, we use controlled experiments to manipulate these mechanisms and measure their contributions to the origins of lexical variation.

One well-established source of lexical variation is passive divergence arising from the pattern and frequency of social interactions[1–4]. Because a core cognitive function of language is referential informativity – making sure a listener correctly identifies an intended concept – people who interact frequently tend to develop shorter and more efficient naming conventions for concepts they need to talk about, often innovating new lexical items as needed[5,6]. This interaction-driven account aligns with evolutionary approaches that model languages as information systems that spread through social contact and evolve through drift, diffusion, and selection[7–11]. Random innovations arise, spread within groups, and persist because there is insufficient cross-group contact to select for shared forms. As a result, geographically or socially separated populations tend to have more dissimilar lexical systems[12–15], producing local variants (e.g., idiosyncratic conventions within households, friend groups, or workplaces) and regional variants (e.g., "soda" versus "pop," or "roundabout" versus "rotary").

The relationship between social interaction patterns and lexical variation is particularly striking in cases of geographic isolation. Papua New Guinea, despite its relatively small land area, contains over 800 distinct languages (nearly 12% of the world's total), a pattern that has been attributed to its mountainous terrain that limits contact between communities[7,16]. Similarly, when populations split, their languages predictably drift apart: after the settlement of Iceland in the 9th century, Icelandic and mainland Scandinavian languages diverged substantially despite their common origin[17,18]. Even within a single country, geographic barriers create systematic variation. Isolated Appalachian communities in the United States preserved linguistic features, like the *a-prefix* con-

struction, as in '*a-hunting* we will go', well after they disappeared from mainstream American dialects[19,20]. These patterns demonstrate how linguistic diversity covaries with patterns of social interaction, suggesting that it can emerge as a side effect of referential pressures operating within isolated social groups.

However, patterns of social contact alone cannot explain all lexical variation[14,21–25]. In many contexts, lexical differences persist or even become more pronounced despite frequent interactions between people who use each variant. For example, even though adolescents and adults interact daily, adolescents often develop and maintain distinctive slang terms that differentiate them (e.g., "aura farming" to mean "trying to look cool") from older generations. Sociolinguistic research suggests that such differentiation reflects the use of linguistic forms to express and maintain group identity, and mark social boundaries[26–28]. A classic case comes from the Gail Valley of Austria, where German and Slovenian speakers have been in contact for roughly a thousand years. Even after German was declared as the official language during the period of German occupation, bilingual Slovenian communities actively retained Slovenian as a symbol of in-group identity[28]. Yet over the past century, the use of Slovenian has declined substantially, potentially because improved intergroup relations have reduced the incentive to use language as a marker of social distinctiveness[28,29]. Similar identity-driven motivations help sustain other minority languages like Basque[30], drive major revitalization efforts such as Hebrew and Hawaiian[31,32], and support the continued use of constructed languages such as Esperanto and Klingon[33,34]. Linguistic markers that signal shared identity can also have social benefits, helping people decide who to trust and cooperate with[22,24,35–37].

Together, these observations indicate that lexical variation arises from at least two competing pressures. Referential informativity, the drive to be clearly understood, pushes towards shared conventions and so reduces variation, while social signaling, the drive to express identity and affiliation, pushes towards distinctive expressions and so increases variation. Referential informativity directly links language change to patterns of interaction and isolation: people who interact frequently converge on arbitrary, group-specific words for novel concepts[38]; while limited contact

allows existing terms to drift apart through the accumulation of random innovations. By contrast, social signaling is an active pressure for the creation or maintenance of distinct linguistic forms, even at the expense of communicative efficiency[39–42]. The tension between these pressures may help explain why linguistic diversity persists despite unprecedented global communication[43,44].

While many historical and corpus-based studies link linguistic variation to passive isolation and active social-signaling motivations, they cannot directly manipulate speakers' goals or interaction patterns and often rely on geographic or demographic proxies for social contact[45,46]. Moreover, in natural settings, these two pressures are usually confounded[24]. Communities with strong identity commitments often also have distinct interaction networks, so their linguistic distinctiveness could reflect isolation, social signaling, or both. As a result, for many real-life examples, the relative contributions of referential versus social-signaling goals remain controversial. For example, after the partition of Korea in 1945, North and South Korea developed different words for the same concepts, such as different terms for "friend," "corn," or "cell phone." The divergence in lexical terms has been attributed both to referential pressures, such as political isolation and contact with distinct languages, and to social-signaling pressures tied to distinct national identities, but observational data alone cannot determine their relative contribution[47].

Controlled laboratory experiments offer a complementary way to isolate and quantify these pressures[48]. Using artificial languages, such experiments have shown that social-selection incentives can drive linguistic divergence when dyads interact frequently[24], and that population structure influences the communication systems that emerge[49–52]. Our study integrates aspects of these approaches but shifts the focus to people's cognitive goals. We adopt the repeated reference game paradigm, where participants use natural language to communicate about abstract shapes[53], into a large microsociety experiment where interaction structure and motivations are simultaneously manipulated. This paradigm has traditionally been used to study how referential motivations are sufficient for explaining the emergence of conventions[6], so inducing a social-signaling motivation allows us to create a direct and measurable tension between social-signaling and referential goals.

In our study, the social interaction patterns in the population create the conditions under which

the balance between each motivation can show its effects. Because the full interaction unfolds within the experiment, we can trace the emergence and use of referential labels step by step, separating the effects of social-signaling goals on the formation of conventions from their effects on later use. Every word (the full study is expected to generate over 100,000 words of referential language across over 24,000 messages) is linked to the speaker's identity, group membership, experimental condition, referential target, full conversational history, and referential outcome – enabling fine-grained analysis that naturalistic approaches cannot support. Furthermore, the use of natural language descriptions enables fine-grained characterization of the lexical system, aided by advancements in natural language processing[54]. In sum, the referential and social-signaling pressures can be manipulated independently, their onset can be precisely controlled, and their effects on the emergent lexical system can be compared directly and analyzed at scale.

In the planned study, groups of participants first develop naming conventions through repeated interaction (Phase 1), and then enter a second phase that manipulates who they interact with (whether they remain separated or mix into a larger population), and what their primary goals are (whether they have an explicit incentive to signal social identity to their original group members). Our design uses a minimal group approach[55]: groups are randomly assigned, participants are strangers, and interaction is limited to an online game. This setup minimizes the influence of aspects that complicate inference in natural settings, like preexisting social ties, demographic differences, and historically established identities; it also allows us to isolate the effect of affiliation-signaling goals on lexical choice rather than on other dimensions of language, such as accent or prosody. If affiliation-signaling goals shape the lexical system even under these minimal conditions, this would show that affiliation signaling is sufficient to influence lexical variation, and speak to its role in natural communities that include richer and more durable social identities and relationships.

Four complementary experimental conditions allow us to target distinct points in the lifecycle of a lexical system where social-signaling goals can exert an influence: during the formation of conventions, during their later use, or both (Figure 2). This design allows us to test four main

hypotheses (see Table 1):

- H1 (Mixing promotes convergence): Under purely referential goals, mixing will lead to greater linguistic convergence than continued separation.
- H2 (Social signaling resists convergence): The explicit incentive to signal group identity resists pressures from mixing and allows the system to maintain more differentiated lexical systems.
- H3 (Anticipated social goals shape convention formation): Expecting a later social-identification incentive will shape Phase 1 convention formation in three ways:
  - H3a: Phase 1 conventions will be more group-specific.
  - H3b: Within-group conventions will converge faster during Phase 1.
  - H3c: Descriptions will be less concrete, more lexically distinctive, and use lower-frequency vocabulary.
- H4 (Anticipated social goals produce downstream consequences): These Phase 1 differences will affect Phase 2 communicative success:
  - H4a: Higher social-identification accuracy.
  - H4b: Lower referential accuracy.

H1 and H2 concern the *use* of established conventions. H1 tests whether, under purely referential goals, mixing leads groups to converge on shared terms. H2 then asks whether an incentive to signal group identity can resist that convergence and preserve lexical differentiation despite contact. H3 and H4 shift from convention *use* to convention *formation*: they ask whether anticipating later social goals changes how conventions emerge, and whether those early differences have downstream consequences.

Support for H1 and H2 would show that linguistic diversity can be induced causally and experimentally under minimal conditions, either through referential goals operating passively through

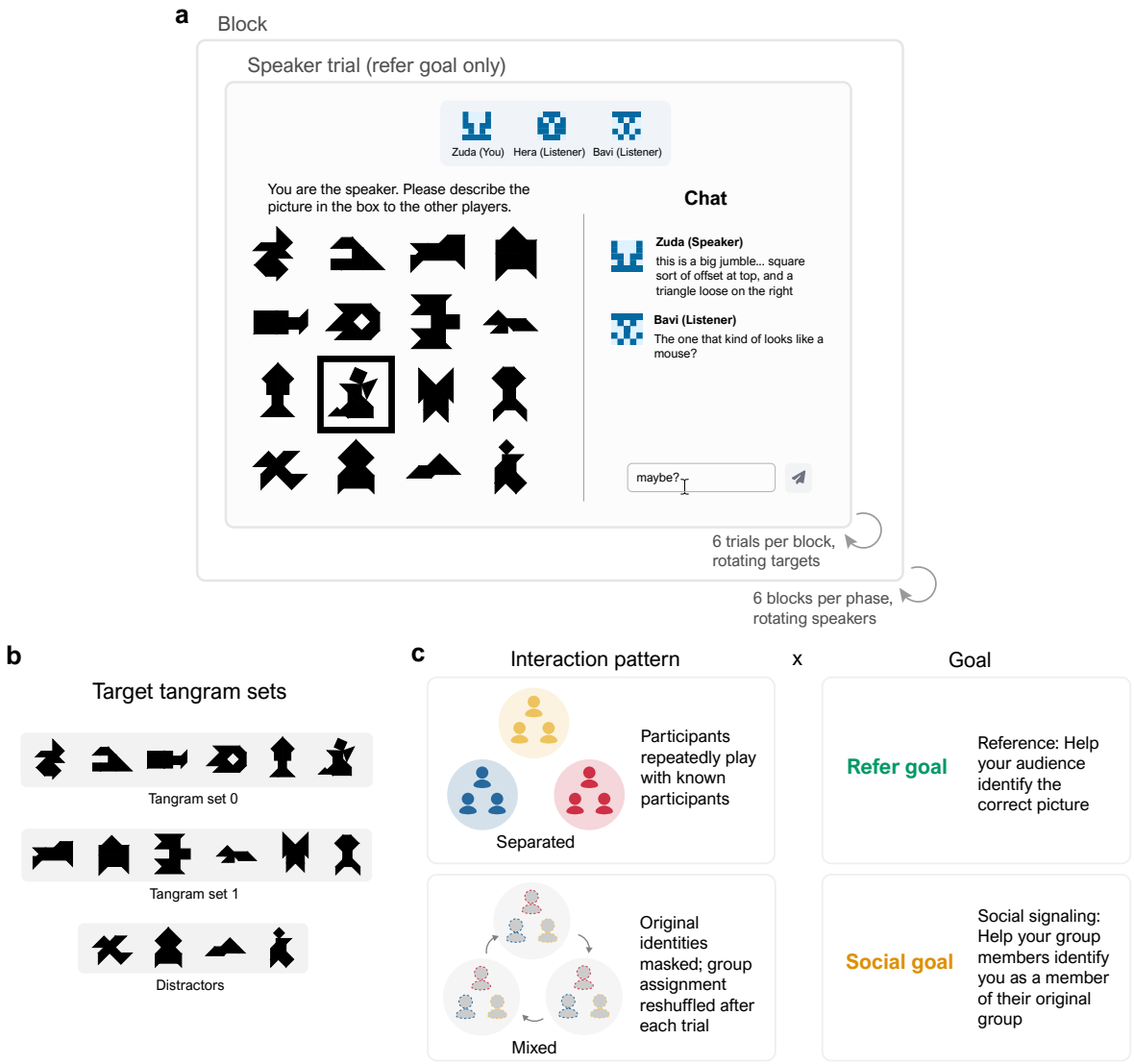


Figure 1: Task schematic and manipulations. (a) On each trial, the speaker sees a highlighted target tangram, and describes it to the listeners. Listeners can respond in the chat until all listeners have made their selection. (b) Each 9-person game is assigned to refer to one of two sets of target tangrams. (c) The experiment manipulates interaction pattern in Phase 2 (who participants play with and whether their identities are masked), and participants' goals.

interaction structure, or through active social motivations. Isolating these contributions in a controlled setting would provide a basis for linking them to factors in natural settings – such as network structure or identity salience – that drive convergence in some cases and sustained distinctiveness in others, potentially enabling predictions of when social contact will produce one outcome or the other.

If H1 is supported but H2 is not, this would suggest that referential pressures operating through interaction patterns alone account for the lexical variation observed in our task, and that social signaling may require stronger identity commitments or higher-stakes consequences than our minimal-group paradigm provides. If H1 is not supported, by contrast, then the expected convergence pressure from mixing is absent, implying that once conventions are established in small groups they may be relatively stable and resistant to change. H2 is most straightforward to interpret when H1 is supported, as it establishes that mixing creates a convergence pressure that social signaling can resist. Still, we will test H2 even if H1 is not supported. If H1 is not supported but H2 is, this would suggest that social identity incentives actively drive the differentiation of terms in the presence of social contact, pushing them further apart even when contact alone leaves them stable.

The H3 sub-hypotheses examine several ways social-signaling goals may shape the emergence of a lexical system, with H3a and H3c focusing on the properties of the resulting conventions and H3b on the dynamics of convention formation. Support for any of the H3 sub-hypotheses and H4 would suggest that anticipating later social-signaling goals alters how conventions emerge, with downstream consequences for communicative success. Support for both H4a and H4b would further indicate a trade-off between referential clarity and social distinctiveness. If H3a and H3c are supported but neither H4a nor H4b is, this would suggest that anticipated social goals shape which conventions emerge without producing measurable consequences for communication. If H3a and H3c are not supported, this would suggest that anticipated social-signaling goals do not affect the emergent conventions at the end of Phase 1 in terms of our metrics, in which case H4 cannot be meaningfully tested.

Methods

Ethics information

All procedures have been approved by the MIT Committee for the Use of Humans as Experimental Subjects (protocol #0602001639). Informed consent will be obtained from all participants. See details about participant recruitment and compensation below.

Pilot data

We built the experiment using Empirica, a React-based framework for real-time multiplayer tasks[56], with custom server-side logic for real-time group reshuffling, dropout detection and recovery, and role reassignment developed specifically for this study to support the complexity of 9-player synchronous experiments. Code is available in the project repository (see Code Availability).

We collected one pilot game per condition, to demonstrate the feasibility of our approach. The pilot data confirmed that (1) participants successfully form group-specific conventions during Phase 1, as evidenced by decreasing description length and increasing listener accuracy over blocks, (2) the outcome-neutral criteria pass, and (3) participants can be successfully recruited and retained with our procedures. Analyses of the pilot data are reported in Section S1.

Design

Materials

In each game, participants communicate about a set of 6 target tangrams (see Figure 1b). The target set is counterbalanced across games so that each game uses one of two tangram sets. We manually selected tangrams with high Shape Naming Divergence (SND) – a measure of label variability across describers – from a large dataset[57]. The tangrams are displayed in a 4×4 grid of 16 items: 6 targets, 6 non-targets from the alternate set, and 4 permanent distractors. To discourage spatial descriptions (e.g., "the one on the top left"), the grid order is randomized across participants

but fixed within each game. This design – high-SND tangrams in a 16-item context, so that the tangrams are difficult to identify but also can be arbitrarily named – encourages the emergence of group-specific conventions and reduction in description length over rounds, which is necessary in Phase 1 to test the main hypotheses in Phase 2.

Procedure

Participants will play a multiparty reference game where they type freely in a chat box to communicate about abstract tangrams[38]. Each 'game' consists of 9 participants who interact in subgroups ('groups') of 3. Upon entering the game, participants are randomly assigned to a group. In each round, one player (the 'speaker') describes the target tangram in a communal chat, and the two other players (the 'listeners') in the group each make a guess by selecting one tangram (Figure 1a). All players receive feedback about listeners' guesses.

The task consists of two phases. In the first phase of the task, participants start by interacting within their assigned groups for 6 blocks. In each block, one speaker describes each of the six target tangrams once. At the end of the block, the speaker role moves to the next participant, and the next block begins. Thus, because there are 3 speakers in a group, each participant is in the speaker role twice in the first phase of the task. This design (participants interact in groups of 3 and everyone gets full feedback) promotes rapid within-group convention formation[38] in Phase 1, while maintaining a reasonable task length. Listeners earn 2 points if they successfully guess the target tangram, and speakers earn up to 2 points based on the proportion of listeners who correctly guessed the target tangram (to account for smaller group size due to potential dropout). After the 6 blocks of Phase 1, participants proceed to Phase 2. The between-game experimental manipulations (Figure 2) determine subsequent interaction partners and participants' incentives:

1. In the 'refer + separated' condition, participants stay in their initially assigned groups throughout the game, and continue playing the game as before, for 6 more blocks.

2. In the 'refer + mixed' condition, after the initial 6 blocks, participants' group assignments are randomly shuffled, and their identities are masked. Groups are reshuffled on every trial

in Phase 2, so that participants interact in a randomly-composed group each round. At each reshuffling, the assignment of participants to groups is randomized with a constraint ensuring that one of the two listeners in each group is from the speaker's original Phase 1 group, with approximate fallbacks if the group size is smaller due to dropout. This constraint ensures that each group always contains at least one listener from the speaker's original group, which is necessary for the social-identification task in the 'social + mixed' condition; applying the same constraint here keeps interaction patterns identical across the two mixed conditions. The speaker rotation follows the same schedule as the 'refer + separated' condition (i.e., each participant is listener twice, then speaker once, and so on).

3. In the 'social + mixed' condition, the interaction patterns match the 'refer + mixed' condition, but with an additional social signaling incentive: on each trial, alongside their tangram selection, listeners also guess whether the speaker was in their originally assigned Phase 1 group. Participants primarily earn bonuses for social signaling: in addition to the standard 2 points for reference, listeners are rewarded six points for a correct response, and speakers are rewarded up to six points for the proportion of in-group listeners who correctly identify the speaker as a member of their original group. In addition to the referential feedback, listeners see whether their social guess was correct or incorrect, and speakers see how many of their original-group listeners recognized them.

4. In the three conditions above, participants learn about the Phase 2 manipulation only at the transition from Phase 1 to Phase 2. By contrast, in the 'social-first + mixed' condition, participants are told before Phase 1 begins that in Phase 2 they will be rewarded primarily based on whether members of their original group can identify them from their descriptions. The comprehension quiz includes a question confirming that participants understood this. This allows the manipulation to shape convention formation during Phase 1. Phase 2 otherwise follows the same structure as the 'social + mixed' condition.

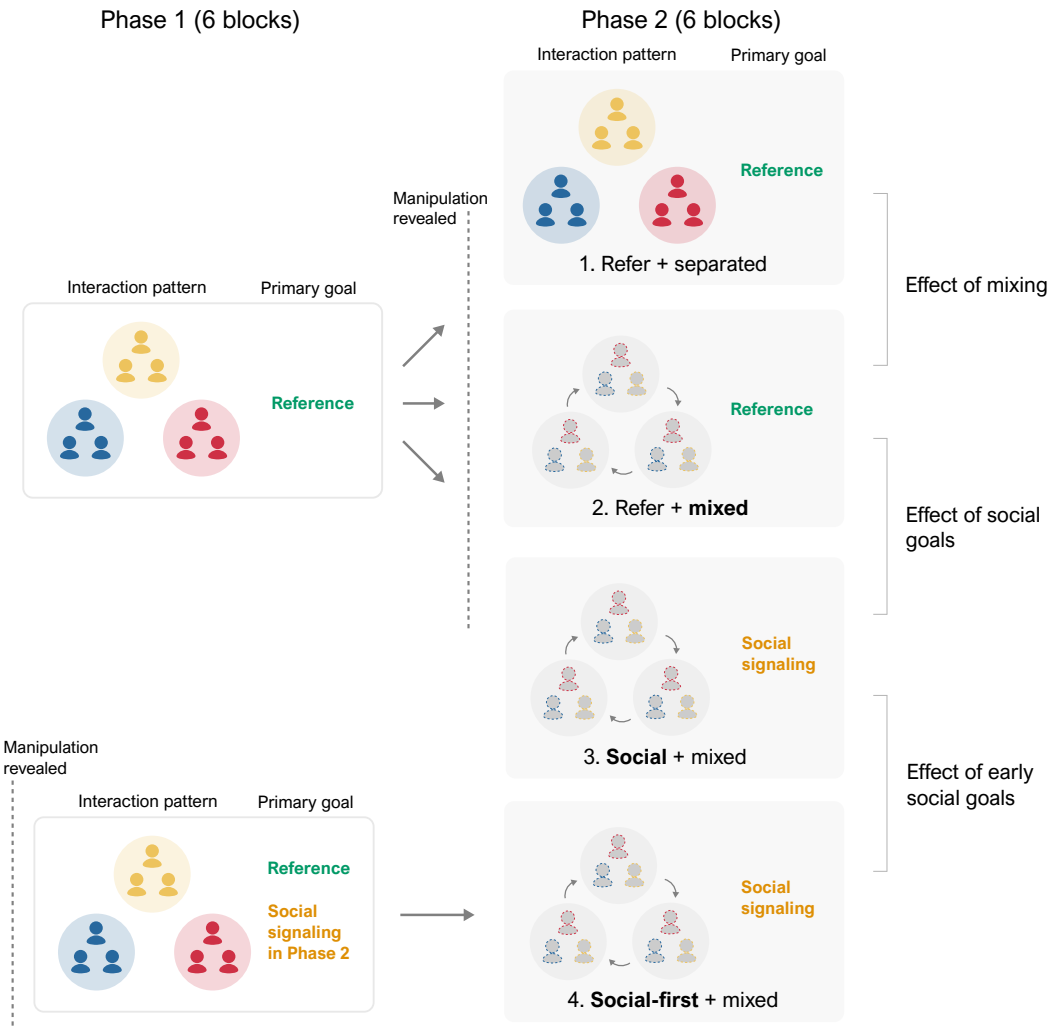


Figure 2: Experimental design. The contrast between successive conditions is designed to isolate the effect of population mixing, the effect of social-signaling goals, and the effect of anticipated social-signaling goals.

Timing Across all conditions, participants play for a total of 12 blocks (6 blocks in Phase 1, 6 blocks in Phase 2). Participants have 45 seconds to select a tangram in Phase 1 and 25 seconds in Phase 2. The shorter time limit in Phase 2 reflects the expectation that established conventions require less deliberation and is also intended to induce a trade-off between goals, so that participants aren't exhaustively optimizing for all goals at the same time.

Compensation The task takes 45-60 minutes. Participants receive a base pay of $12 for completing the task, with a bonus of up to $8 based on the number of points they earned. Participants

receive partial pay proportional to time spent and points earned if the game ends early for reasons that are out of their control (e.g., their Phase 1 partners drop out). Participants' points are translated into a bonus at the end of the experiment. For the 'refer + separated' and 'refer + mixed' conditions, participants earn \$0.056 per point. For the 'social + mixed' and 'social-first + mixed' conditions, participants earn \$0.023 per point. The lower per-point rate for social conditions accounts for the additional scoring opportunities from social guessing, keeping maximum bonuses similar across conditions.

Our goal is to measure the balance of referential and social signaling goals in the production of referring expressions. Thus, participants are instructed that to receive their pay, they need to restrict chat messages to tangram-related discussion and to avoid discussing other topics. This instruction is designed to prevent the use of alternative strategies such as explicitly discussing one's group identity or coming up with an unrelated 'codeword' for social signaling. Messages violating this rule will be flagged, and persistent violators will be excluded (see Exclusion criteria).

Sampling plan

Sample size We will aim to recruit 20 full games per condition, for a total of approximately 720 participants (20 games $\times$ 9 players/game $\times$ 4 conditions). Based on pilot rates of approximately 300 messages and 200 speaker utterances per game, the full study is expected to yield at minimum 24,000 messages comprising over 100,000 words of referential language across 5,760 rounds.

The sample size was determined using a power analysis using existing data from a corpus of multi-party reference games[38] (Experiment 2b; 6 person full-feedback condition). We used the data to simulate hypothetical 'games' to test the contrast between the 'refer + mixed' and the 'refer + separated' conditions (the largest expected effect in our design). We did this by (1) treating 6-person games as-is and randomly partitioning groups to approximate a 'mixed' condition, and (2) randomly pairing these partitions from different games to approximate the 'separated' condition. We computed the main analysis target, the group-specificity metric, (see Analysis plan) for these simulated games. The effect size was Cohen's $d = 2.42$ (two-sample t-test between mixed and

separate). Because this is a 'ceiling' effect – our primary comparison between 'refer + mixed' and 'social + mixed' should yield a smaller effect – we calculated the sample size needed to detect 50% of this effect at 95% power at $\alpha = 0.05$, resulting in 20 games per condition. The analysis notebooks and outputs for the power analysis are available in the project repository.

Because we will recruit until 20 *complete* 9-person games are obtained per condition (see Dropout handling below), games in which players drop out but that still run to completion under the dropout-handling rules will yield additional usable data. The effective sample size per condition will therefore be at least 20 games and may vary slightly across conditions.

We acknowledge that no prior experiments have manipulated anticipation of social-signaling goals in this paradigm, so reliable effect-size estimates for H3 and H4 are not available. However, even with a single pilot game per condition, we observed emerging differences in the predicted directions: conventions in the 'social-first + mixed' condition were more group-specific from the outset, and the H3c description-property measures (concreteness, lexical uniqueness, word frequency) showed directional separation between conditions (Section S1; Figures S3-S5). While these single-game observations cannot yield stable effect-size estimates, the fact that predicted patterns were detectable with minimal data suggests the true effects are unlikely to be negligibly small. With $N = 20$ games per condition, a two-sample t-test at $\alpha = 0.05$ (two-tailed) has 80% power to detect effects of $d \geq 0.91$. To distinguish true null effects from insufficient power, we will use Bayes factors for all non-significant results (see below).

Recruitment We will recruit adult fluent English speakers from the United States through Prolific. Because the game depends on synchronous participation, we will use a two-stage recruiting process to guarantee that enough participants log on at the same time to fill games. Participants will first complete a short form assessing their eligibility for the task and their availability at a scheduled time. Participants are paid $0.40 for completing the form. Then, we deliver the study to eligible participants at the scheduled time. After participants complete the consent form and the instructions, participants will be taken to a waiting room where they wait up to 10 minutes to

be assigned to a game. Participants who are not assigned to a game after 10 minutes are removed from the experiment and compensated $2 for their time. The nature of online recruitment means that some variability in attendance is expected, so that more participants need to be recruited than will be ultimately assigned to games. This, and the possibility of during-game dropout, means that the actual number of games per condition might differ slightly across conditions.

Exclusion criteria Participants will be removed from the experiment in real time if they (1) fail to pass a comprehension quiz about the task instructions after three attempts, or (2) idle for 3 consecutive rounds. Additionally, at the end of Phase 1, groups where fewer than two-thirds of participants achieved at least two-thirds referential accuracy during the last three blocks will be removed from the game and compensated for their time, as their conventions are insufficiently established for a meaningful test of the Phase 2 hypotheses. We will also exclude participants who consistently send inappropriate, adversarial, or task-irrelevant messages (determined through post-hoc manual inspection of chat logs). We will also exclude participants who showed an increase in description length in Phase 1, as this suggests AI use (see below).

AI detection AI agent use is a potential concern in online studies involving open-ended text production[58] (but see[59]). Several features of our design reduce this risk. First, recruitment uses a two-stage process on Prolific: participants must first complete a screening and scheduling step, and then return at a specific time for a synchronous multi-party session. This procedure likely discourages low-effort or automated participation relative to a single-stage asynchronous study. Second, we monitor and screen for multiple indicators of genuine task engagement, including comprehension-quiz performance, sustained participation across rounds, referential accuracy during Phase 1, and post-hoc inspection of chat logs for responses that are off-task, formulaic, inconsistent with the game state, or otherwise suggestive of automated assistance. Also, the post-test survey asks participants whether they felt they were playing with other humans; groups in which any participant reports otherwise will be flagged for detailed inspection.

As an additional benchmark, we simulated the task with a state-of-the-art multimodal language

model (Gemini 3.1 Pro) using prompts designed to strongly support success (see Section S3 for details). Accuracy increased over blocks, and 15/20 simulated groups (75%) met the group viability criterion. However, the LLM agents achieved this accuracy through a fundamentally different mechanism than humans: descriptions grew longer across blocks (24.3 to 45.7 words) rather than shorter (14.2 to 4.0 words in pilot data), and description similarity saturated near 1.0 through near-verbatim copying from history (Figure S6). These divergent signatures provide a basis for distinguishing AI-generated from human-generated game data through post-hoc inspection. While this does not rule out AI use, it suggests that participants relying on such models would produce qualitatively distinguishable behavioral profiles, informing our exclusion criteria above.

Dropout handling To avoid losing all data for a 9-person game when one player idles or disconnects, we will implement several dropout-handling rules. Participants who disconnect or idle for 3 rounds are removed, with speaker roles reassigned as necessary. If two members of a participant's initial group remain active, gameplay continues; if two group members drop out, the remaining member is removed and compensated for their time. The game continues as long as at least two original groups each retain at least two active members. The same logic applies in Phase 2: gameplay continues as long as two original groups each retain at least one active member, for consistency between the 'mixed' and 'separated' conditions. In the mixed conditions, if a dropout would leave a participant as the sole active member of their new group, we will trigger a mid-block reshuffle to maintain the minimum group size. Our primary analyses will include all games that ran to completion under the dropout-handling rules above. We will report attrition rates by condition and check whether dropout is differential across conditions. As a robustness check, we will confirm that the results hold when restricted to complete 9-person games.

In pilot data (4 games, 36 participants), all games ran to completion. 1 participant was removed for inactivity (in the 'social-first + mixed' game), and 1 group was removed for failing the accuracy threshold (in the 'refer + mixed' game). In 3 of 4 games, all three groups were active through both phases; the remaining game continued with 2 active groups, yielding usable data under the rules

above. The median session duration was 42.7 minutes (range: 32.5–49.1). These rates suggest that the synchronous format and compensation structure are sufficient to maintain engagement.

Analysis plan

We will conduct our analyses with linear mixed-effects regression models using the lmer and lmerTest packages in R[60,61]. We will use the maximal random effects structure supported by the design, simplifying if models fail to converge by first removing correlations between random effects, then removing random slopes starting with the smallest variance component, following the procedure recommended by Barr et al.[62]. For models with binary outcomes (e.g., trial-level accuracy), we will use logistic mixed-effects models (`glmer` with `family = binomial`). The models will be fit using maximum likelihood. For all analyses, we will report standardized effect sizes (Cohen's *d* or equivalent) alongside test statistics and confidence intervals. Visualizations of the below metrics and comparisons using pilot data are reported in Section S1.

Dependent variables and preprocessing

We will collect the following dependent measures:

- **Speaker utterances.** Our primary dependent variable is the descriptions speakers produce for the target tangram. To analyze these descriptions, we will concatenate speakers' messages within a trial by separating each message by a comma. We will filter out non-referential messages (e.g., "thanks", "good job") by using a LLM-based classifier that has been calibrated and validated against a manually labeled subset, and manually checking the resulting classifications (see Section S2 for details). This obtains, in a sentence format, the referential information that the speaker provides to the listeners on each trial[1].

[1] In some trials, listeners ask yes/no questions and speakers provide brief answers. Our primary analysis excludes these listener turns, acknowledging that this may omit potentially informative context. As a secondary analysis, we will explore representations that incorporate the full conversational exchange. One possibility is to use SBERT to embed the full dialogue, acknowledging that this representation reflects the overall interaction rather than the speaker's lexical information specifically. Another option is to use non-embedding-based similarity measures (e.g., a LLM-as-judge approach) that can operate on the full conversational context.

  - To obtain a measure of description length (a proxy for cognitive cost), we will count the number of words in speakers' utterances. This metric allows us to analyze and reproduce the phenomena shown in classic reference game studies of reduction in description lengths over repeated interaction[6,53,63].
  - To obtain a measure of the semantic content of speakers' descriptions, we will embed them in SBERT embedding space using the `paraphrase-MiniLM-L12-v2` model[64], which transforms each description into a 384-dimensional vector. Cosine similarity between these vectors provides a measure of semantic similarity between descriptions.

- **Listener accuracy.** We will track both referential and social signaling accuracy, at the trial, participant, group, and game levels:
  - Referential accuracy: Operationalized as the proportion of listeners that correctly select the target tangram on each trial.
  - Social signaling accuracy: Operationalized as the proportion of listeners that correctly identify whether the speaker was in their original group.

Our primary dependent metric is **group-specificity**. If the emergent lexicons are differentiated based on original group assignment, then a description a participant uses for a given tangram should be more similar to descriptions produced by members of the same group, compared to descriptions produced by members of other groups. Thus, for each game, we will compute pairwise cosine similarities between SBERT embeddings of all utterances for each tangram produced during the relevant time window. We will predict similarity between pairs from whether the pair was produced by members of the same group, with random effects for tangram and crossed random effects for the two participants in each pair (Formula: `similarity` $\sim$ `same_group + (1 | tangram) + (1 | participant_1) + (1 | participant_2)`), where pairs are ordered consistently by participant ID and each unique pair appears once. The crossed participant random effects account for the non-independence introduced by participants appearing in multiple pairs.

The coefficient for `same_group` corresponds to the group-specificity score for the game: a positive coefficient indicates that within-group descriptions are more similar than between-group descriptions. We will also extract the standard error of this coefficient, which will be used as a measure of uncertainty in the game-level analysis. We will compute this metric separately for the final three blocks of Phase 1 (baseline group-specificity after convention formation) and the final three blocks of Phase 2 (primary outcome variable). Three blocks is the minimum needed for each participant to produce one expression for each tangram, providing a complete "lexicon." We will also compute group-specificity for the most recent set of utterances at each block (resetting at the Phase 1–Phase 2 boundary), beginning once at least two participants per group have produced descriptions for a given tangram. This allows us to examine the temporal trajectory of group-specificity as conventions emerge and evolve (see Figure S3b).

Because this coefficient reflects the difference between within-group and between-group similarity, a positive value could arise from high within-group convergence, low between-group similarity, or both. To disambiguate these contributions, we will also report the within-group and between-group cosine similarities separately, and how it evolves over blocks (see Figure S3a).

Outcome-neutral criteria

To ensure the basic experimental paradigm functions as intended, several outcome-neutral criteria must be satisfied before testing our primary hypotheses. Because the first phase of our task replicates the structure of classic reference game experiments and is identical across all conditions, we expect these criteria to be satisfied based on extensive prior work characterizing the dynamics of convention formation in repeated reference games[6,38,53,54]. All of the outcome-neutral criteria were satisfied for each game in the pilot data (Section S1).

First, if participants are successfully coordinating on conventions, then description length should decrease, while listener accuracy should increase, over blocks. Furthermore, the emergent conventions should be stable (once established, conventions should persist; operationalized by increasing cosine similarity between successive utterances for each tangram, across speakers and

within a group). We will fit the following linear mixed-effects models, and we require significant main effects of block number in all cases:

- `utt_length` ~ `block_num` + (`block_num` | `participant`) + (`block_num` | `group`) + (`block_num` | `tangram`)
- `ref_accuracy` ~ `block_num` + (`block_num` | `group`) + (`block_num` | `tangram`)
- `sim_adjacent` ~ `block_num` + (`block_num` | `participant`) + (`block_num` | `group`) + (`block_num` | `tangram`)

Second, group-specificity scores at the end of Phase 1 must exceed chance levels. We will conduct a permutation test: For each game, we will permute the assignment of participants to groups 1000 times and recalculate the group-specificity metric. We will compare the observed group-specificity to this null distribution to obtain a p-value for each game's group-specificity value. We require that at least 80% of games (16/20 per condition) show significant group-specificity ($p < .05$).

If any of the criteria fail, this indicates a fundamental problem with the paradigm, and we will not proceed to test the primary hypotheses without first identifying and addressing the issue (which would require resubmission as a new Stage 1 protocol).

Primary analyses

Our primary analyses test four hypotheses using planned directional contrasts. See Table 1 for a summary of hypotheses and interpretations given to different outcomes. For all contrasts, we will use two-tailed tests with $\alpha = 0.05$, to remain sensitive to unexpected effects in the opposite direction. Because these contrasts test distinct, theoretically motivated hypotheses rather than a family of exploratory comparisons, we will not apply correction for multiple comparisons. For each contrast, we will report the estimated difference, its 95% confidence interval, and the exact p-value. A contrast will be considered significant if $p < .05$ and the direction of the effect matches our prediction.

If we obtain non-significant p-values, we will compute Bayes factors to evaluate evidence for the null hypothesis (i.e., no difference between contrasts). We will refit the relevant model in brms[65] and compute Bayes Factors. Following standard recommendations in psychological research[66], we will standardize the estimates and use a Cauchy $\left(0, \frac{\sqrt{2}}{2}\right)$ prior (corresponds to a small to medium effect) for the effect size under the alternative. We will interpret Bayes factors > 10 as strong evidence for the alternative hypothesis, BF > 3 as moderate evidence for the alternative hypothesis, BF < 0.1 as strong evidence for the null hypothesis, BF $< 1/3$ as moderate evidence for the null hypothesis, and values between 1/3 and 3 as inconclusive evidence.

We also note that there are two possible tangram sets. Any set-level differences in difficulty are absorbed by the tangram random effects in trial-level models and by residual variance in the game-level regressions. As a robustness check, we will confirm that results are unchanged when tangram set is included as a covariate.

H1 and H2 (Effects of mixing and social signaling on Phase 2 group-specificity) H1 and H2 both concern the group-specificity of lexical systems at the end of the experiment. Because group-specificity is a game-level summary statistic extracted from a within-game model (see above), we test these hypotheses using weighted least-squares (WLS) regression predicting Phase 2 group-specificity from condition, with Phase 1 group specificity as a predictor to account for variation in the lexical systems participants formed before the manipulation (Formula: `group_specificity_phase2` $\sim$ `condition` + `group_specificity_phase1`). The observations will be weighted by the inverse variance of the Phase 2 group-specificity estimates (i.e., $1/\text{SE}^2$, where SE is the standard error of the `same_group` coefficient extracted from each game's similarity model). We note that Phase 1 group specificity is also estimated with uncertainty; treating it as a fixed predictor may cause us to under-control for baseline differences, making our tests more (rather than less) conservative.

We will test two planned contrasts by comparing estimated marginal means:

- **H1 (Effect of mixing):** 'refer + separated' $>$ 'refer + mixed'. In the 'refer + separated' condition, participants continue interacting within their original three-person groups throughout

the game. Their final lexicons should therefore remain efficient, stable, and group-specific, closely resembling those formed during Phase 1. By contrast, in the 'refer + mixed' condition, players must communicate across the full nine-person population. We predict that under referential goals, population mixing will lead the game to converge on a common set of descriptions, leading to reduced group-specificity.

- **H2 (Effect of social signaling):** 'social + mixed' $>$ 'refer + mixed'. If communication systems are shaped only by referential informativity, both mixed conditions should converge toward similar, population-wide lexicons with low group specificity. However, if participants strategically use referential language to signal group identity, then speakers should maintain their original group-specific conventions despite interacting with the full population. Therefore, we predict that the lexicons in the 'social + mixed' condition will be more group-specific than in the 'refer + mixed' condition.

H3 (Effect of anticipated social goal on Phase 1 conventions) H3 tests whether anticipating social signaling shapes convention formation in Phase 1. In the 'social-first + mixed' condition, participants are told before Phase 1 that they will later need to identify group members based on their language. We predict that anticipating social-signaling goals will shape conventions in three ways:

H3a (Group-specificity). We predict that Phase 1 conventions will be more group-specific in the 'social-first + mixed' condition than in the 'social + mixed' condition. In other words, we predict that participants actively choose utterances that are not likely to be used by other groups. We will fit a weighted least-squares regression predicting the final Phase 1 group-specificity from condition (Formula: `group_specificity_phase1` $\sim$ `condition`), restricted to these two conditions, weighted by the inverse variance of the Phase 1 group-specificity estimates. We will test the directional contrast 'social-first + mixed' $>$ 'social + mixed'.

H3b (Convergence rate). We predict that within-group conventions will converge faster in the 'social-first + mixed' condition. For each group, we will compute within-group cosine similarity

at each Phase 1 block and fit a mixed-effects model: `within_group_similarity` ∼ `condition` * `block_num` + (`block_num` | `group`) + (`block_num` | `tangram`), restricted to the two conditions. A significant `condition` * `block` interaction, with faster convergence in 'social-first + mixed', would support H3b.

H3c (Description properties). While H3a tests whether the overall lexical systems are more differentiated between groups, H3c examines specific properties of the descriptions that may drive this differentiation. We predict that descriptions in the 'social-first + mixed' condition will be less concrete and geometric, more lexically distinctive, use lower-frequency vocabulary, because participants anticipate that these properties of expressions are more effective group markers. We will compare the following properties of the final Phase 1 descriptions. To compute all three measures, we will first extract the content words by alphabetic sequences, removing standard English stopwords from the NLTK list[67] (e.g., *and*, *the*), and retaining only words longer than one character.

- **Concreteness**: the proportion of content words in each description that match concrete term stems, using prefix matching on a keyword list from Boyce *et al.* [38]. The list comprises four categories: geometric terms (e.g., *triangle*, *square*, *diamond*, *line*), body-part terms (e.g., *arm*, *head*, *leg*, *foot*), positional terms (e.g., *left*, *right*, *top*, *bottom*, *side*), and posture terms (e.g., *kick*, *kneel*, *stand*, *sit*, *lean*). Prefix matching allows stems like *squar* to match both *square* and *squares*.

- **Lexical uniqueness**: the proportion of content words in a description that do not appear in any other group's descriptions for the same tangram, across games.

- **Word frequency**: the mean Zipf frequency of content words in each description, computed using the `wordfreq` package[68], which draws on a variety of contemporary corpora (Wikipedia, Google Books, movie subtitles). Zipf frequency is a logarithmic scale where common words score higher (e.g., "the" ≈ 7.7) and rare words score lower.

For each measure, we will fit a mixed-effects model with condition as predictor (Formula: `measure` ∼ `condition` + (`1` | `participant`) + (`condition` | `tangram`)), restricted to the two condi-

tions. We predict lower concreteness and higher lexical uniqueness, and lower word frequency, in the 'social-first + mixed' condition.

H4 (Downstream consequences of anticipated social goal) H4 tests whether differences in convention formation under anticipated social goals produce downstream consequences for communicative success. If anticipating a social-identification goal leads participants to form more distinctive but potentially less transparent conventions (as predicted by H3), their language should carry stronger group-identity cues in Phase 2. The resulting utterances may also be harder for out-group listeners to interpret[39–41].

We will fit logistic mixed-effects models restricted to Phase 2 in the 'social + mixed' and 'social-first + mixed' conditions, predicting social and referential accuracy on each trial. We include block number as a predictor to account for adaptation over the course of Phase 2, as participants may improve at both tasks through repeated interaction with the mixed population. Note that H4 is logically contingent on H3a/c: if there is no evidence that anticipated social goals shape the Phase 1 conventions, then there is no basis for downstream consequences.

H4a (Social-identification accuracy). We predict that participants in the 'social-first + mixed' condition will be more accurate at identifying group members during Phase 2. If these participants formed more distinctive conventions during Phase 1 (as predicted by H3), their language should carry stronger group-identity cues, making it easier for listeners to determine which group the speaker belongs to. We will fit a logistic mixed-effects model: `social_correct` $\sim$ `condition` + `block_num` + `(block_num | game)` + `(block_num | listener)` + `(block_num | speaker)` + `(condition | tangram)`. We will test the directional contrast 'social-first + mixed' $>$ 'social + mixed'.

H4b (Referential accuracy). We predict that referential accuracy will be lower in the 'social-first + mixed' condition. More distinctive, less concrete conventions may be effective group markers but harder for out-group listeners to interpret. We will fit a logistic mixed-effects model: `ref_correct` $\sim$ `condition` + `block_num` + `(block_num | game)` + `(block_num | listener)`

+ (block_num | speaker) + (condition | tangram). We will test the directional contrast 'social + mixed' > 'social-first + mixed'.

Secondary analyses

The following analyses are pre-registered as secondary: they test directional predictions derived from the same framework as our primary hypotheses, but address questions that are not the central focus of the study. We distinguish these from purely exploratory analyses, which we will conduct and report separately in the Stage 2 manuscript (e.g., analyses of individual differences, post-hoc clustering of description strategies, or unanticipated patterns in the data).

Semantic change To quantify how much lexical change occurs between phases, we will compute the cosine similarity between each participant's last Phase 1 description and their last Phase 2 description for each tangram (i.e., the description from the last block in which that participant served as speaker for that tangram within each phase). This captures the degree to which individuals' conventions change due to the manipulation. We will fit the model sim_phase1_phase2 $\sim$ condition + (1 | participant) + (condition | tangram). We predict:

- 'refer + separated' > 'refer + mixed': Participants who continue interacting with the same people will show higher Phase 1–Phase 2 similarity (i.e., less semantic change) than those who interact with a broader population.
- 'social + mixed' > 'refer + mixed': A social-signaling goal will resist the semantic shift created by mixing, leading participants to retain their original group-specific expressions (i.e., higher Phase 1–Phase 2 similarity).

The 'refer + separated' condition also allows us to assess baseline lexical drift. We expect the similarity between Phase 1 and Phase 2 utterances in this condition to be close to ceiling because participants continue interacting in their original groups. Any deviations from ceiling reflect drift due to random innovation or memory errors.

Referential accuracy We will analyze Phase 2 referential accuracy at the trial level (whether each listener correctly identified the target tangram on each trial). We will fit a logistic mixed-effects model: `ref_correct` $\sim$ `condition` * `block_num` + `(block_num | game)` + `(block_num | participant)` + `(condition * block_num | tangram)`. We predict the following:

- Referential accuracy will be reliably above chance (6.25%, i.e., $1/16$ tangrams) across all conditions, serving as a quality check that participants are engaged in the task.
- Main effect of block: Referential accuracy will generally increase over blocks, across conditions (though it may plateau if it has already reached ceiling in Phase 1).
- Averaged across blocks, referential accuracy will be highest in the 'refer + separated' condition, intermediate in the 'refer + mixed' condition, lower in the 'social + mixed' condition, and lowest in the 'social-first + mixed' condition. This pattern would suggest that social-signaling goals compete with referential informativity: when speakers must both identify the target and signal group membership, the resulting utterances may be harder for out-group members to understand.

Social-signaling accuracy For the social conditions ('social + mixed' and 'social-first + mixed'), we will conduct several additional analyses concerning the relationship between social signaling and referential success:

- **In-group vs. out-group referential accuracy.** We will test whether referential accuracy is lower for out-group versus in-group listeners using a logistic mixed-effects model: `ref_correct` $\sim$ `in_group_speaker` * `condition` + `(1 | speaker)` + `(1 | listener)` + `(condition | tangram)`. We predict that listeners will be more referentially accurate when the speaker is from the same group. The condition interaction tests whether the anticipated-goal manipulation amplifies this in-group advantage.
- **Social accuracy and group-specificity.** We will test whether games with higher social-signaling accuracy also have more group-specific expressions using a weighted least-squares

regression: `group_specif_phase2` ~ `social_accuracy` + `group_specif_phase1`, weighted by the inverse variance of the Phase 2 group-specificity estimates (as in the H1/H2 analysis). A positive effect of `social_accuracy` would provide direct evidence that distinctive, group-specific language helps with social identification.

Description length Description length (word count) serves two purposes: (a) replicating the classic finding that descriptions shorten over repeated interaction[6,53,63,69], and (b) disambiguating whether group-specificity differences arise from description content or simply from description length (since descriptions of a similar length may be more similar to one another in embedding space). If description length differs across conditions in a way that tracks group-specificity, we will include description length as a covariate in the group-specificity models as a robustness check.

We will fit: `utt_length` ~ `condition` * `block_num` + `(1 | participant)` + `(1 | tangram)`. We predict that in the 'refer + separated' condition, description length will continue to decrease or plateau as conventions stabilize. In the 'refer + mixed' condition, description length may initially increase as speakers accommodate unfamiliar listeners[69], then potentially decrease as new population-wide conventions emerge. In the 'social + mixed' and 'social-first + mixed' conditions, description length may remain relatively short, similar to the 'refer + separated' condition, as speakers maintain their group-specific conventions.

Utterance properties Both the 'refer + separated' and 'social + mixed' conditions are predicted to retain group-specific expressions, but the kinds of expressions that support within-group reference in isolation may differ systematically from those that function as social markers in mixed populations. We will apply the concreteness, lexical uniqueness, and word frequency measures defined above (see H3) across all four conditions to explore whether group-specific language that is maintained passively differs qualitatively from language that is maintained actively under social-signaling goals.

Temporal dynamics To visualize the evolution of lexical systems over time, we will track description length, accuracy, and semantic change across blocks. We will also visualize the trajectory of semantic change by projecting utterances into a 2D representational space using UMAP dimensionality reduction on SBERT embeddings. This will allow us to animate how descriptions drift, converge, or remain distinct across conditions over time.

Post-test survey and group identification check A potential concern is that participants in a brief online game with strangers may not develop a sufficiently meaningful sense of group identity for social-signaling goals to shape behavior. We will assess how much participants actually felt like they were part of a group. Participants will complete a post-test survey including questions about (a) perceived group identification strength ("How much did you feel a sense of being a part of your Phase 1 group?" on a 7-point Likert scale), (b) perceived closeness ("How close did you feel to the people in your Phase 1 group?" on a 7-point Likert scale), (c) awareness of group-specific language ("Did you notice your group developing its own way of describing the pictures?" yes/no), (d) strategy ("What was your strategy in the game?" open-ended), and (e) a check for perceived bot use ("Did you feel like you were playing with other humans?" yes/no). For the 'refer + separated' condition, where there is no Phase 2 group change, the questions refer to "your group" rather than "your Phase 1 group."

Pilot data (Section S1) did not include questions (a), (b), (c), and (e), but in open-ended responses, pilot participants spontaneously reported identity-signaling strategies (e.g., "I tried to speak in a different manner so my teammates would know me"; "my strategy was to describe the items as my team did so they could identify each other later in the game"), suggesting that the linguistic conventions participants develop can themselves act as social markers. These markers may generate a sense of affiliation even in the absence of pre-existing social ties[22,24,40,48].

Timeline for data collection

The experimental software, analysis pipeline, and recruitment workflow have all been developed and tested during pilot data collection, so data collection can begin immediately upon in-principle acceptance of the Stage 1 manuscript. We anticipate the following timeline:

- **Week 1-4: Data collection.** We will collect 80 games (20 per condition) via Prolific. Each game requires 9 concurrent participants and takes 45–60 minutes. Based on pilot recruitment rates, we can run multiple concurrent sessions and expect to complete data collection within approximately one month. We will recruit additional participants as needed to replace games lost to dropout or technical issues.
- **Week 5: Pre-registered analyses.** Our preprocessing and analysis pipeline is fully implemented, so running all primary and secondary analyses requires minimal additional effort.
- **Weeks 6-12: Exploratory analyses and manuscript preparation.** We will conduct the exploratory analyses, prepare figures, and write the Stage 2 manuscript.

We estimate submitting the Stage 2 manuscript within 3 months of in-principle acceptance. If unforeseen delays arise, we will negotiate a revised timeline with the handling editor.

Data availability

The pilot data for our study are available at https://github.com/aliciamchen/lexical-variation/. All anonymized data will be available in a public repository upon acceptance of the Stage 2 manuscript.

Code availability

The code for the power analyses, pilot experiments, and analyses are available at https://github.com/aliciamchen/lexical-variation/. All code and materials will be available in a public repository upon acceptance of the Stage 2 manuscript.

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Author contributions

A.M.C., R.S., and R.D.H. designed the study. A.M.C. developed the experimental software, collected and analyzed the data, and wrote the manuscript. R.S. and R.D.H. supervised the project and edited the manuscript. All authors approved the final version of the manuscript.

AI disclosure

Claude Code was used to assist with software development (experiment code, analysis pipeline, and tests) and editing the paper. All AI-generated content was reviewed, verified, and edited by the authors, who take full responsibility for the final work.

Competing interests

The authors declare no competing interests.

Table 1: Design table. For all contrasts, non-significant results will be evaluated with Bayes factors to distinguish true null effects from insufficient power (see Analysis plan).

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Q1: How do referential pressures shape lexical systems under isolation versus mixing?	H1: The 'refer + separated' condition will show higher group-specificity than the 'refer + mixed' condition.	We simulated the number of groups per condition needed to detect 50% of the maximal effect at 95% power at $\alpha = 0.05$, resulting in 20 games per condition.	Weighted least-squares regression predicting Phase 2 group-specificity from condition, with Phase 1 group-specificity as a predictor, followed by two-tailed planned contrasts.	*Support for H1* would indicate that referential pressures, operating through interaction patterns, causally shape lexical systems: when groups that have developed distinct conventions begin interacting, their lexicons converge toward a shared system. This would provide experimental evidence for the passive isolation mechanism documented in historical linguistics and dialect geography. *Evidence against H1* would suggest that conventions, once established, are resistant to change even when speakers encounter partners who use different expressions.
Q2: What pressures do social-signaling motivations exert when there is mixing?	H2: The 'social + mixed' condition will show higher group-specificity than the 'refer + mixed' condition.	Same as above.	Same as above, two-tailed planned contrast. H2 is most interpretable when H1 is supported, as H1 establishes a convergence pressure for social signaling to counteract. We will evaluate the H2 contrast regardless; see main text for interpretation if H1 is not supported.	*Support for H2* would demonstrate that explicit social-signaling goals produce greater lexical diversity under mixing. If H1 is also supported, this would indicate that identity motivations can counteract contact-driven convergence, providing causal evidence that social signaling sustains linguistic distinctiveness. If H1 is not supported, this would suggest that identity incentives actively drive differentiation in the presence of intergroup contact, even when contact alone leaves conventions stable. *Evidence against H2* would suggest that referential pressures dominate social signaling pressures, at least in our minimal-group paradigm. This could indicate that social signaling effects on language require stronger group identities, higher stakes, or longer timescales than our experiment provides. It could also suggest that participants pursue social signaling through means other than lexical choice.
Q3: Do social-signaling goals shape convention formation?	H3a: higher Phase 1 group-specificity; H3b: faster within-group convergence; H3c: less concrete, more lexically distinctive, lower-frequency vocabulary in 'social-first + mixed' vs. 'social + mixed'.	20 games per condition ($\sim$360 participants total).	H3a: WLS regression on Phase 1 group-specificity; H3b: condition * block interaction on within-group similarity; H3c: mixed-effects models predicting concreteness, lexical uniqueness, and word frequency.	*Support for H3* would indicate that anticipating social goals shapes convention formation even before the social-signaling occurs, suggesting that social-signaling motivations can influence the language that groups develop in the first place. *Evidence against H3* would suggest that convention formation is driven by referential pressures during interaction regardless of anticipated goals, and that social-signaling effects require the actual presence of out-group interlocutors.
Q4: Do differences in convention formation from social-signaling goals produce downstream consequences?	H4a: higher social-identification accuracy; H4b: lower referential accuracy in 'social-first + mixed' vs. 'social + mixed' in Phase 2.	Same as H3.	Logistic mixed-effects models predicting Phase 2 social (H4a) and referential (H4b) accuracy from condition and block.	*Support for both H4a and H4b* would demonstrate that anticipated social goals create functionally different conventions: better for identification but worse for reference, revealing a trade-off between communicative clarity and social distinctiveness. *Support for H4a but not H4b* would indicate that distinctive conventions aid social identification without referential cost. *Support for H4b but not H4a* would suggest conventions become harder to interpret without aiding social identification. *Evidence against both* would suggest that convention differences do not translate into measurable consequences for communicative success.

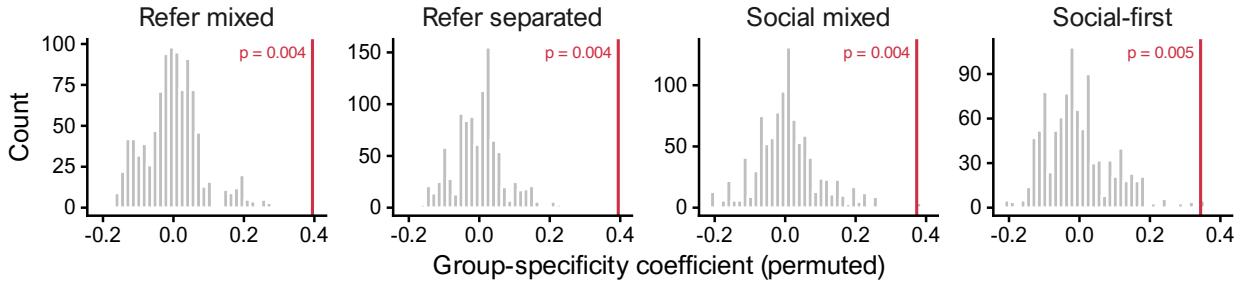


Figure S1: Permutation test for group-specificity in each pilot game. Histograms show the null distribution of the group-specificity coefficient under 1,000 random permutations of group labels. Red vertical lines indicate the observed coefficient. Group-specificity was significant ($p < .05$) in 4 of 4 games.

Supplementary information

S1 Pilot data

We conducted 4 pilot sessions (one per condition, $N = 36$ players) to verify that the experimental paradigm satisfies the outcome-neutral criteria described above and to identify any technical issues. All 4 pilot games used tangram set 0 as the target set. The median session duration was 42.7 minutes (range: 32.5–49.1).

Completion rates All games ran to completion. In 3 of 4 games, all three groups completed the game. In the remaining game (*refer* + *mixed*), 1 group was removed at the Phase 1–Phase 2 transition for failing the accuracy threshold, but the game continued with 2 active groups, yielding usable data under the dropout-handling rules described in the main text. 1 participant in the 'social-first + mixed' game was removed for idling.

Non-referential utterances Across the 4 pilot games, participants produced 1,215 chat messages and 288 unique rounds (72 per game). After concatenating speaker messages within each round, this yielded 820 speaker utterances containing 5,219 words total. A non-referential message classifier (Section S2) flagged 36 of 1,215 messages (3.0%) as non-referential. We manually scanned the classified utterances and refined the classifier prompt for later use. After filtering, the

analysis dataset contained 820 speaker utterances with 5,141 words.

Outcome-neutral criteria

The outcome-neutral criteria were all satisfied in the pilot data:

Convention formation Participants successfully formed conventions during Phase 1, replicating the classic patterns observed in reference game studies (Figure S2). We ran the linear models testing the outcome-neutral criteria as described in the main text. Description length decreased significantly over blocks ($b = -1.890$, $SE = 0.328$, $t = -5.76$, $p = .017$), from an average of 14.2 words in the first block to 4.0 words in the last. Listener accuracy increased over blocks ($b = 0.094$, $SE = 0.010$, $t = 9.75$, $p < .001$), rising from 0.39 in the first block to 0.92 in the last, well above chance ($1/16 \approx .06$). Adjacent-utterance cosine similarity also increased ($b = 0.062$, $SE = 0.011$, $t = 5.40$, $p = .003$), indicating that speakers converged on stable referring expressions within their groups.

Group-specificity Group-specificity scores at the end of Phase 1 exceeded chance levels in all pilot games (Figure S1), satisfying the criterion that $\geq 80\%$ of games show significant group-specificity.

Primary analysis previews

We visualized the group-specificity trajectories in the four pilot games (Figure S3). Even though there was just one game per condition, the visualizations show emerging distinctions between conditions. Note that in the plots, the error regions (95% bootstrapped confidence intervals) are computed over observations rather than games (since we only have one game per condition).

Figure S3a shows how within- and between-group similarity evolved over the course of the four games. Over the course of Phase 1, the descriptions within groups converge and the descriptions between groups diverge, across the 'refer + separated', 'refer + mixed', and 'social + mixed' games. In the 'refer + separated' game, these metrics plateaued. Mixing led between-group similarity to

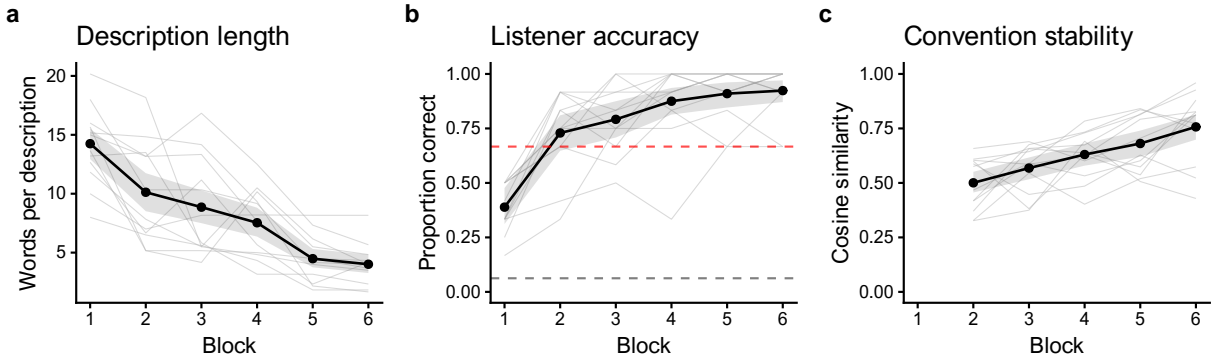


Figure S2: Convention formation during Phase 1 pilot sessions ($N = 36$ players across 4 games). (a) Mean description length (word count) decreased over blocks. (b) Listener accuracy (proportion correct) increased, exceeding chance (dashed line, $1/16$) from the first block. (c) Adjacent-utterance cosine similarity increased, indicating convergence on stable referring expressions. In all panels, thin gray lines show per-group mean trajectories (4 games $\times$ 3 groups), bold lines and points show the overall mean, and ribbons show 95% bootstrapped CIs (resampling individual observations).

increase but within-group similarity to stay the same, suggesting that mixing leads the groups to adopt each other's labels. The social-signaling goal counteracted this effect. Knowing about a social-signaling goal before Phase 1 led the conventions to start out more differentiated.

Figure S3b shows the difference between within- and between-group similarity, across conditions. Figure S3c shows the game-wide coefficients estimated using the pre-registered linear model in the main text. These values correspond directly to the primary pre-registered dependent variable for H1 and H2.

Secondary analysis previews

Figure S4 show how referential accuracy, description length, social-signaling accuracy, and convention stability evolved in the pilot games. Figure S5 previews the H3c description property measures for the two social conditions.

Technical issues and design changes. Pilot testing revealed a few issues that we plan to change in the full sample. First, there was a situation in which a listener's tangram selection submitted immediately before the stage timer expired could fail to register server-side, causing the participant

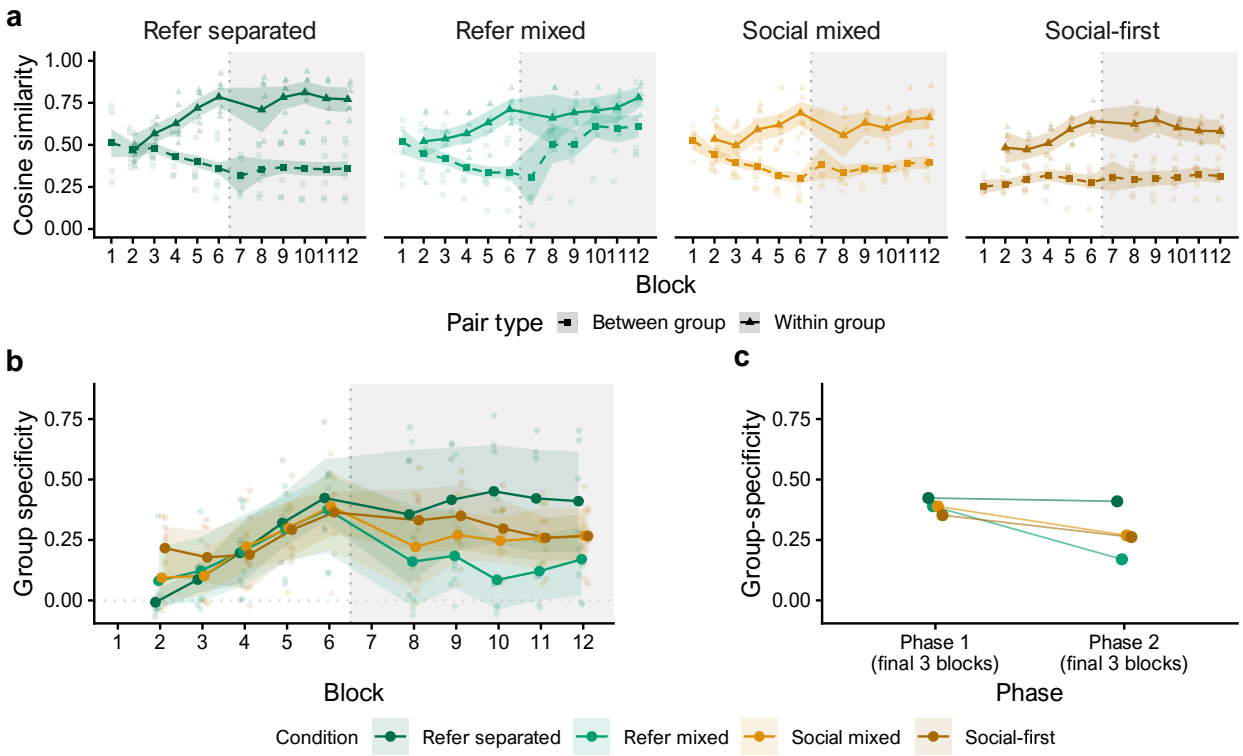


Figure S3: Group-specificity pilot results. (a) Within-group vs. between-group pairwise cosine similarity across Phase 1 and Phase 2, faceted by condition. Faded points show per-tangram means; lines and ribbons show condition means $\pm$ 95% bootstrapped CI. Gray shading marks Phase 2. (b) Group-specificity trajectory (within minus between similarity) across conditions. Faded points show per-tangram specificity scores. (c) Group-specificity coefficient for the final three blocks of Phase 1 vs. Phase 2. Each color is one pilot game; lines connect the same game across phases.

to be incorrectly marked as inactive. This was fixed by moving the auto-submit logic from the client to the server, ensuring that all selections made before the deadline are recorded regardless of network latency.

Additionally, the most frequent complaint in the post-test survey (raised independently by 6 of 33 respondents who provided open-ended feedback) was that speakers sometimes used most of the allotted time to type their description, leaving listeners only 1-2 seconds to read the message and click. In response, we added clearer language in the instructions about the time pressure and inactivity rules, and idle-round threshold was also adjusted from 2 to 3 rounds before removal. One participant reported a laptop crash during the session but was able to rejoin; no server-side disconnections were observed.

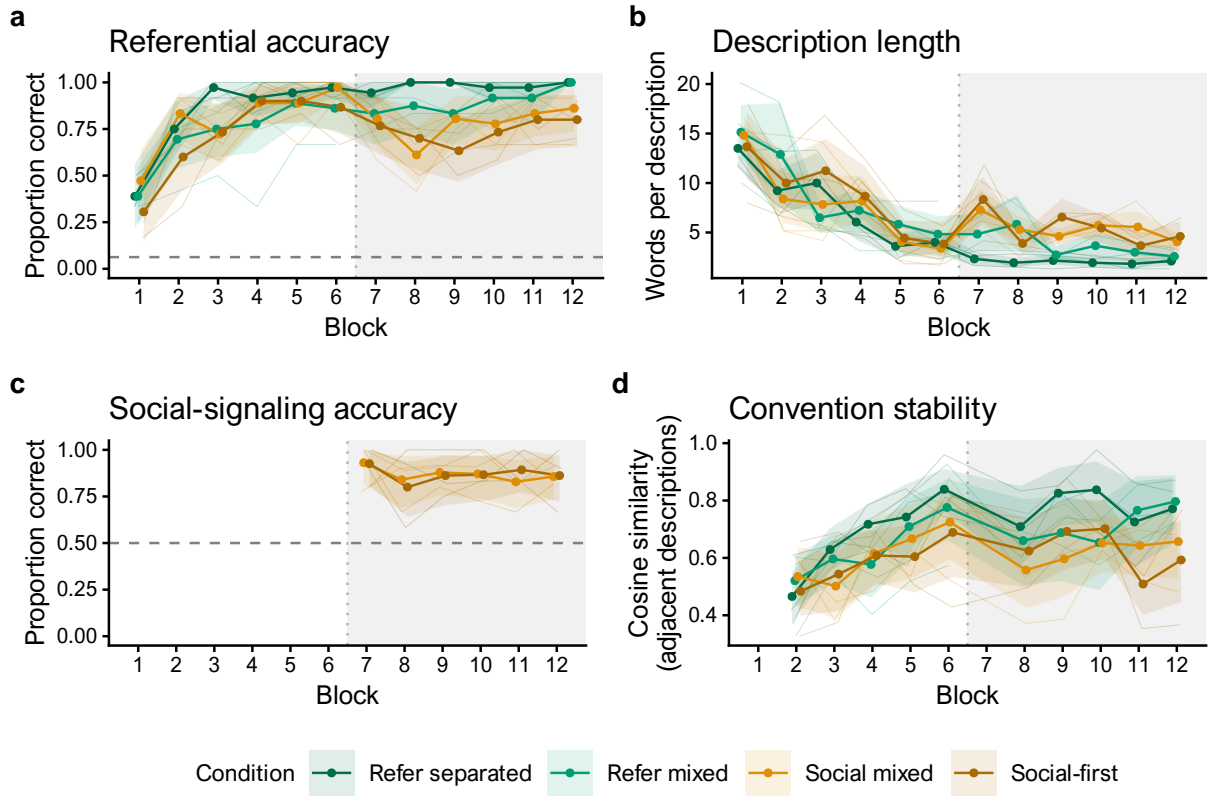


Figure S4: Secondary analysis previews from pilot data ($N = 4$ games). (a) Referential accuracy (proportion of listeners selecting the correct tangram) across both phases, by condition. Dashed line marks chance (1/16). (b) Speaker description length (word count) across both phases, by condition. (c) Social-signaling accuracy (proportion of correct group-membership guesses; social conditions only). Dashed line marks chance (50%). (d) Convention stability (cosine similarity between successive descriptions of the same tangram, across speakers within a group) across both phases, by condition. In all panels, gray shading marks Phase 2, thin lines show per-group trajectories, bold lines and points show condition means, and ribbons show 95% bootstrapped CIs (resampling individual observations).

Qualitative responses. Participants responded to two open-ended questions at the end of the task: one about their strategy and one for general feedback. Overall sentiment was strongly positive: participants described the task as enjoyable (e.g., "This was the most fun I've had on Prolific, and the most deeply invested I've been in a survey"; "THIS WAS SO FUN!"; "excellent study, I welcome contact if any future ones come up").

Strategically, participants reported converging on shared labels for the tangrams within their groups (e.g., "bunny," "birdhouse," "fireball," "whale"), consistent with the convention formation

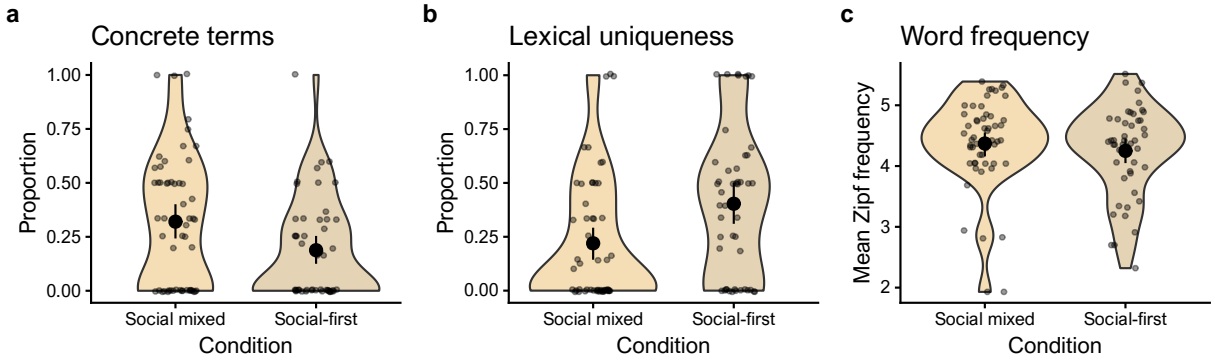


Figure S5: H3c description properties from pilot data ($N = 1$ game per social condition). (a) Concreteness: proportion of content words matching concrete term stems (geometric, body-part, positional, and posture terms). (b) Lexical uniqueness: proportion of content words not appearing in any other group's descriptions of the same tangram, across games. (c) Word frequency: mean Zipf frequency of content words (higher values indicate more common words). In all panels, violins show the distribution of utterance-level values, individual points are single utterances (final Phase 1 description per speaker per tangram), and black markers show bootstrapped means with 95% CIs.

patterns observed in the behavioral data. Several participants in the mixed conditions spontaneously described adapting their language when interacting with members of other groups: "My group started giving names to the shapes and we started recognizing them that way, and when we got mixed in the second phase, I started recognizing what other groups named shapes as well"; "I tried to use the same descriptions for my first phase partner and mix it up for the wider group of phase 2."

Notably, multiple participants – particularly in the social conditions – reported a sense of group membership and described deliberate strategies to signal their identity. Some participants explicitly marked their messages to be individually recognizable: one wrote "I used the same description every time, and got into the habit of using the word 'again' at the end, which gave me a unique label so that people could tie the description to me" ('social + mixed'), while another noted that a teammate began signing their messages and they adopted the same practice: "I noticed one of my teammates was signing his name in every description, so I started doing the same. I think it helped!" ('social-first + mixed'). Others described deliberately modulating their language depending on the audience: one participant reported trying "to speak in a different manner so my

teammates would know me" ('social-first + mixed'), while another described using "the same descriptions for my first phase partner and mix[ing] it up for the wider group of phase 2" ('social-first + mixed'). Several participants framed convention loyalty itself as an identity strategy: one wrote "my strategy was to describe the items as my team did. I was not the first to call a shape by a name, so I just used the same words my teammates used to describe the items. This way they knew we were all on the same page, and could identify each other later in the game" ('social-first + mixed'), and another reported trying to "find common phrasing and maintain consistency to alert initial group" ('social + mixed'). Even in the 'refer + mixed' condition, where no social identification incentive was present, one participant described trying "to pay attention to clues, use same ones when I was the speaker," and another reported trying to pick "words that coordinated to the picture across all groups." These spontaneous identity-signaling behaviors provide preliminary qualitative evidence that the experimental paradigm elicits the social-signaling motivations targeted by our design, with the strongest and most explicit signaling strategies emerging in the social conditions where group identification is incentivized.

S2 Non-referential message classifier

As described in the main text, we filter non-referential messages (e.g., "thanks", "good job") from speaker utterances before analysis, retaining only messages that contain information helpful for identifying the target tangram.

Classification approach We use a zero-shot LLM-based classifier (`gemini-2.0-flash` via Vertex AI) to label each speaker message as referential (R) or non-referential (NR). Messages are classified in batches of 30 with temperature $= 0$ for deterministic output. The classifier uses a detailed prompt specifying what counts as referential (visual descriptions, tangram labels, metaphorical references, typos of referential content, references to other players' conventions) and what counts as non-referential (social pleasantries, meta-game comments, greetings, pure acknowledgments with no tangram content). When in doubt, the classifier is instructed to default to referential, making the NR category narrow and conservative.

Examples Messages classified as non-referential include: "thanks", "Hi", "yes", "Get it", "sorry pc crashed", "We're grooving now", "Did you see that?". Messages classified as referential include all visual descriptions ("birdhouse", "triangle on top"), established conventions ("the dancer", "asteroid"), vague shape descriptions ("jumbled picture", "this shape is weird"), and references to other players' labels ("Or chess piece I heard one say").

Application Non-referential messages are removed before concatenating speaker messages into per-trial utterances. The filtered utterances (`speaker_utterances_filtered.csv`) are used for all downstream analyses, including SBERT embedding, description length, and group-specificity computation. The unfiltered data are preserved for comparison. Full prompt text and classification code are available in the project repository (`analysis/filter_nonreferential.py`).

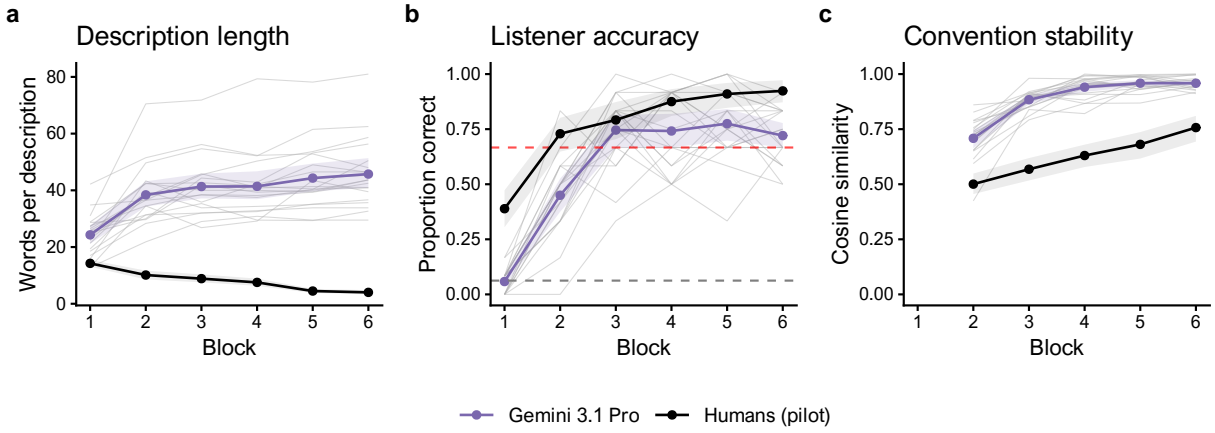


Figure S6: LLM simulation results compared to human pilot data. Purple: $N = 20$ groups of 3 `gemini-3.1-pro-preview` agents; black: human pilot participants ($N = 4$ games). Ribbons show 95% bootstrap CIs; gray lines show individual LLM groups. (a) LLM descriptions, but not human descriptions, became longer over blocks. (b) LLM groups, but not humans, plateaued after rapid early improvement; red dashed line shows the 2/3 exclusion threshold and the gray dashed line shows chance (1/16). (c) LLM groups' conventions were more stable (through near-verbatim copying)

S3 LLM agent convention formation

As described in the main text, we tested whether a state-of-the-art vision-language model could match the behavioral profile of human convention formation in the Phase 1 reference game. We evaluated both whether LLM groups could meet the group viability criterion (≥ 2 of 3 group members achieving $\geq 66.7\%$ referential accuracy across the last 3 blocks) and whether the trajectory of performance over blocks was qualitatively similar to that of human participants.

Simulation setup We simulated $N = 20$ independent groups of 3 `gemini-3.1-pro-preview` agents playing the Phase 1 reference game (6 blocks $\times$ 6 tangrams $= 36$ rounds per group). Each API call included all 16 tangram images (rendered as 400×400 px PNGs from the experiment SVGs) alongside a text prompt. The runs used temperature $= 1.0$, and nucleus sampling with top-$p = 0.95$. Each group used a distinct random seed for reproducibility.

The simulation matched the structure of the experiment:

- 3 agents per group; 1 speaker and 2 listeners per round, with the speaker role rotating each

block.

- Each agent saw the same 16 tangrams in a fixed, individually randomized grid order throughout the game (matching the experiment's "randomized across participants but fixed within each game" design).
- All agents had access to the full shared game history (descriptions and listener feedback from all previous rounds).
- Each listener independently selected a tangram from their own grid based on the speaker's text description. All agents received full feedback.

Prompts The speaker prompt instructed the model to describe the target tangram so that both listeners could identify it, to be concise, to avoid referencing grid position (since listeners see different orders), and to build on shared understanding from previous rounds. The listener prompt presented the speaker's description and asked the model to respond with only the number of the matching image. Both prompts were designed to strongly support success by providing clear task framing and encouraging convention formation. We deliberately did not include more aggressive instructions that explicitly direct the model to shorten descriptions by extracting salient tokens from prior messages and converging on a fixed expression – a strategy shown to elicit surface-level length reduction in simpler setups[70]. We expected that a participant using an LLM during a synchronous, time-pressured game is unlikely to engineer task-specific meta-instructions for adaptation. The prompt text is available in the project repository (`analysis/llm_simulation/llm_prompts.yaml`).

Results 15/20 groups (75%) met the exclusion criterion, with mean listener accuracy of 74.6% (SD = 12.1%) across the final 3 blocks. The model thus achieved sufficient referential accuracy to pass the group viability threshold. However, the behavioral profile of LLM agents was qualitatively distinguishable from human performance across all three trajectory measures (Figure S6).

Description length. Descriptions grew longer as the game progressed (block 1: 24.3 words; block 6: 45.7 words), the opposite of the shortening trajectory characteristic of human convention

formation[6,53]. Human pilot participants reduced descriptions from 14.2 to 4.0 words over the same period. Rather than developing compressed shorthand (e.g., "the dancer"), LLM agents produced increasingly verbose feature lists, often copying near-verbatim from history. This is consistent with evaluations showing that language models do not show strong reduction in message length across repetitions under standard prompting[70–72]. Note that our setup is substantially more challenging than the configurations used in prior LLM reference game evaluations: 16 images per grid (vs. 4[70,73] and 10[72]), 3-player groups with rotating speakers (vs. dyads), and tangram images rather than photographs.

Accuracy. LLM groups showed rapid early improvement (5.8% in block 1 to 74.6% in block 3) but then plateaued in the 72–78% range across blocks 4–6. Human pilot participants improved slower but did not plateau.

Description similarity. Cosine similarity between adjacent descriptions of the same tangram (computed using `paraphrase-MiniLM-L12-v2`) was 0.89 (SD = 0.04), substantially higher than in human pilot data. However, this high similarity reflected a different mechanism: LLM agents converged by copying previous descriptions with minor elaboration (see outputs available in the project repository). Human groups' descriptions, instead, became shorter and more idiosyncratic over time.